$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}}_{b}$$

CX1104: **Linear Algebra for Computing**

Chap. No : **6.1.1**

Lecture : **Orthogonality**

Topic : **Dot Product**

Concept : **Review of Vectors**

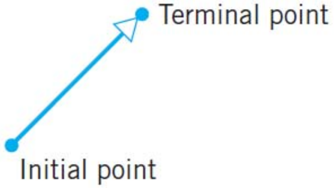
Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su, Vishal Choudhari**

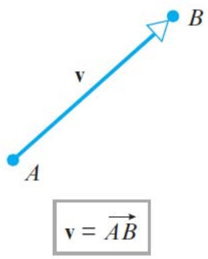
Rev: 25th June 2020

Geometric Vectors

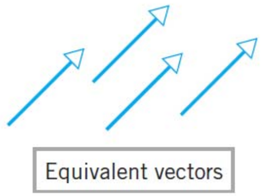
Geometric Vectors



▲ Figure 3.1.1



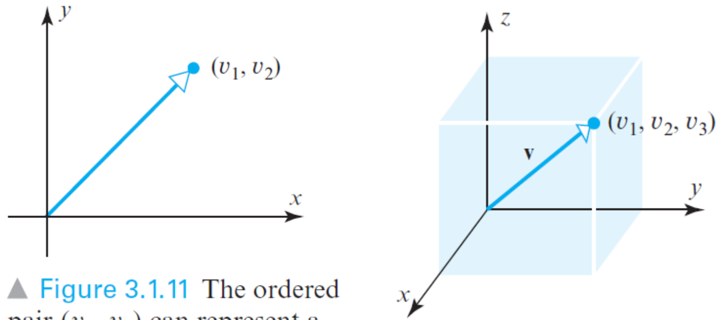
▲ Figure 3.1.2



▲ Figure 3.1.3

Vectors with the same length and direction, such as those in Figure 3.1.3, are said to be ***equivalent***. Since we want a vector to be determined solely by its length and direction, equivalent vectors are regarded as the same vector even though they may be in different positions. Equivalent vectors are also said to be ***equal***, which we indicate by writing

$$\mathbf{v} = \mathbf{w}$$



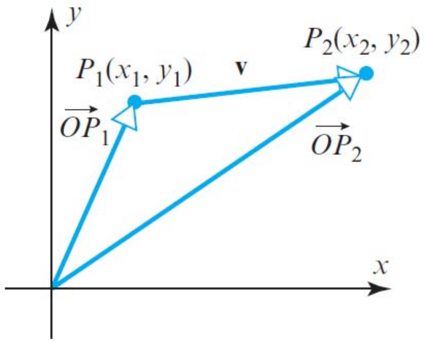
▲ Figure 3.1.11 The ordered pair (v_1, v_2) can represent a point or a vector.

If a vector $\mathbf{v}$ in 2-space or 3-space is positioned with its initial point at the origin of a rectangular coordinate system, then the vector is completely determined by the coordinates of its terminal point (Figure 3.1.11). We call these coordinates the ***components*** of $\mathbf{v}$ relative to the coordinate system. We will write $\mathbf{v} = (v_1, v_2)$ to denote a vector $\mathbf{v}$ in 2-space with components (v_1, v_2), and $\mathbf{v} = (v_1, v_2, v_3)$ to denote a vector $\mathbf{v}$ in 3-space with components (v_1, v_2, v_3).

Remark It may have occurred to you that an ordered pair (v_1, v_2) can represent either a vector with *components* v_1 and v_2 or a point with *coordinates* v_1 and v_2 (and similarly for ordered triples). Both are valid geometric interpretations, so the appropriate choice will depend on the geometric viewpoint that we want to emphasize (Figure 3.1.11).

Geometric Vectors

Vectors Whose Initial Point Is Not at the Origin



$$\boxed{\mathbf{v} = \overrightarrow{P_1P_2} = \overrightarrow{OP_2} - \overrightarrow{OP_1}}$$

▲ Figure 3.1.12

It is sometimes necessary to consider vectors whose initial points are not at the origin. If $\overrightarrow{P_1P_2}$ denotes the vector with initial point $P_1(x_1, y_1)$ and terminal point $P_2(x_2, y_2)$, then the components of this vector are given by the formula

$$\overrightarrow{P_1P_2} = (x_2 - x_1, y_2 - y_1) \tag{4}$$

As you might expect, the components of a vector in 3-space that has initial point $P_1(x_1, y_1, z_1)$ and terminal point $P_2(x_2, y_2, z_2)$ are given by

$$\overrightarrow{P_1P_2} = (x_2 - x_1, y_2 - y_1, z_2 - z_1) \tag{5}$$

▶ **EXAMPLE 1 Finding the Components of a Vector**

The components of the vector $\mathbf{v} = \overrightarrow{P_1P_2}$ with initial point $P_1(2, -1, 4)$ and terminal point $P_2(7, 5, -8)$ are

$$\mathbf{v} = (7 - 2, 5 - (-1), (-8) - 4) = (5, 6, -12)$$ ◀

DEFINITION 1 If n is a positive integer, then an ***ordered n-tuple*** is a sequence of n real numbers $(v_1, v_2, \ldots, v_n)$. The set of all ordered n-tuples is called ***n-space*** and is denoted by R^n.

Remark You can think of the numbers in an n-tuple $(v_1, v_2, \ldots, v_n)$ as either the coordinates of a *generalized point* or the components of a *generalized vector*, depending on the geometric image you want to bring to mind—the choice makes no difference mathematically, since it is the algebraic properties of n-tuples that are of concern.

Differences between a set and a tuple:

Set: An **unordered** collection of **distinct** objects.
Tuple: An **ordered** collection of objects.

1. A tuple may contain multiple instances of the same element, so tuple $(1, 2, 2, 3) \neq (1, 2, 3)$; but set $\{1, 2, 2, 3\} = \{1, 2, 3\}$.
2. Tuple elements are ordered: tuple $(1, 2, 3) \neq (3, 2, 1)$, but set $\{1, 2, 3\} = \{3, 2, 1\}$.

Ref: https://qr.ae/pNKbD4

Vectors in an Euclidean Space

Euclidean space

A. The basic vector space

We shall denote by $\mathbb{R}$ the field of real numbers. Then we shall use the Cartesian product $\mathbb{R}^n = \mathbb{R} \times \mathbb{R} \times \ldots \times \mathbb{R}$ of ordered n-tuples of real numbers (n factors). Typical notation for $x \in \mathbb{R}^n$ will be

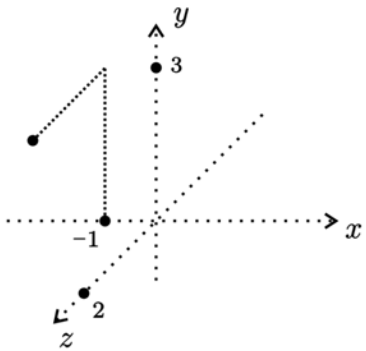
$$x = (x_1, x_2, \ldots, x_n).$$

Here x is called a *point* or a *vector*, and $x_1, x_2, \ldots, x_n$ are called the *coordinates* of x. The natural number n is called the *dimension* of the space. Often when speaking about $\mathbb{R}^n$ and its vectors, real numbers are called *scalars*.

Special notations:

$$\begin{array}{ll} \mathbb{R}^1 & x \\ \mathbb{R}^2 & x = (x_1, x_2) \quad \text{or} \quad p = (x, y) \\ \mathbb{R}^3 & x = (x_1, x_2, x_3) \quad \text{or} \quad p = (x, y, z). \end{array}$$

We like to draw pictures when $n = 1, 2, 3$; e.g. the point $(-1, 3, 2)$ might be depicted as



Ref: http://www.owlnet.rice.edu/~fjones/chap1.pdf

Different ways to represent a vector in R^n

Up to now we have been writing vectors in R^n using the notation

$$\mathbf{v} = (v_1, v_2, \ldots, v_n) \tag{15}$$

We call this the ***comma-delimited*** form. However, since a vector in R^n is just a list of its n components in a specific order, any notation that displays those components in the correct order is a valid way of representing the vector. For example, the vector in (15) can be written as

$$\mathbf{v} = [v_1 \quad v_2 \quad \cdots \quad v_n] \tag{16}$$

which is called ***row-vector*** form, or as

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \tag{17}$$

which is called ***column-vector*** form.

Ref: Elementary Linear Algebra (11th edition), Howard Anton, Chris Rorres (pg. 139), Wiley 2019

Vector Spaces

In this course, we deal with R^n, C^n

R^n: *Real coordinate space of dimension* n
R^n: *Also, set of all* $n-$ *tuples of real numbers*

C^n: *Complex coordinate space of dimension* n
C^n: *Also, set of all* $n-$ *tuples of complex numbers*

Examples

Some examples of vector spaces are

$$\mathbb{R}^2 = \left\{ \begin{pmatrix} a \\ b \end{pmatrix} \middle| a, b \in \mathbb{R} \right\} \quad \text{or} \quad \{(a,b) \mid a, b \in \mathbb{R}\}$$

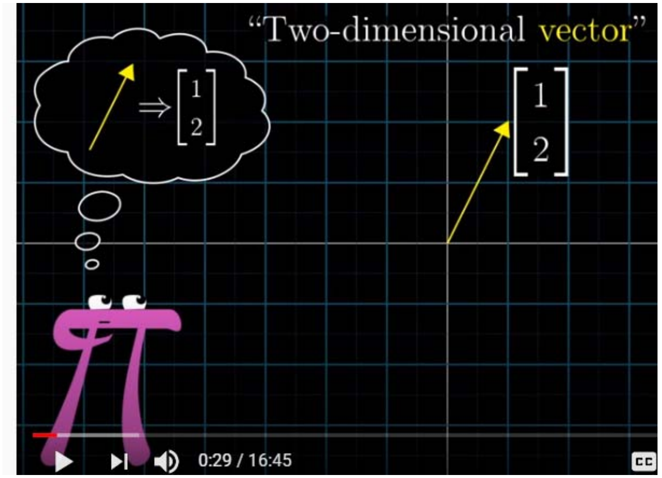
$$\mathbb{R}^3 = \left\{ \begin{pmatrix} a \\ b \\ c \end{pmatrix} \middle| a, b, c \in \mathbb{R} \right\} \quad \text{or} \quad \{(a,b,c) \mid a, b, c \in \mathbb{R}\}$$

$$\mathbb{R}^n = \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \middle| x_1, \ldots, x_n \in \mathbb{R} \right\} \quad \text{or} \quad \{(x_1,\ldots,x_n) \mid x_1,\ldots,x_n \in \mathbb{R}\}$$

$$\mathbb{C}^n = \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \middle| x_1, \ldots, x_n \in \mathbb{C} \right\} \quad \text{or} \quad \{(x_1,\ldots,x_n) \mid x_1,\ldots,x_n \in \mathbb{C}\}$$

Ref: Elementary Linear Algebra by James R. Kirkwood, Bessie H. Kirkwood

3Blue1Brown: "What are Vectors?"



Ref: https://www.youtube.com/watch?v=TgKwz5Ikpc8

Ref: https://www.youtube.com/watch?v=ozwodzD5bJM

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}}_{b}$$

CX1104: **Linear Algebra for Computing**

Chap. No : **6.1.2**

Lecture : **Orthogonality**

Topic : **Dot Product**

Concept : **Norm of a Vector and Unit Vectors**

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su, Vishal Choudhari**

Rev: 25th June 2020

Norm of a Vector

- Indicates the **length** or the **magnitude** of the vector.

In this text we will denote the length of a vector $\mathbf{v}$ by the symbol $\|\mathbf{v}\|$, which is read as the *norm* of $\mathbf{v}$, the *length* of $\mathbf{v}$, or the *magnitude* of $\mathbf{v}$ (the term "norm" being a common mathematical synonym for length). As suggested in Figure 3.2.1*a*, it follows from the Theorem of Pythagoras that the norm of a vector (v_1, v_2) in R^2 is

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2} \tag{1}$$

Similarly, for a vector (v_1, v_2, v_3) in R^3, it follows from Figure 3.2.1*b* and two applications of the Theorem of Pythagoras that

$$\|\mathbf{v}\|^2 = (OR)^2 + (RP)^2 = (OQ)^2 + (QR)^2 + (RP)^2 = v_1^2 + v_2^2 + v_3^2$$

and hence that

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2} \tag{2}$$

Motivated by the pattern of Formulas (1) and (2), we make the following definition.

DEFINITION 1 If $\mathbf{v} = (v_1, v_2, \dots, v_n)$ is a vector in R^n, then the ***norm*** of $\mathbf{v}$ (also called the ***length*** of $\mathbf{v}$ or the ***magnitude*** of $\mathbf{v}$) is denoted by $\|\mathbf{v}\|$, and is defined by the formula

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2} \tag{3}$$

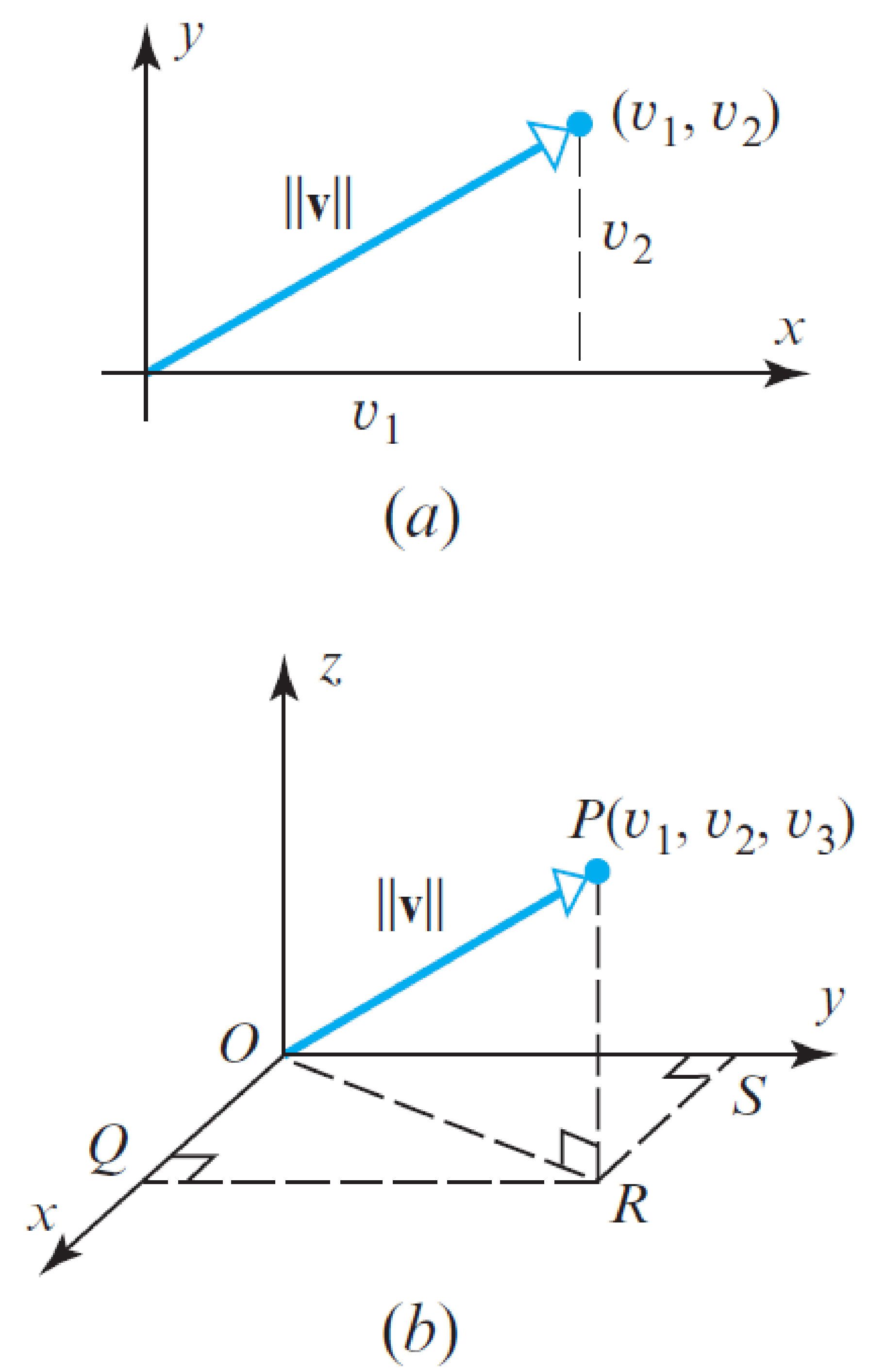


Figure 3.2.1

▶ **EXAMPLE 1 Calculating Norms**

It follows from Formula (2) that the norm of the vector $\mathbf{v} = (-3, 2, 1)$ in R^3 is

$$\|\mathbf{v}\| = \sqrt{(-3)^2 + 2^2 + 1^2} = \sqrt{14}$$

and it follows from Formula (3) that the norm of the vector $\mathbf{v} = (2, -1, 3, -5)$ in R^4 is

$$\|\mathbf{v}\| = \sqrt{2^2 + (-1)^2 + 3^2 + (-5)^2} = \sqrt{39}$$ ◀

Norm of a Vector

THEOREM 3.2.1 *If* $\mathbf{v}$ *is a vector in* R^n, *and if* k *is any scalar, then:*

(*a*) $\|\mathbf{v}\| \geq 0$

(*b*) $\|\mathbf{v}\| = 0$ *if and only if* $\mathbf{v} = \mathbf{0}$

(*c*) $\|k\mathbf{v}\| = |k|\|\mathbf{v}\|$

We will prove part (*c*) and leave (*a*) and (*b*) as exercises.

Proof (c) If $\mathbf{v} = (v_1, v_2, \ldots, v_n)$, then $k\mathbf{v} = (kv_1, kv_2, \ldots, kv_n)$, so

$$\begin{aligned}
\|k\mathbf{v}\| &= \sqrt{(kv_1)^2 + (kv_2)^2 + \cdots + (kv_n)^2} \\
&= \sqrt{(k^2)(v_1^2 + v_2^2 + \cdots + v_n^2)} \\
&= |k|\sqrt{v_1^2 + v_2^2 + \cdots + v_n^2} \\
&= |k|\|\mathbf{v}\| \quad \blacktriangleleft
\end{aligned}$$

Unit Length Vector

A vector of norm 1 is called a ***unit vector***. Such vectors are useful for specifying a direction when length is not relevant to the problem at hand. You can obtain a unit vector in a desired direction by choosing any *nonzero* vector $\mathbf{v}$ in that direction and multiplying $\mathbf{v}$ by the reciprocal of its length. For example, if $\mathbf{v}$ is a vector of length 2 in R^2 or R^3, then $\frac{1}{2}\mathbf{v}$ is a unit vector in the same direction as $\mathbf{v}$. More generally, if $\mathbf{v}$ is any nonzero vector in R^n, then

$$\mathbf{u} = \frac{1}{\|\mathbf{v}\|}\mathbf{v} \tag{4}$$

defines a unit vector that is in the same direction as $\mathbf{v}$. We can confirm that (4) is a unit vector by applying part (*c*) of Theorem 3.2.1 with $k = 1/\|\mathbf{v}\|$ to obtain

$$\|\mathbf{u}\| = \|k\mathbf{v}\| = |k|\|\mathbf{v}\| = k\|\mathbf{v}\| = \frac{1}{\|\mathbf{v}\|}\|\mathbf{v}\| = 1$$

The process of multiplying a nonzero vector by the reciprocal of its length to obtain a unit vector is called ***normalizing*** $\mathbf{v}$.

EXAMPLE 2 Normalizing a Vector

Find the unit vector $\mathbf{u}$ that has the same direction as $\mathbf{v} = (2, 2, -1)$.

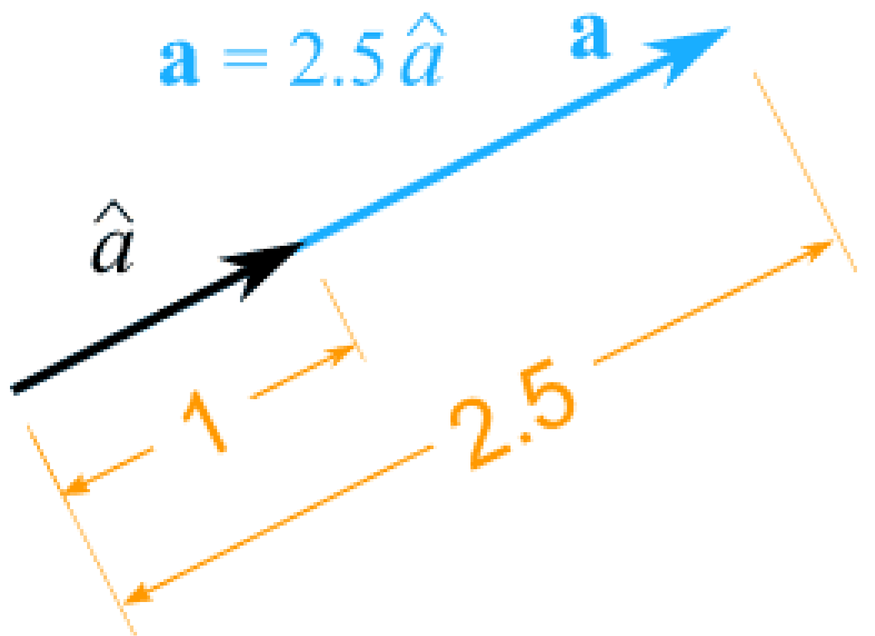
Solution The vector $\mathbf{v}$ has length

$$\|\mathbf{v}\| = \sqrt{2^2 + 2^2 + (-1)^2} = 3$$

Thus, from (4)

$$\mathbf{u} = \tfrac{1}{3}(2, 2, -1) = \left(\tfrac{2}{3}, \tfrac{2}{3}, -\tfrac{1}{3}\right)$$

As a check, you may want to confirm that $\|\mathbf{u}\| = 1$.



$\hat{a}$ is a unit vector with a direction that of vector a

Vector a is normalised to obtain the unit vector $\hat{a}$

Ref: https://www.mathsisfun.com/algebra/vector-unit.html

Examples

A vector whose length is 1 is called a **unit vector**. If we *divide* a nonzero vector **v** by its length—that is, multiply by $1/\|\mathbf{v}\|$—we obtain a unit vector **u** because the length of **u** is $(1/\|\mathbf{v}\|)\|\mathbf{v}\|$. The process of creating **u** from **v** is sometimes called **normalizing v**, and we say that **u** is *in the same direction* as **v**.

Several examples that follow use the space-saving notation for (column) vectors.

EXAMPLE 2 Let $\mathbf{v} = (1, -2, 2, 0)$. Find a unit vector **u** in the same direction as **v**.

SOLUTION First, compute the length of **v**:

$$\|\mathbf{v}\|^2 = \mathbf{v}\cdot\mathbf{v} = (1)^2 + (-2)^2 + (2)^2 + (0)^2 = 9$$
$$\|\mathbf{v}\| = \sqrt{9} = 3$$

Then, multiply **v** by $1/\|\mathbf{v}\|$ to obtain

$$\mathbf{u} = \frac{1}{\|\mathbf{v}\|}\mathbf{v} = \frac{1}{3}\mathbf{v} = \frac{1}{3}\begin{bmatrix} 1 \\ -2 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/3 \\ -2/3 \\ 2/3 \\ 0 \end{bmatrix}$$

To check that $\|\mathbf{u}\| = 1$, it suffices to show that $\|\mathbf{u}\|^2 = 1$.

$$\|\mathbf{u}\|^2 = \mathbf{u}\cdot\mathbf{u} = \left(\tfrac{1}{3}\right)^2 + \left(-\tfrac{2}{3}\right)^2 + \left(\tfrac{2}{3}\right)^2 + (0)^2$$
$$= \tfrac{1}{9} + \tfrac{4}{9} + \tfrac{4}{9} + 0 = 1$$

Note: $||v||^2 = v.v$

Derived in Slide 6 of Lecture 6.1.3 on Dot Product

Lay's Linear Algebra and Applications:

DEFINITION For **u** and **v** in $\mathbb{R}^n$, the **distance between u and v**, written as dist(**u**, **v**), is the length of the vector $\mathbf{u} - \mathbf{v}$. That is,

$$\text{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

In $\mathbb{R}^2$ and $\mathbb{R}^3$, this definition of distance coincides with the usual formulas for the Euclidean distance between two points, as the next two examples show.

EXAMPLE 4 Compute the distance between the vectors $\mathbf{u} = (7, 1)$ and $\mathbf{v} = (3, 2)$.

SOLUTION Calculate

$$\mathbf{u} - \mathbf{v} = \begin{bmatrix} 7 \\ 1 \end{bmatrix} - \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$
$$\|\mathbf{u} - \mathbf{v}\| = \sqrt{4^2 + (-1)^2} = \sqrt{17}$$

The vectors **u**, **v**, and $\mathbf{u} - \mathbf{v}$ are shown in Fig. 4. When the vector $\mathbf{u} - \mathbf{v}$ is added to **v**, the result is **u**. Notice that the parallelogram in Fig. 4 shows that the distance from **u** to **v** is the same as the distance from $\mathbf{u} - \mathbf{v}$ to **0**. ■

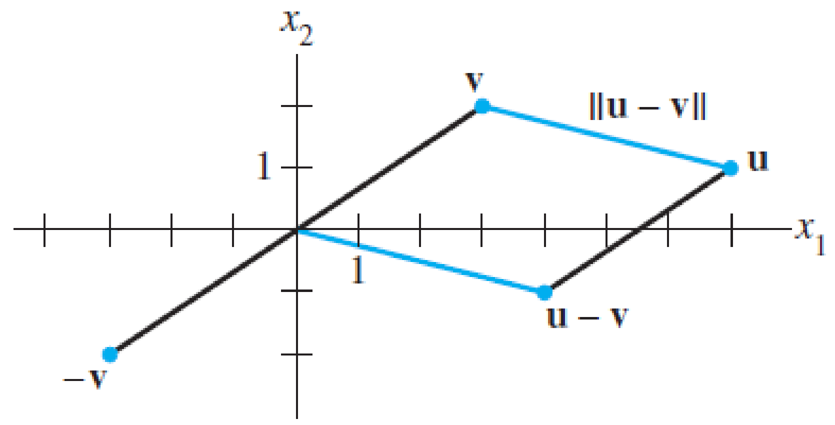


FIGURE 4 The distance between **u** and **v** is the length of $\mathbf{u} - \mathbf{v}$.

EXAMPLE 5 If $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$, then

$$\text{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{(\mathbf{u} - \mathbf{v})\cdot(\mathbf{u} - \mathbf{v})}$$
$$= \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + (u_3 - v_3)^2}$$

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}}_{b}$$

CX1104: **Linear Algebra for Computing**

Chap. No : **6.1.3**

Lecture : **Orthogonality**

Topic : **Dot Product**

Concept : **The Dot Product**

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su, Vishal Choudhari**

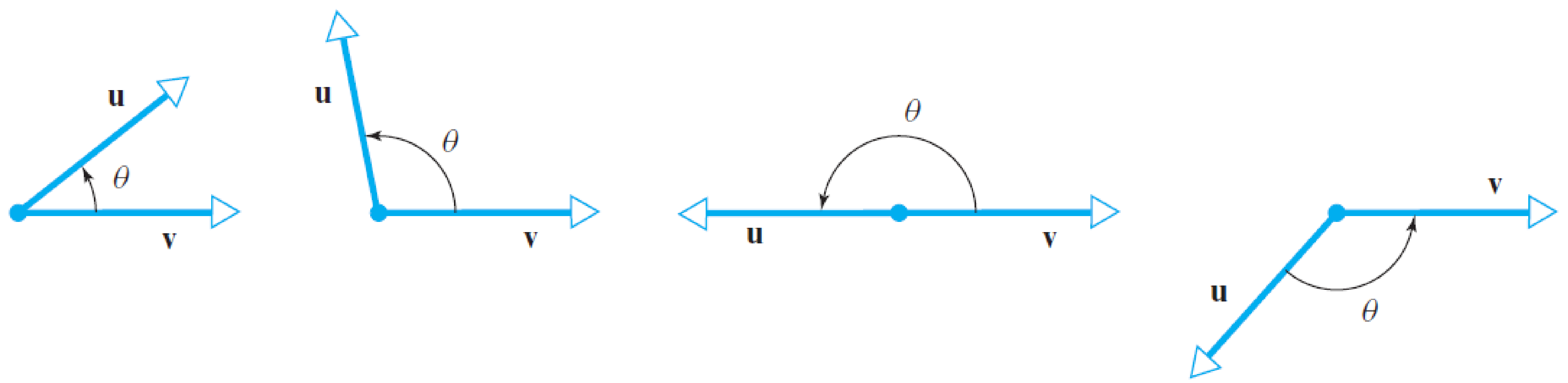
Rev: 25th June 2020

Dot Product

DEFINITION 3 If **u** and **v** are nonzero vectors in R^2 or R^3, and if θ is the angle between **u** and **v**, then the ***dot product*** (also called the ***Euclidean inner product***) of **u** and **v** is denoted by $\mathbf{u} \cdot \mathbf{v}$ and is defined as

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\| \cos\theta \tag{12}$$

If $\mathbf{u} = \mathbf{0}$ or $\mathbf{v} = \mathbf{0}$, then we define $\mathbf{u} \cdot \mathbf{v}$ to be 0.



The angle θ between **u** and **v** satisfies $0 \le \theta \le \pi$.

How to define "angle" between two vectors in R^2 or R^3? For this purpose, let **u** and **v** be nonzero vectors in R^2 or R^3 that have been positioned so that their initial points coincide. We define the ***angle between* u *and* v** to be the angle θ determined by **u** and **v** that satisfies the inequalities $0 \le \theta \le \pi$ (Figure 3.2.4).

The sign of the dot product reveals information about the angle θ that we can obtain by rewriting Formula (12) as

$$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} \tag{13}$$

Since $0 \le \theta \le \pi$, it follows from Formula (13) and properties of the cosine function studied in trigonometry that

- θ is acute if $\mathbf{u} \cdot \mathbf{v} > 0$.
- θ is obtuse if $\mathbf{u} \cdot \mathbf{v} < 0$.
- $\theta = \pi/2$ if $\mathbf{u} \cdot \mathbf{v} = 0$.

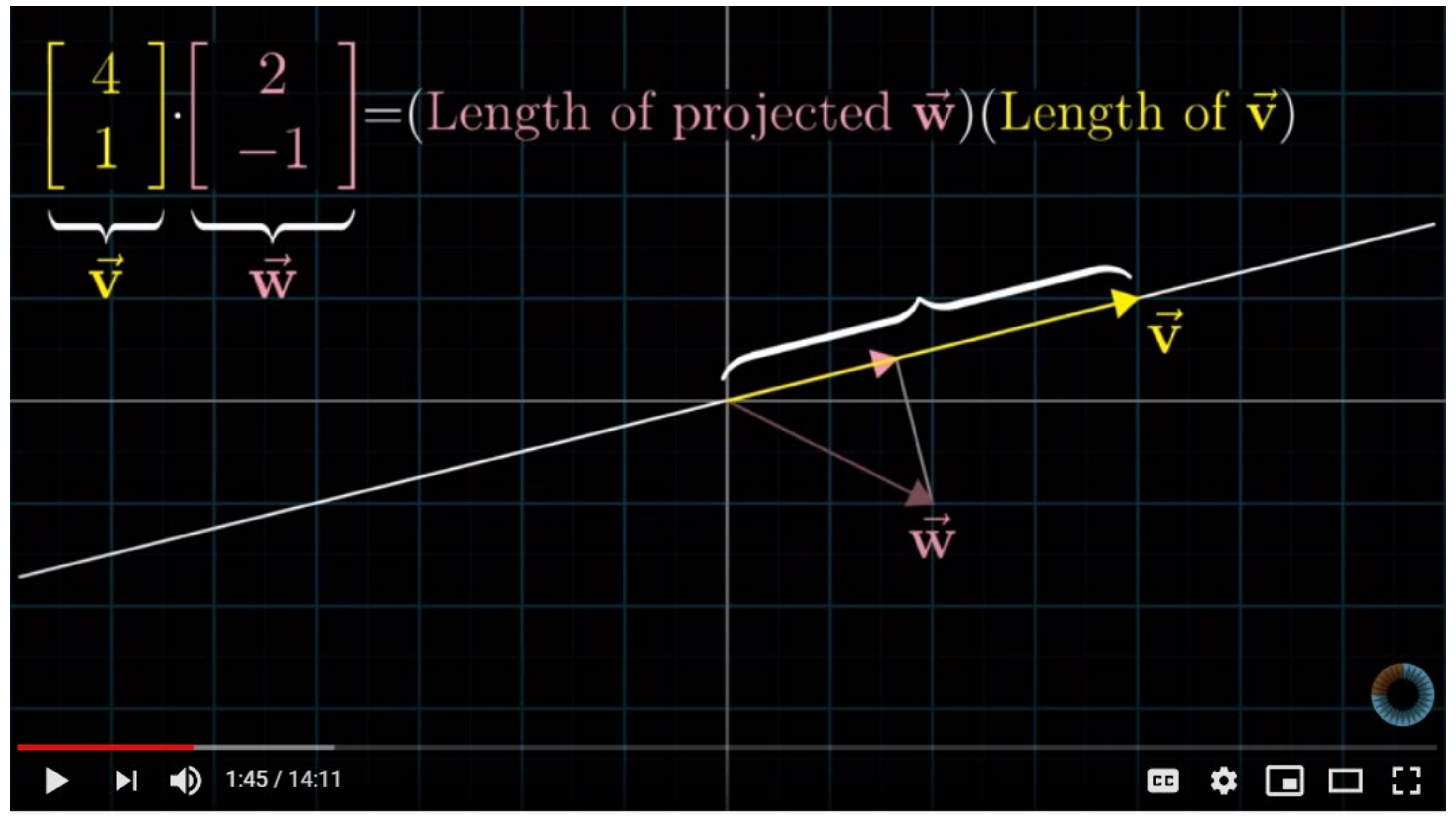
Dot product relates the length of two vectors and the angle (θ) between them.

If vectors u and v are unit vectors,
$i.e, ||v|| = ||u|| = 1,$
the dot product is $u.v = cos\theta$.

Ref:

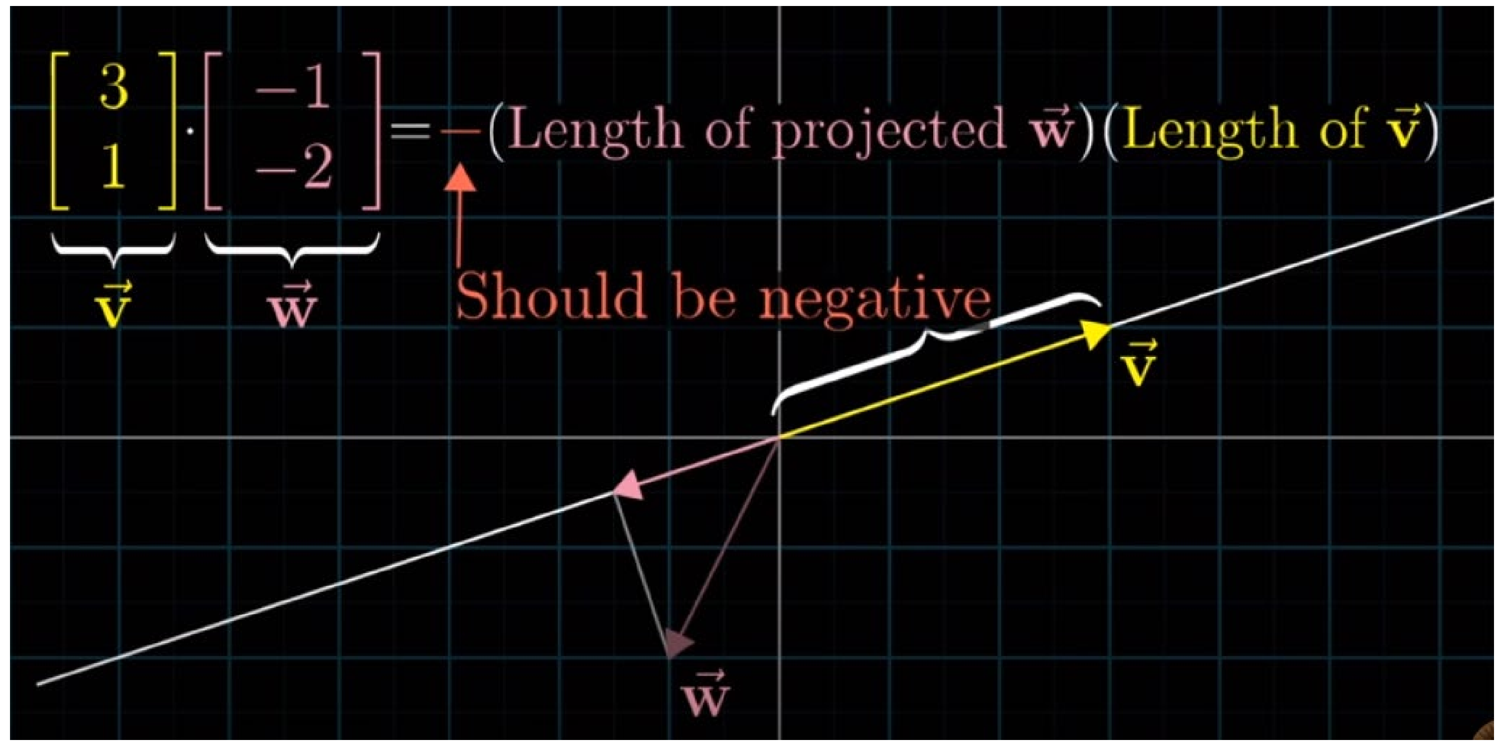
1. Stack Exchange
2. Khan Academy: https://www.youtube.com/watch?v=KDHuWxy53uM
3. 3Blue1Brown, Dot Product and Duality: https://www.youtube.com/watch?v=LyGKycYT2v0
4. MathsTheBeautiful: https://www.youtube.com/watch?v=QPkKWGq_V0U

Dot Product Interpretation



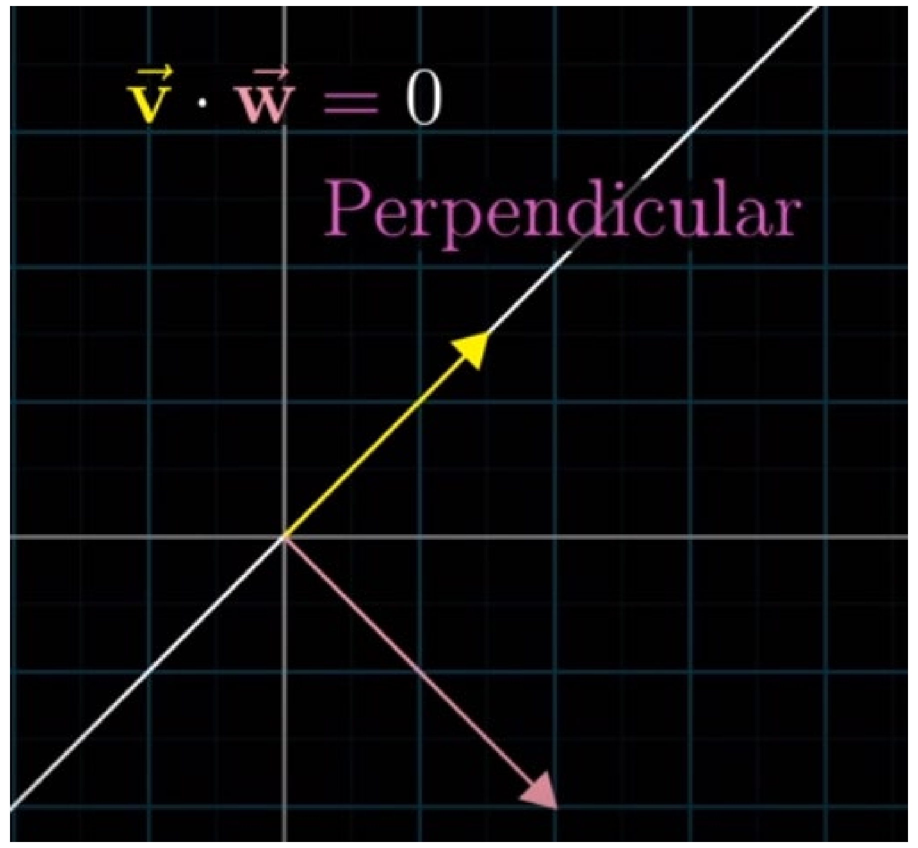
Case 1: When two vectors are pointing nearly towards the same direction, their dot product is +ve.

Angle between vectors: acute.



Case 2: When two vectors are pointing away from one another, their dot product is -ve.

Angle between vectors: obtuse.



Case 3: When two vectors are perpendicular, their dot product is zero.

$$v.w = \sum_{i=1}^{n} v_i w_i = v_1 w_1 + v_2 w_2 + \cdots + v_n$$

$$= ||v|| \times ||w|| \times cos\theta$$

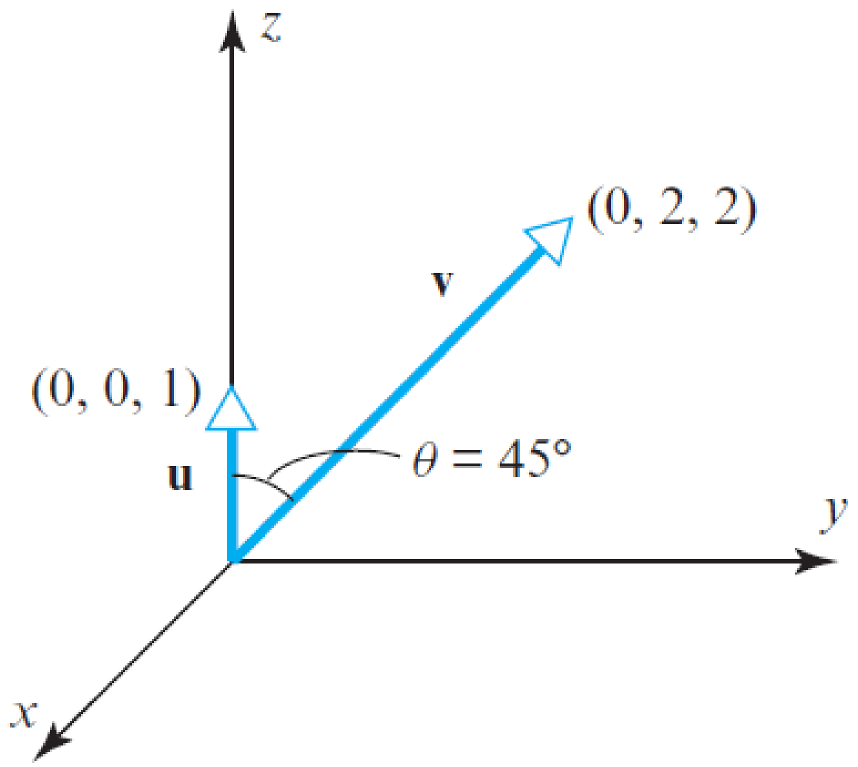
Derived in Slide 5

$$v.w = ||v|| \times ||w|| \times cos\theta$$

$$v.w = \underbrace{(||w|| \times cos\theta)}_{\text{Length of projection of } w \text{ onto } v} \times \underbrace{||v||}_{\text{Length of vector } v}$$

$$v.w = \underbrace{(||v|| \times cos\theta)}_{\text{Length of projection of } v \text{ onto } w} \times \underbrace{||w||}_{\text{Length of vector } w}$$

Example



▲ Figure 3.2.5

▶ EXAMPLE 5 Dot Product

Find the dot product of the vectors shown in Figure 3.2.5.

Solution The lengths of the vectors are

$$\|\mathbf{u}\| = 1 \quad \text{and} \quad \|\mathbf{v}\| = \sqrt{8} = 2\sqrt{2}$$

and the cosine of the angle θ between them is

$$\cos(45^\circ) = 1/\sqrt{2}$$

Thus, it follows from Formula (12) that

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\| \cos\theta = (1)(2\sqrt{2})(1/\sqrt{2}) = 2$$

Component Form of the Dot Product

Component Form of the Dot Product

For computational purposes it is desirable to have a formula that expresses the dot product of two vectors in terms of components. We will derive such a formula for vectors in 3-space; the derivation for vectors in 2-space is similar.

Let $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$ be two nonzero vectors. If, as shown in Figure 3.2.6, θ is the angle between $\mathbf{u}$ and $\mathbf{v}$, then the law of cosines yields

$$\|\overrightarrow{PQ}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\|\cos\theta \tag{14}$$

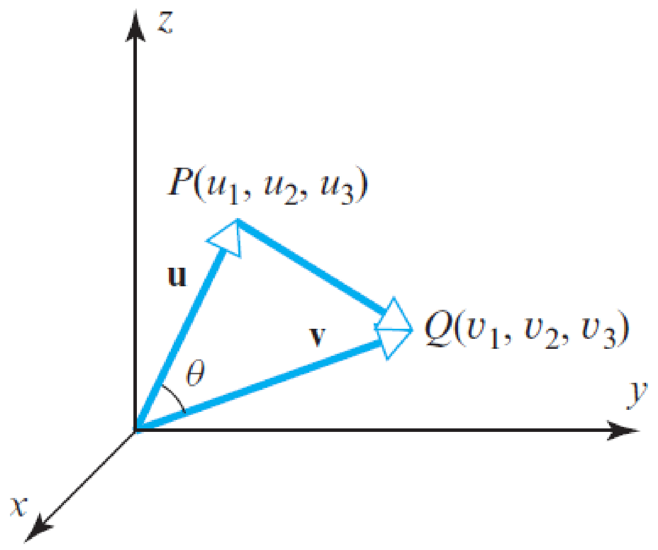


Figure 3.2.6

Since $\overrightarrow{PQ} = \mathbf{v} - \mathbf{u}$, we can rewrite (14) as

$$\|\mathbf{u}\|\|\mathbf{v}\|\cos\theta = \tfrac{1}{2}(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - \|\mathbf{v} - \mathbf{u}\|^2)$$

or

$$\mathbf{u} \cdot \mathbf{v} = \tfrac{1}{2}(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - \|\mathbf{v} - \mathbf{u}\|^2)$$

Substituting

$$\|\mathbf{u}\|^2 = u_1^2 + u_2^2 + u_3^2, \qquad \|\mathbf{v}\|^2 = v_1^2 + v_2^2 + v_3^2$$

and

$$\|\mathbf{v} - \mathbf{u}\|^2 = (v_1 - u_1)^2 + (v_2 - u_2)^2 + (v_3 - u_3)^2$$

we obtain, after simplifying,

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3 \tag{15}$$

The companion formula for vectors in 2-space is

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 \tag{16}$$

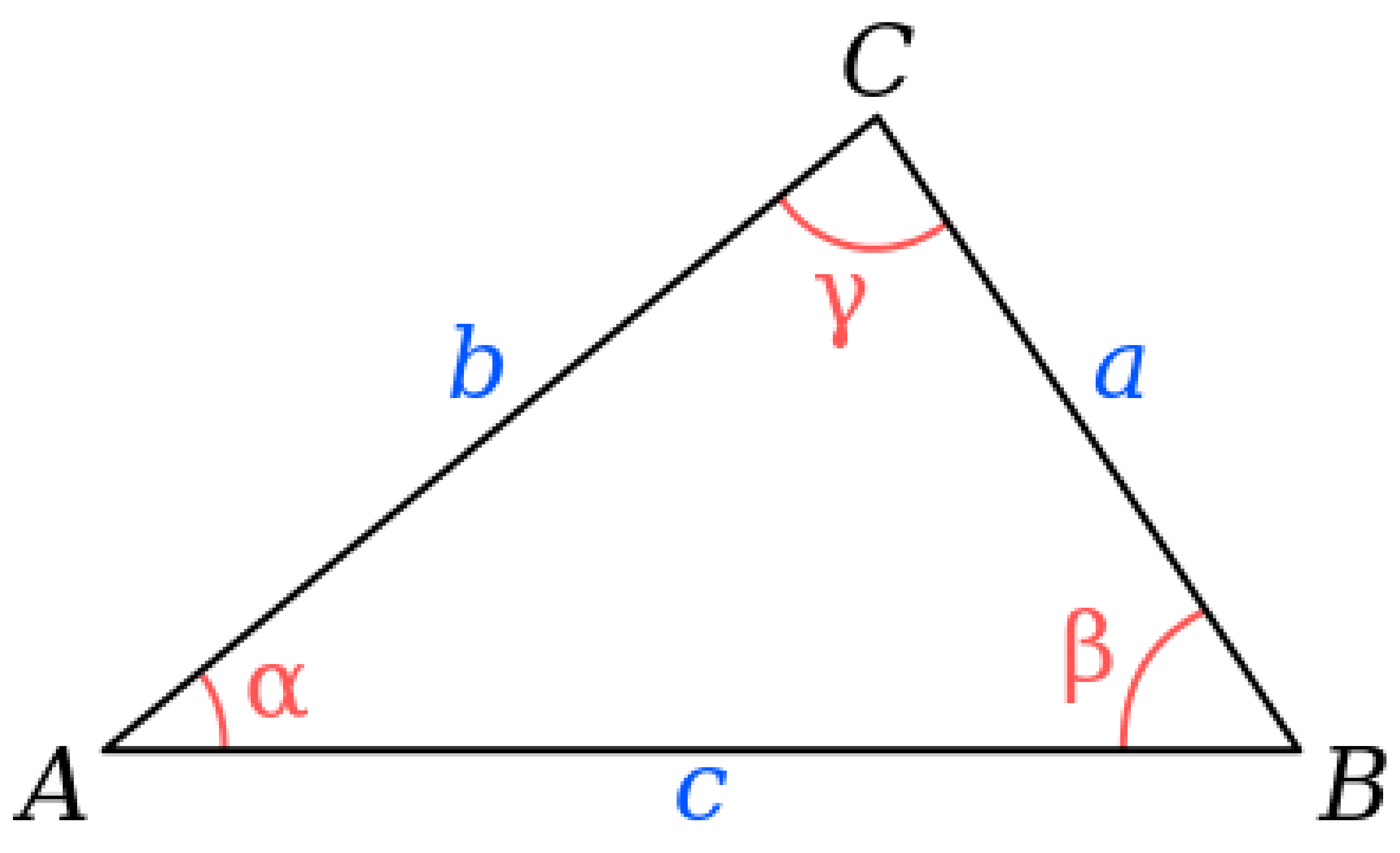
Motivated by the pattern in Formulas (15) and (16), we make the following definition.

Although we derived Formula (15) and its 2-space companion under the assumption that $\mathbf{u}$ and $\mathbf{v}$ are nonzero, it turned out that these formulas are also applicable if $\mathbf{u} = \mathbf{0}$ or $\mathbf{v} = \mathbf{0}$ (verify).

DEFINITION 4 If $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ and $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ are vectors in R^n, then the ***dot product*** (also called the ***Euclidean inner product***) of $\mathbf{u}$ and $\mathbf{v}$ is denoted by $\mathbf{u} \cdot \mathbf{v}$ and is defined by

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + \cdots + u_nv_n \tag{17}$$

Reviewing Law of Cosines



$$c^2 = a^2 + b^2 - 2ab\cos\gamma,$$

Example

▶ EXAMPLE 6 **Calculating Dot Products Using Components**

(a) Use Formula (15) to compute the dot product of the vectors **u** and **v** in Example 5.

(b) Calculate $\mathbf{u} \cdot \mathbf{v}$ for the following vectors in R^4:

$$\mathbf{u} = (-1, 3, 5, 7), \quad \mathbf{v} = (-3, -4, 1, 0)$$

Solution (a) The component forms of the vectors are $\mathbf{u} = (0, 0, 1)$ and $\mathbf{v} = (0, 2, 2)$. Thus,

$$\mathbf{u} \cdot \mathbf{v} = (0)(0) + (0)(2) + (1)(2) = 2$$

which agrees with the result obtained geometrically in Example 5.

Solution (b)

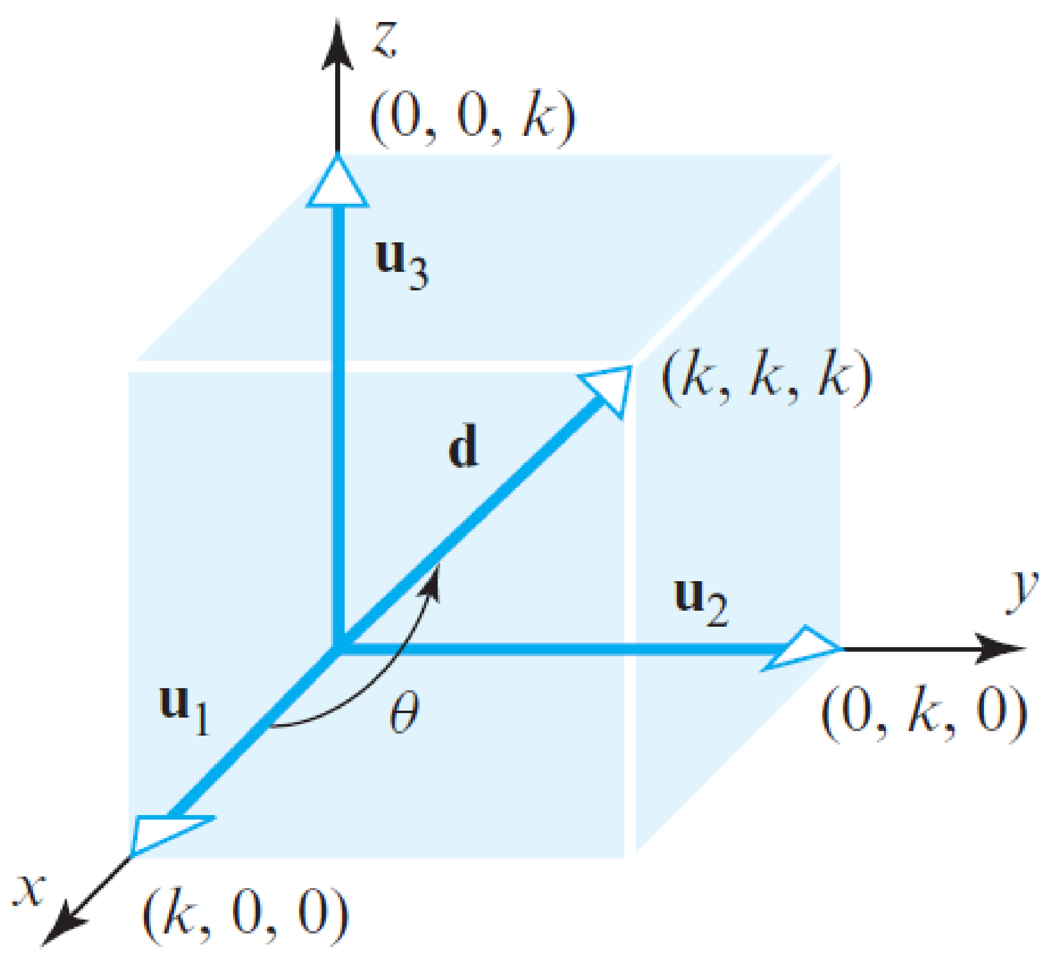
$$\mathbf{u} \cdot \mathbf{v} = (-1)(-3) + (3)(-4) + (5)(1) + (7)(0) = -4$$

In the special case where $\mathbf{u} = \mathbf{v}$ in Definition 4, we obtain the relationship

$$\mathbf{v} \cdot \mathbf{v} = v_1^2 + v_2^2 + \cdots + v_n^2 = \|\mathbf{v}\|^2 \tag{18}$$

This yields the following formula for expressing the length of a vector in terms of a dot product:

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} \tag{19}$$



▲ Figure 3.2.7

Note that the angle θ obtained in Example 7 does not involve k. Why was this to be expected?

▶ EXAMPLE 7 **A Geometry Problem Solved Using Dot Product**

Find the angle between a diagonal of a cube and one of its edges.

Solution Let k be the length of an edge and introduce a coordinate system as shown in Figure 3.2.7. If we let $\mathbf{u}_1 = (k, 0, 0)$, $\mathbf{u}_2 = (0, k, 0)$, and $\mathbf{u}_3 = (0, 0, k)$, then the vector

$$\mathbf{d} = (k, k, k) = \mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_3$$

is a diagonal of the cube. It follows from Formula (13) that the angle θ between **d** and the edge $\mathbf{u}_1$ satisfies

$$\cos\theta = \frac{\mathbf{u}_1 \cdot \mathbf{d}}{\|\mathbf{u}_1\|\|\mathbf{d}\|} = \frac{k^2}{(k)(\sqrt{3k^2})} = \frac{1}{\sqrt{3}}$$

With the help of a calculator we obtain

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right) \approx 54.74° \blacktriangleleft$$

Properties of Dot Product

Dot products have many of the same algebraic properties as products of real numbers.

THEOREM 3.2.2 *If* $\mathbf{u}$, $\mathbf{v}$, *and* $\mathbf{w}$ *are vectors in* R^n, *and if* k *is a scalar, then:*

(*a*) $\mathbf{u}\cdot\mathbf{v}=\mathbf{v}\cdot\mathbf{u}$ [Symmetry property]

(*b*) $\mathbf{u}\cdot(\mathbf{v}+\mathbf{w})=\mathbf{u}\cdot\mathbf{v}+\mathbf{u}\cdot\mathbf{w}$ [Distributive property]

(*c*) $k(\mathbf{u}\cdot\mathbf{v})=(k\mathbf{u})\cdot\mathbf{v}$ [Homogeneity property]

(*d*) $\mathbf{v}\cdot\mathbf{v}\geq 0$ *and* $\mathbf{v}\cdot\mathbf{v}=0$ *if and only if* $\mathbf{v}=\mathbf{0}$ [Positivity property]

Proof (c) Let $\mathbf{u}=(u_1,u_2,\ldots,u_n)$ and $\mathbf{v}=(v_1,v_2,\ldots,v_n)$. Then

$$\begin{aligned}k(\mathbf{u}\cdot\mathbf{v})&=k(u_1v_1+u_2v_2+\cdots+u_nv_n)\\&=(ku_1)v_1+(ku_2)v_2+\cdots+(ku_n)v_n=(k\mathbf{u})\cdot\mathbf{v}\end{aligned}$$

Proof (d) The result follows from parts (*a*) and (*b*) of Theorem 3.2.1 and the fact that

$$\mathbf{v}\cdot\mathbf{v}=v_1v_1+v_2v_2+\cdots+v_nv_n=v_1^2+v_2^2+\cdots+v_n^2=\|\mathbf{v}\|^2 \blacktriangleleft$$

THEOREM 3.2.3 *If* $\mathbf{u}$, $\mathbf{v}$, *and* $\mathbf{w}$ *are vectors in* R^n, *and if* k *is a scalar, then:*

(*a*) $\mathbf{0}\cdot\mathbf{v}=\mathbf{v}\cdot\mathbf{0}=0$

(*b*) $(\mathbf{u}+\mathbf{v})\cdot\mathbf{w}=\mathbf{u}\cdot\mathbf{w}+\mathbf{v}\cdot\mathbf{w}$

(*c*) $\mathbf{u}\cdot(\mathbf{v}-\mathbf{w})=\mathbf{u}\cdot\mathbf{v}-\mathbf{u}\cdot\mathbf{w}$

(*d*) $(\mathbf{u}-\mathbf{v})\cdot\mathbf{w}=\mathbf{u}\cdot\mathbf{w}-\mathbf{v}\cdot\mathbf{w}$

(*e*) $k(\mathbf{u}\cdot\mathbf{v})=\mathbf{u}\cdot(k\mathbf{v})$

Proof (b)

$$\begin{aligned}(\mathbf{u}+\mathbf{v})\cdot\mathbf{w}&=\mathbf{w}\cdot(\mathbf{u}+\mathbf{v}) && \text{[By symmetry]}\\&=\mathbf{w}\cdot\mathbf{u}+\mathbf{w}\cdot\mathbf{v} && \text{[By distributivity]}\\&=\mathbf{u}\cdot\mathbf{w}+\mathbf{v}\cdot\mathbf{w} && \text{[By symmetry]} \blacktriangleleft\end{aligned}$$

▶ EXAMPLE 8 Calculating with Dot Products

$$\begin{aligned}(\mathbf{u}-2\mathbf{v})\cdot(3\mathbf{u}+4\mathbf{v})&=\mathbf{u}\cdot(3\mathbf{u}+4\mathbf{v})-2\mathbf{v}\cdot(3\mathbf{u}+4\mathbf{v})\\&=3(\mathbf{u}\cdot\mathbf{u})+4(\mathbf{u}\cdot\mathbf{v})-6(\mathbf{v}\cdot\mathbf{u})-8(\mathbf{v}\cdot\mathbf{v})\\&=3\|\mathbf{u}\|^2-2(\mathbf{u}\cdot\mathbf{v})-8\|\mathbf{v}\|^2 \blacktriangleleft\end{aligned}$$

Property and Example

Table 1

Form	Dot Product	Example	
$\mathbf{u}$ a column matrix and $\mathbf{v}$ a column matrix	$\mathbf{u}\cdot\mathbf{v}=\mathbf{u}^T\mathbf{v}=\mathbf{v}^T\mathbf{u}$	$\mathbf{u}=\begin{bmatrix}1\\-3\\5\end{bmatrix}$ $\mathbf{v}=\begin{bmatrix}5\\4\\0\end{bmatrix}$	$\mathbf{u}^T\mathbf{v}=[1\quad -3\quad 5]\begin{bmatrix}5\\4\\0\end{bmatrix}=-7$ $\mathbf{v}^T\mathbf{u}=[5\quad 4\quad 0]\begin{bmatrix}1\\-3\\5\end{bmatrix}=-7$

If A is an $n\times n$ matrix and $\mathbf{u}$ and $\mathbf{v}$ are $n\times 1$ matrices, then it follows from the first row in Table 1 and properties of the transpose that

$$A\mathbf{u}\cdot\mathbf{v}=\mathbf{v}^T(A\mathbf{u})=(\mathbf{v}^TA)\mathbf{u}=(A^T\mathbf{v})^T\mathbf{u}=\mathbf{u}\cdot A^T\mathbf{v}$$
$$\mathbf{u}\cdot A\mathbf{v}=(A\mathbf{v})^T\mathbf{u}=(\mathbf{v}^TA^T)\mathbf{u}=\mathbf{v}^T(A^T\mathbf{u})=A^T\mathbf{u}\cdot\mathbf{v}$$

The resulting formulas

$$A\mathbf{u}\cdot\mathbf{v}=\mathbf{u}\cdot A^T\mathbf{v} \tag{26}$$

$$\mathbf{u}\cdot A\mathbf{v}=A^T\mathbf{u}\cdot\mathbf{v} \tag{27}$$

provide an important link between multiplication by an $n\times n$ matrix A and multiplication by A^T.

EXAMPLE 9 Verifying that $A\mathbf{u}\cdot\mathbf{v}=\mathbf{u}\cdot A^T\mathbf{v}$

Suppose that

$$A=\begin{bmatrix}1&-2&3\\2&4&1\\-1&0&1\end{bmatrix},\quad \mathbf{u}=\begin{bmatrix}-1\\2\\4\end{bmatrix},\quad \mathbf{v}=\begin{bmatrix}-2\\0\\5\end{bmatrix}$$

Then

$$A\mathbf{u}=\begin{bmatrix}1&-2&3\\2&4&1\\-1&0&1\end{bmatrix}\begin{bmatrix}-1\\2\\4\end{bmatrix}=\begin{bmatrix}7\\10\\5\end{bmatrix}$$

$$A^T\mathbf{v}=\begin{bmatrix}1&2&-1\\-2&4&0\\3&1&1\end{bmatrix}\begin{bmatrix}-2\\0\\5\end{bmatrix}=\begin{bmatrix}-7\\4\\-1\end{bmatrix}$$

from which we obtain

$$A\mathbf{u}\cdot\mathbf{v}=7(-2)+10(0)+5(5)=11$$
$$\mathbf{u}\cdot A^T\mathbf{v}=(-1)(-7)+2(4)+4(-1)=11$$

Thus, $A\mathbf{u}\cdot\mathbf{v}=\mathbf{u}\cdot A^T\mathbf{v}$ as guaranteed by Formula (26). We leave it for you to verify that Formula (27) also holds.

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}}_{b}$$

CX1104: **Linear Algebra for Computing**

Chap. No : **6.1.4**
Lecture : **Orthogonality**
Topic : **Dot Product**
Concept : **Important Inequalities**

Instructor: **A/P Chng Eng Siong**
TAs: **Zhang Su, Vishal Choudhari**

Rev: 25th June 2020

Cauchy-Schwarz Inequality

DEFINITION 3 If $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors in R^2 or R^3, and if θ is the angle between $\mathbf{u}$ and $\mathbf{v}$, then the ***dot product*** (also called the ***Euclidean inner product***) of $\mathbf{u}$ and $\mathbf{v}$ is denoted by $\mathbf{u} \cdot \mathbf{v}$ and is defined as

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\| \cos\theta \tag{12}$$

If $\mathbf{u} = \mathbf{0}$ or $\mathbf{v} = \mathbf{0}$, then we define $\mathbf{u} \cdot \mathbf{v}$ to be 0.

$$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} \Rightarrow \theta = \cos^{-1}\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}\right)$$

Formula (20) is not defined unless its argument satisfies the inequalities

$$-1 \le \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} \le 1 \tag{21}$$

Fortunately, these inequalities *do* hold for all nonzero vectors in R^n as a result of the following fundamental result known as the ***Cauchy–Schwarz inequality***.

THEOREM 3.2.4 Cauchy–Schwarz Inequality

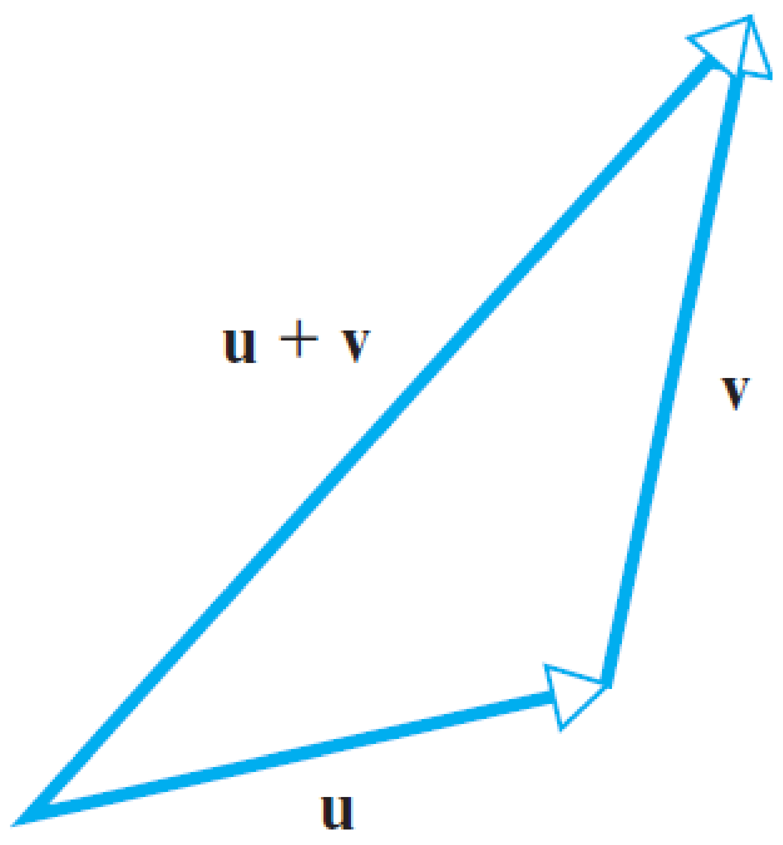
If $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ *and* $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ *are vectors in* R^n*, then*

$$|\mathbf{u} \cdot \mathbf{v}| \le \|\mathbf{u}\|\|\mathbf{v}\| \tag{22}$$

or in terms of components

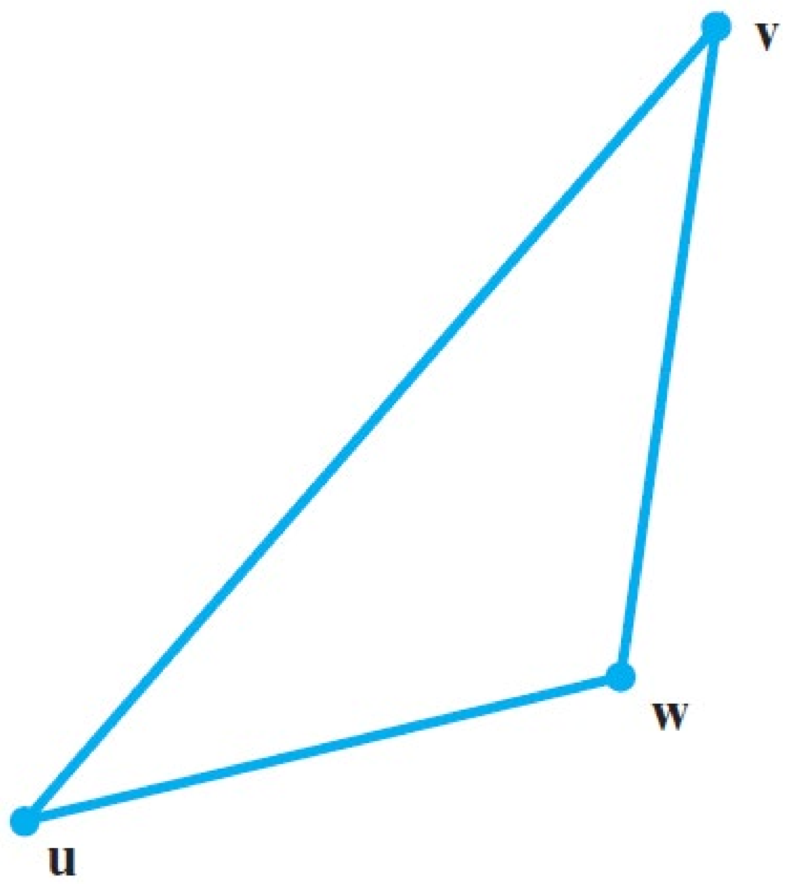
$$|u_1v_1 + u_2v_2 + \cdots + u_nv_n| \le (u_1^2 + u_2^2 + \cdots + u_n^2)^{1/2}(v_1^2 + v_2^2 + \cdots + v_n^2)^{1/2} \tag{23}$$

Triangle Inequality



$$\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$$

▲ Figure 3.2.8



$$d(\mathbf{u}, \mathbf{v}) \le d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$$

▲ Figure 3.2.9

THEOREM 3.2.5 *If* **u**, **v**, *and* **w** *are vectors in* R^n, *then:*

(*a*) $\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$ **[Triangle inequality for vectors]**

(*b*) $d(\mathbf{u}, \mathbf{v}) \le d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$ **[Triangle inequality for distances]**

Proof:

Proof (a)

$$\begin{aligned}
\|\mathbf{u} + \mathbf{v}\|^2 &= (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) = (\mathbf{u} \cdot \mathbf{u}) + 2(\mathbf{u} \cdot \mathbf{v}) + (\mathbf{v} \cdot \mathbf{v}) \\
&= \|\mathbf{u}\|^2 + 2(\mathbf{u} \cdot \mathbf{v}) + \|\mathbf{v}\|^2 \\
&\le \|\mathbf{u}\|^2 + 2|\mathbf{u} \cdot \mathbf{v}| + \|\mathbf{v}\|^2 && \longleftarrow \text{Property of absolute value} \\
&\le \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}\| + \|\mathbf{v}\|^2 && \longleftarrow \text{Cauchy–Schwarz inequality} \\
&= (\|\mathbf{u}\| + \|\mathbf{v}\|)^2
\end{aligned}$$

This completes the proof since both sides of the inequality in part (*a*) are nonnegative.

Proof (b) It follows from part (*a*) and Formula (11) that

$$\begin{aligned}
d(\mathbf{u}, \mathbf{v}) &= \|\mathbf{u} - \mathbf{v}\| = \|(\mathbf{u} - \mathbf{w}) + (\mathbf{w} - \mathbf{v})\| \\
&\le \|\mathbf{u} - \mathbf{w}\| + \|\mathbf{w} - \mathbf{v}\| = d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v}) \quad ◀
\end{aligned}$$

Parallelogram Equation for Vectors

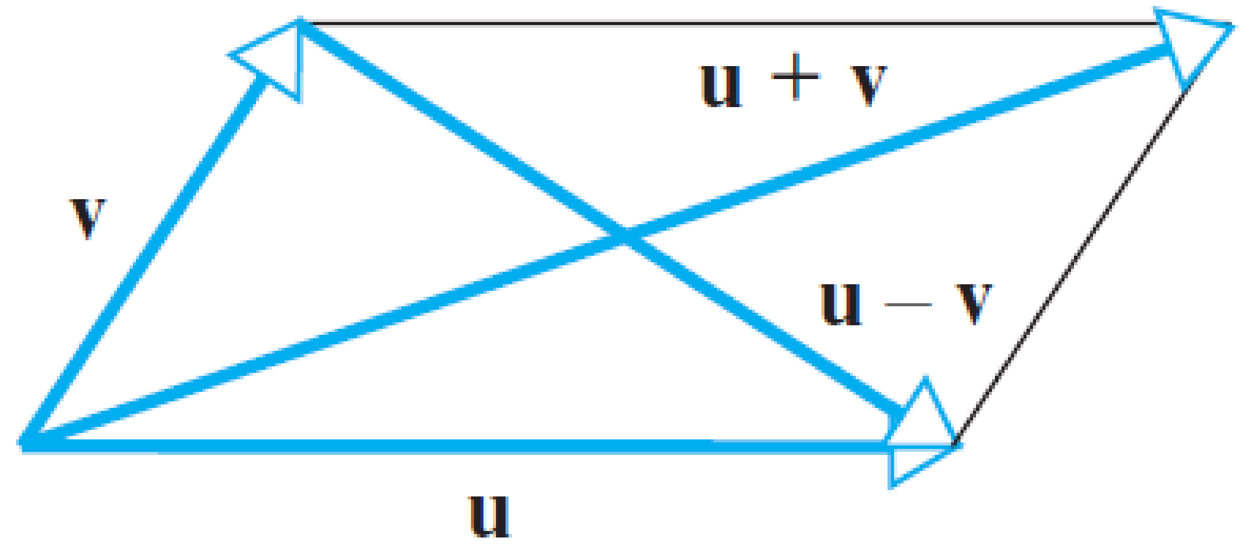


Figure 3.2.10

THEOREM 3.2.6 **Parallelogram Equation for Vectors**

If **u** *and* **v** *are vectors in* R^n, *then*

$$\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2\left(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2\right) \tag{24}$$

Proof:

THEOREM 3.2.6 **Parallelogram Equation for Vectors**

If **u** *and* **v** *are vectors in* R^n, *then*

$$\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2\left(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2\right) \tag{24}$$

Proof

$$\begin{aligned}\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 &= (\mathbf{u}+\mathbf{v})\cdot(\mathbf{u}+\mathbf{v}) + (\mathbf{u}-\mathbf{v})\cdot(\mathbf{u}-\mathbf{v}) \\ &= 2(\mathbf{u}\cdot\mathbf{u}) + 2(\mathbf{v}\cdot\mathbf{v}) \\ &= 2\left(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2\right) \blacktriangleleft\end{aligned}$$

THEOREM 3.2.7 *If* **u** *and* **v** *are vectors in* R^n *with the Euclidean inner product, then*

$$\mathbf{u}\cdot\mathbf{v} = \tfrac{1}{4}\|\mathbf{u}+\mathbf{v}\|^2 - \tfrac{1}{4}\|\mathbf{u}-\mathbf{v}\|^2 \tag{25}$$

Proof

$$\|\mathbf{u}+\mathbf{v}\|^2 = (\mathbf{u}+\mathbf{v})\cdot(\mathbf{u}+\mathbf{v}) = \|\mathbf{u}\|^2 + 2(\mathbf{u}\cdot\mathbf{v}) + \|\mathbf{v}\|^2$$
$$\|\mathbf{u}-\mathbf{v}\|^2 = (\mathbf{u}-\mathbf{v})\cdot(\mathbf{u}-\mathbf{v}) = \|\mathbf{u}\|^2 - 2(\mathbf{u}\cdot\mathbf{v}) + \|\mathbf{v}\|^2$$

from which (25) follows by simple algebra. ◀

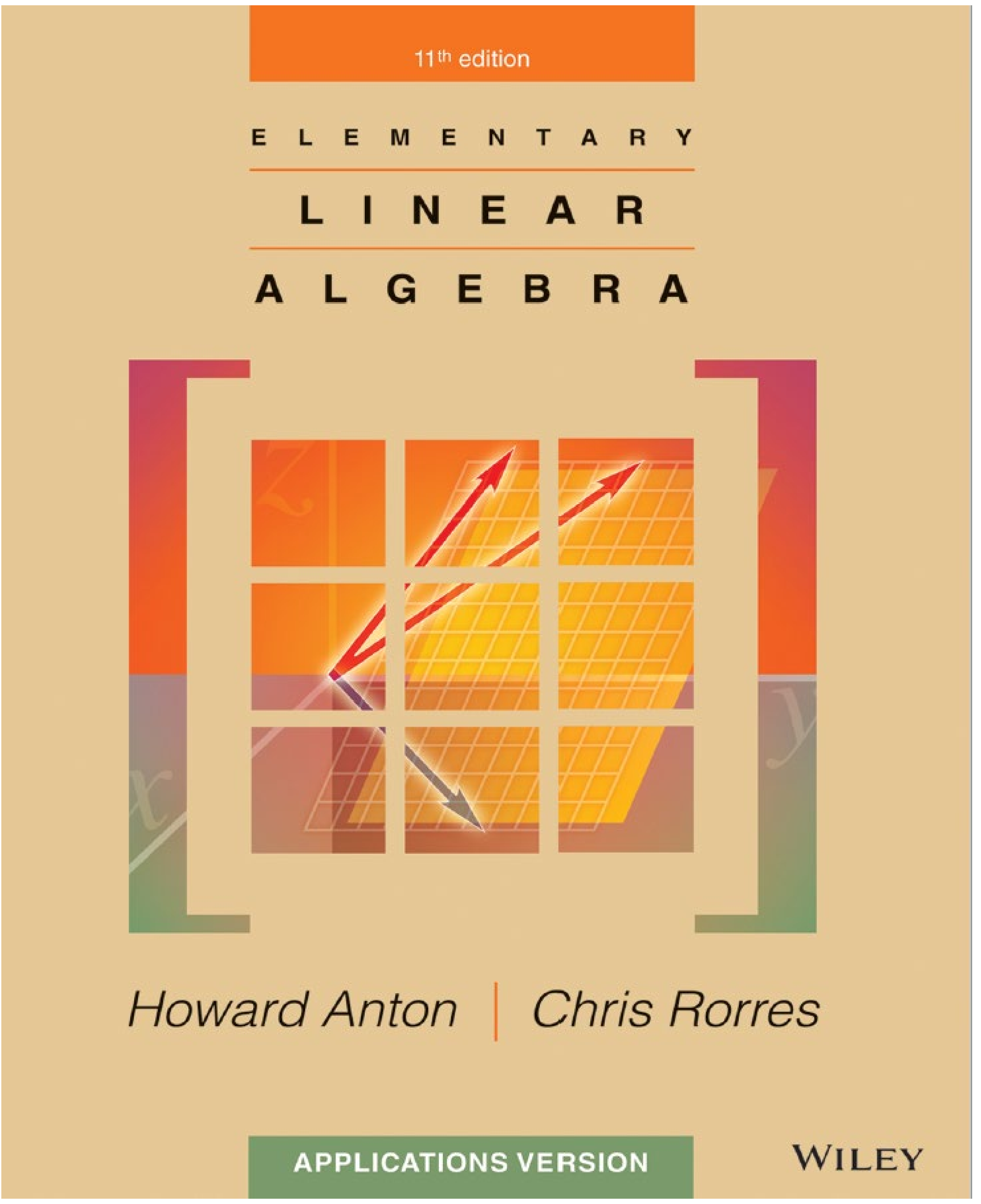
Reference

Materials in these slides have been taken from:

Anton and Rorres, “Linear Algebra”, 11th edition , Wiley.

Chapter: 3.1, 3.2

Euclidean Vector Spaces



$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}}_{b}$$

CX1104: **Linear Algebra for Computing**

Chap. No : **6.2.1**

Lecture : **Orthogonality**

Topic : **Orthogonality**

Concept : **Definition of Orthogonality and Orthogonal Complements**

Instructor: **A/P Chng Eng Siong**

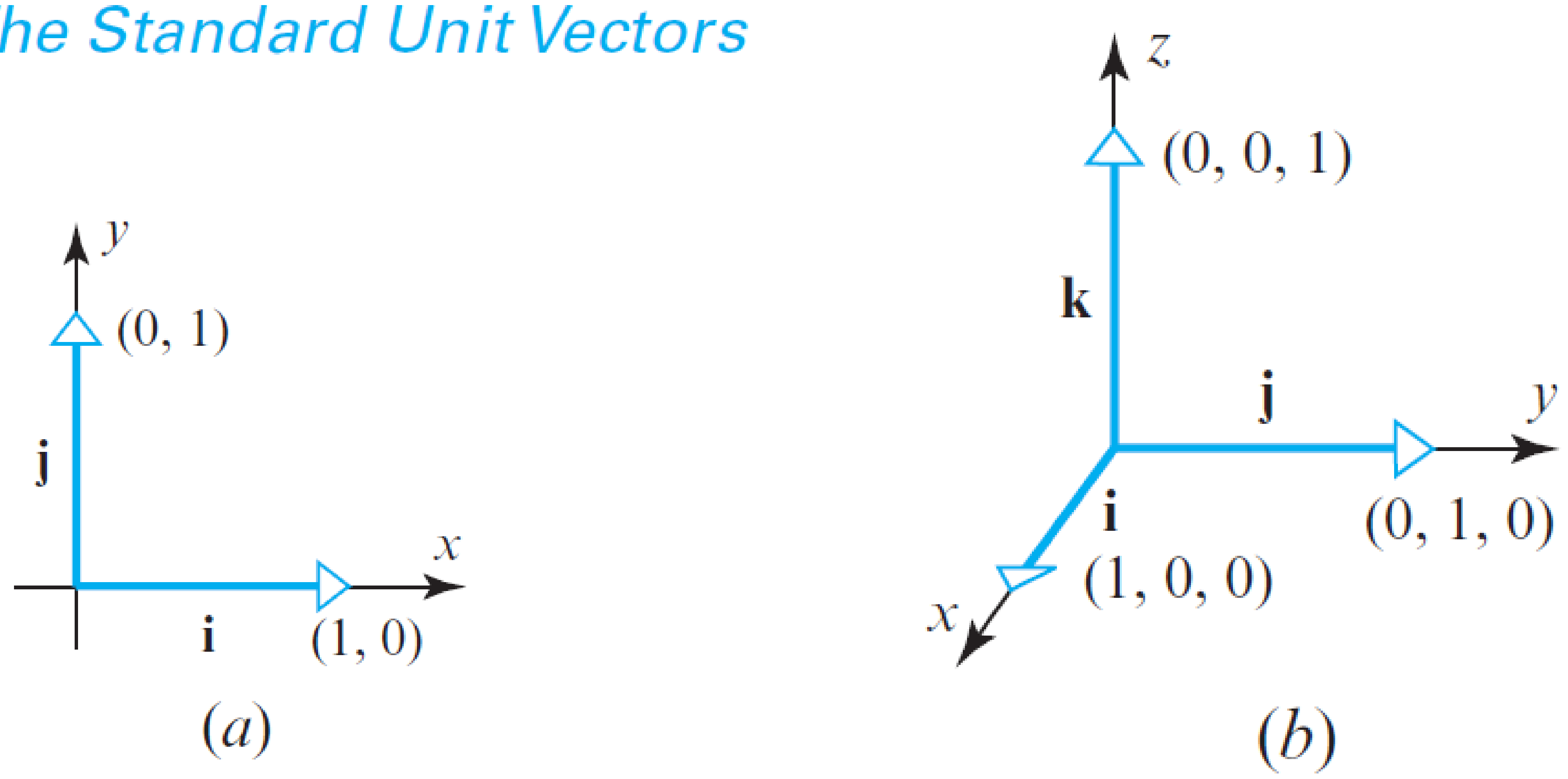
TAs: **Zhang Su, Vishal Choudhari**

LAST REVISED: 25 Sep 2021

Rev: 26th June 2020

Orthogonality Definition

The Standard Unit Vectors



▲ Figure 3.2.2

When a rectangular coordinate system is introduced in R^2 or R^3, the unit vectors in the positive directions of the coordinate axes are called the ***standard unit vectors***. In R^2 these vectors are denoted by

$$\mathbf{i} = (1, 0) \quad \text{and} \quad \mathbf{j} = (0, 1)$$

and in R^3 by

$$\mathbf{i} = (1, 0, 0), \quad \mathbf{j} = (0, 1, 0), \quad \text{and} \quad \mathbf{k} = (0, 0, 1)$$

(Figure 3.2.2). Every vector $\mathbf{v} = (v_1, v_2)$ in R^2 and every vector $\mathbf{v} = (v_1, v_2, v_3)$ in R^3 can be expressed as a linear combination of standard unit vectors by writing

$$\mathbf{v} = (v_1, v_2) = v_1(1, 0) + v_2(0, 1) = v_1\mathbf{i} + v_2\mathbf{j}$$

$$\mathbf{v} = (v_1, v_2, v_3) = v_1(1, 0, 0) + v_2(0, 1, 0) + v_3(0, 0, 1) = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$$

DEFINITION 1 Two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ in R^n are said to be ***orthogonal*** (or ***perpendicular***) if $\mathbf{u} \cdot \mathbf{v} = 0$. We will also agree that the zero vector in R^n is orthogonal to *every* vector in R^n.

Recall from Formula (20) in the previous section that the angle θ between two *nonzero* vectors $\mathbf{u}$ and $\mathbf{v}$ in R^n is defined by the formula

$$\theta = \cos^{-1}\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}\right)$$

It follows from this that $\theta = \pi/2$ if and only if $\mathbf{u} \cdot \mathbf{v} = 0$. Thus, we make the following definition.

▶ **EXAMPLE 1 Orthogonal Vectors**

(a) Show that $\mathbf{u} = (-2, 3, 1, 4)$ and $\mathbf{v} = (1, 2, 0, -1)$ are orthogonal vectors in R^4.

(b) Let $S = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ be the set of standard unit vectors in R^3. Show that each ordered pair of vectors in S is orthogonal.

Solution (a) The vectors are orthogonal since

$$\mathbf{u} \cdot \mathbf{v} = (-2)(1) + (3)(2) + (1)(0) + (4)(-1) = 0$$

Solution (b) It suffices to show that

$$\mathbf{i} \cdot \mathbf{j} = \mathbf{i} \cdot \mathbf{k} = \mathbf{j} \cdot \mathbf{k} = \mathbf{0}$$

because it will follow automatically from the symmetry property of the dot product that

$$\mathbf{j} \cdot \mathbf{i} = \mathbf{k} \cdot \mathbf{i} = \mathbf{k} \cdot \mathbf{j} = \mathbf{0}$$

Although the orthogonality of the vectors in S is evident geometrically from Figure 3.2.2, it is confirmed algebraically by the computations

$$\mathbf{i} \cdot \mathbf{j} = (1, 0, 0) \cdot (0, 1, 0) = 0$$
$$\mathbf{i} \cdot \mathbf{k} = (1, 0, 0) \cdot (0, 0, 1) = 0$$
$$\mathbf{j} \cdot \mathbf{k} = (0, 1, 0) \cdot (0, 0, 1) = 0 \; ◀$$

Lines and Planes Determined by Points and Normals

One learns in analytic geometry that a line in R^2 is determined uniquely by its slope and one of its points, and that a plane in R^3 is determined uniquely by its "inclination" and one of its points. One way of specifying slope and inclination is to use a *nonzero* vector $\mathbf{n}$, called a ***normal***, that is orthogonal to the line or plane in question. For example, Figure 3.3.1 shows the line through the point $P_0(x_0, y_0)$ that has normal $\mathbf{n} = (a, b)$ and the plane through the point $P_0(x_0, y_0, z_0)$ that has normal $\mathbf{n} = (a, b, c)$. Both the line and the plane are represented by the vector equation

$$\mathbf{n} \cdot \overrightarrow{P_0P} = 0 \tag{1}$$

where P is either an arbitrary point (x, y) on the line or an arbitrary point (x, y, z) in the plane. The vector $\overrightarrow{P_0P}$ can be expressed in terms of components as

$$\overrightarrow{P_0P} = (x - x_0, y - y_0) \quad \textbf{[line]}$$

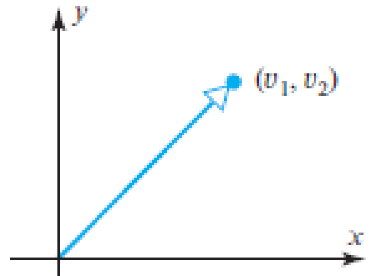
$$\overrightarrow{P_0P} = (x - x_0, y - y_0, z - z_0) \quad \textbf{[plane]}$$

Thus, Equation (1) can be written as

$$a(x - x_0) + b(y - y_0) = 0 \quad \textbf{[line]} \tag{2}$$

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0 \quad \textbf{[plane]} \tag{3}$$

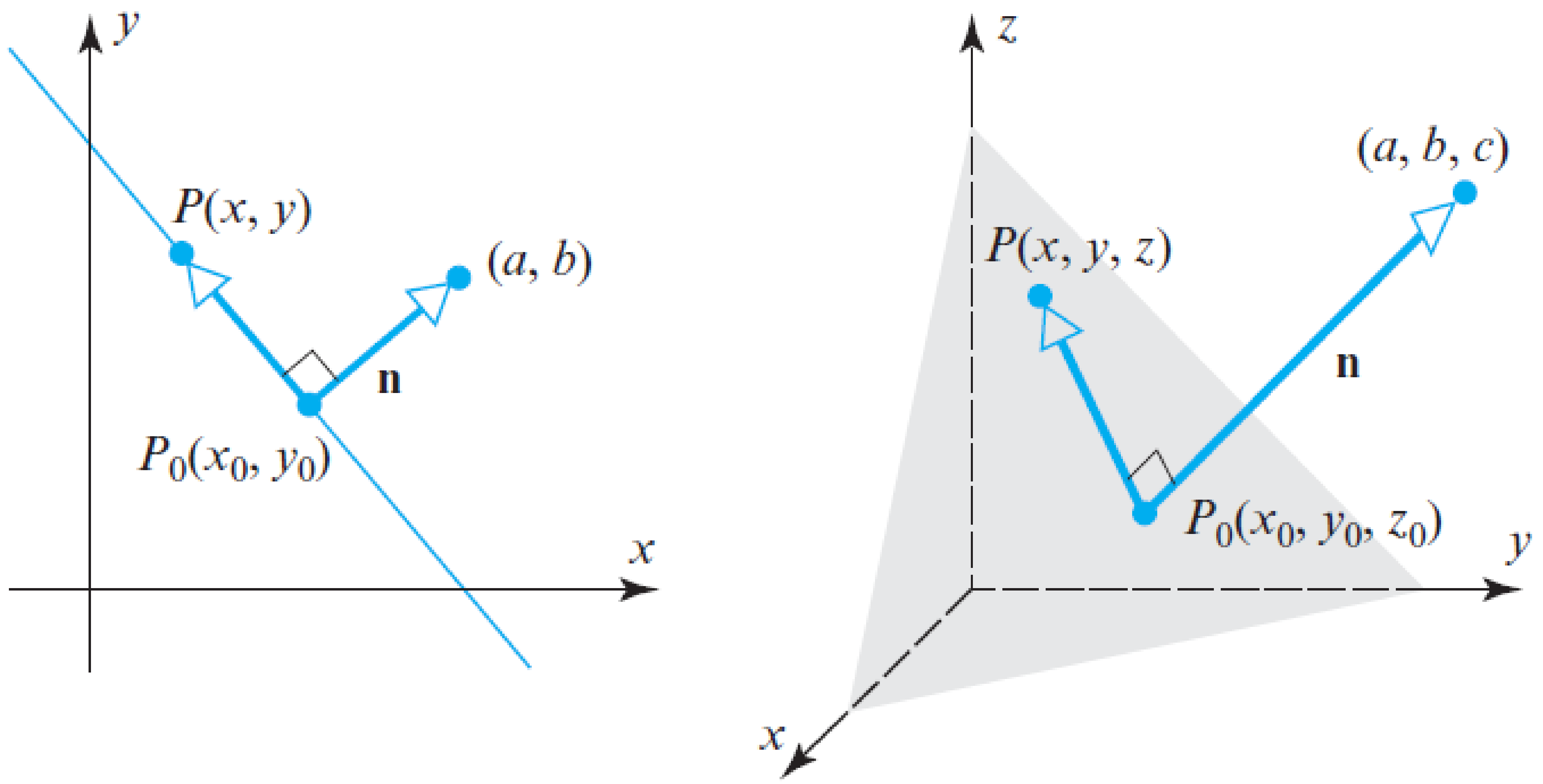
These are called the ***point-normal*** equations of the line and plane.



▲ Figure 3.1.11 The ordered pair (v_1, v_2) can represent a point or a vector.

Note, **n** above represents components of the normal vector and not coordinates.

Remark It may have occurred to you that an ordered pair (v_1, v_2) can represent either a vector with *components* v_1 and v_2 or a point with *coordinates* v_1 and v_2 (and similarly for ordered triples). Both are valid geometric interpretations, so the appropriate choice will depend on the geometric viewpoint that we want to emphasize (Figure 3.1.11).



▶ Figure 3.3.1

▶ EXAMPLE 2 Point-Normal Equations

It follows from (2) that in R^2 the equation

$$6(x - 3) + (y + 7) = 0$$

represents the line through the point $(3, -7)$ with normal $\mathbf{n} = (6, 1)$; and it follows from (3) that in R^3 the equation

$$4(x - 3) + 2y - 5(z - 7) = 0$$

represents the plane through the point $(3, 0, 7)$ with normal $\mathbf{n} = (4, 2, -5)$. ◀

Lines and Planes Determined by Points and Normals

$$\mathbf{n} \cdot \overrightarrow{P_0P} = 0 \tag{1}$$

Thus, Equation (1) can be written as

$$a(x - x_0) + b(y - y_0) = 0 \quad \textbf{[line]} \tag{2}$$

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0 \quad \textbf{[plane]} \tag{3}$$

These are called the ***point-normal*** equations of the line and plane.

THEOREM 3.3.1

(*a*) *If* a *and* b *are constants that are not both zero, then an equation of the form*

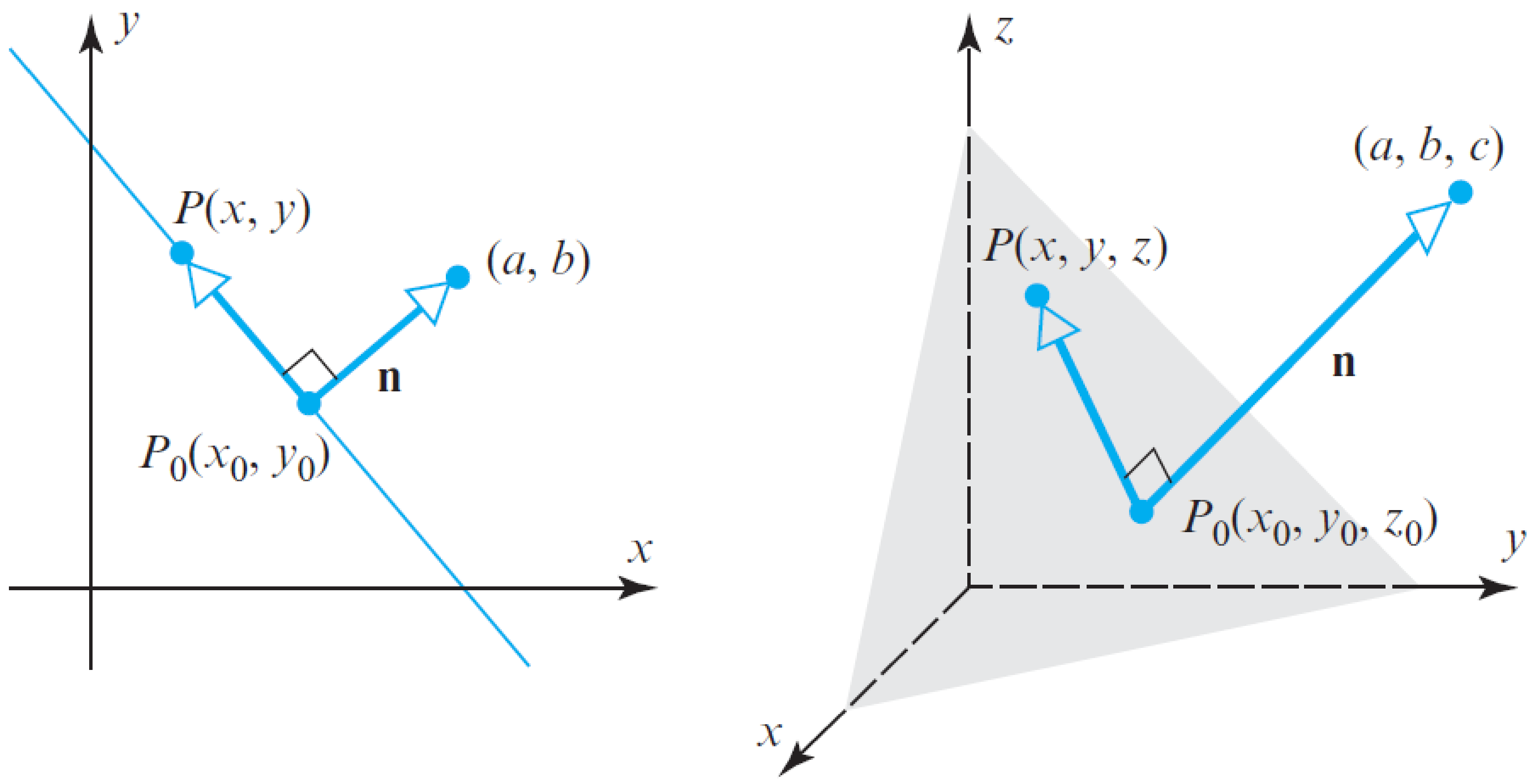
$$ax + by + c = 0 \tag{4}$$

represents a line in R^2 *with normal* $\mathbf{n} = (a, b)$.

(*b*) *If* a, b, *and* c *are constants that are not all zero, then an equation of the form*

$$ax + by + cz + d = 0 \tag{5}$$

represents a plane in R^3 *with normal* $\mathbf{n} = (a, b, c)$.



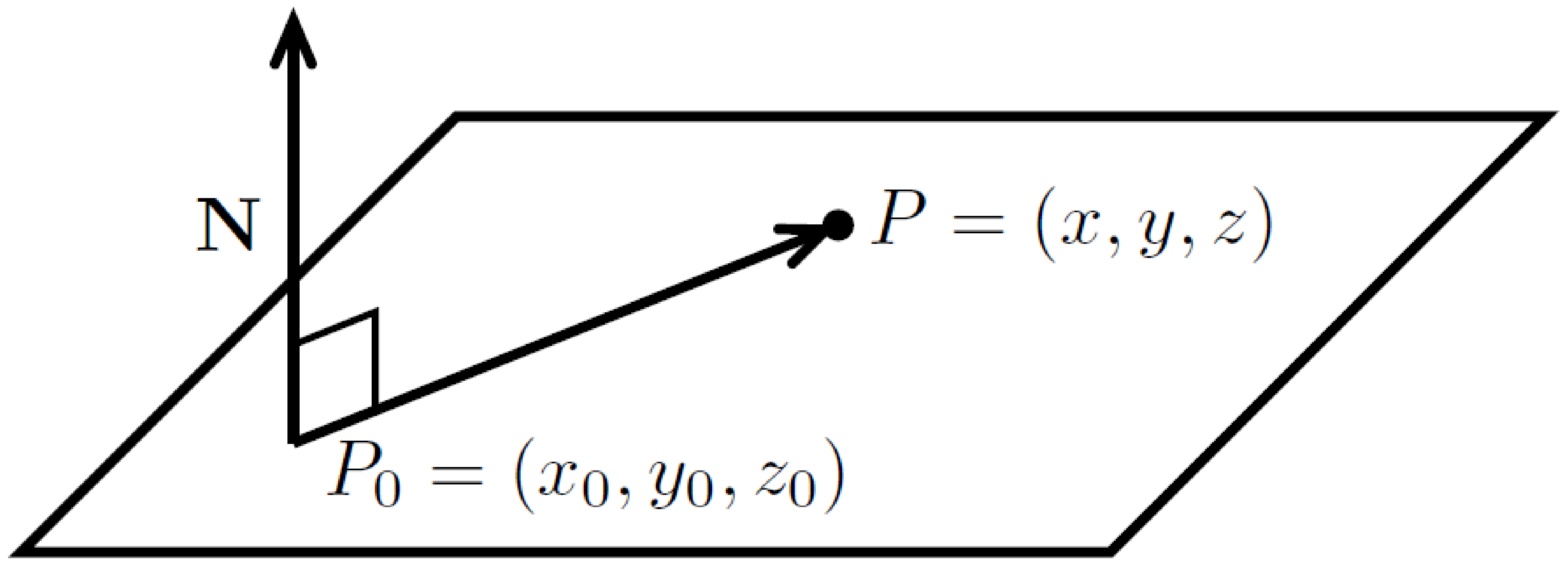
▶ Figure 3.3.1

Lines and Planes Determined by Points and Normals

Let $P = (x, y, z)$ be an arbitrary point in the plane. Then the vector $\overrightarrow{\mathbf{P_0P}}$ is in the plane and therefore orthogonal to $\mathbf{N}$. This means

$$\begin{aligned} & \mathbf{N} \cdot \overrightarrow{\mathbf{P_0P}} = 0 \\ \Leftrightarrow \quad & \langle a, b, c\rangle \cdot \langle x - x_0, y - y_0, z - z_0\rangle = 0 \\ \Leftrightarrow \quad & a(x - x_0) + b(y - y_0) + c(z - z_0) = 0 \end{aligned}$$

We call this last equation the point-normal form for the plane.



Example 1: Find the plane through the point (1,4,9) with normal $\langle 2, 3, 4\rangle$.

__**Answer:**__ Point-normal form of the plane is $2(x-1)+3(y-4)+4(z-9) = 0$. We can also write this as $2x + 3y + 4z = 50$.

Orthogonal Complements

[redacted] If a vector $\mathbf{z}$ is orthogonal to every vector in a subspace W of $\mathbb{R}^n$, then $\mathbf{z}$ is said to be **orthogonal to** W. The set of all vectors $\mathbf{z}$ that are orthogonal to W is called the **orthogonal complement** of W and is denoted by $W^\perp$ (and read as "W perpendicular" or simply "W perp").

1. A vector $\mathbf{x}$ is in $W^\perp$ if and only if $\mathbf{x}$ is orthogonal to every vector in a set that spans W.
2. $W^\perp$ is a subspace of $\mathbb{R}^n$.

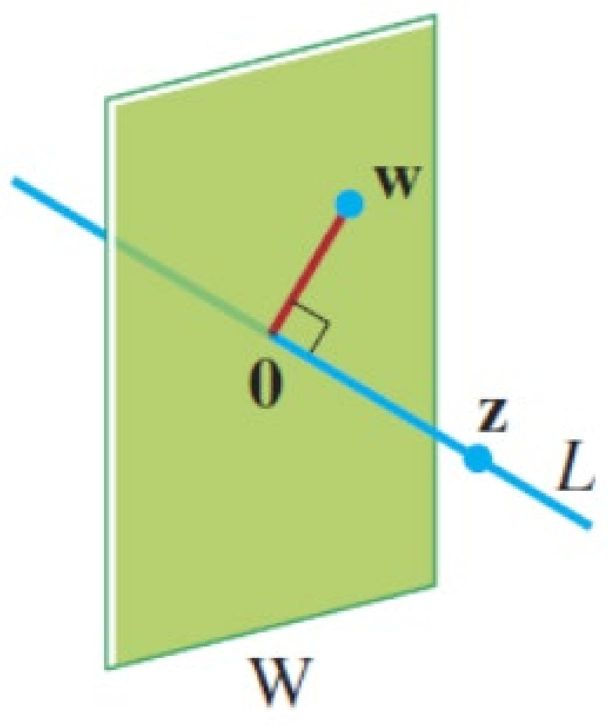


FIGURE 7

A plane and line through $\mathbf{0}$ as orthogonal complements.

EXAMPLE 6 Let W be a plane through the origin in $\mathbb{R}^3$, and let L be the line through the origin and perpendicular to W. If $\mathbf{z}$ and $\mathbf{w}$ are nonzero, $\mathbf{z}$ is on L, and $\mathbf{w}$ is in W, then the line segment from $\mathbf{0}$ to $\mathbf{z}$ is perpendicular to the line segment from $\mathbf{0}$ to $\mathbf{w}$; that is, $\mathbf{z}\cdot\mathbf{w} = 0$. See Figure 7. So each vector on L is orthogonal to every $\mathbf{w}$ in W. In fact, L consists of *all* vectors that are orthogonal to the $\mathbf{w}$'s in W, and W consists of all vectors orthogonal to the $\mathbf{z}$'s in L. That is,

$$L = W^\perp \quad \text{and} \quad W = L^\perp$$ ■

Ref: Lay 5e, pg 336

Orthogonal Complements

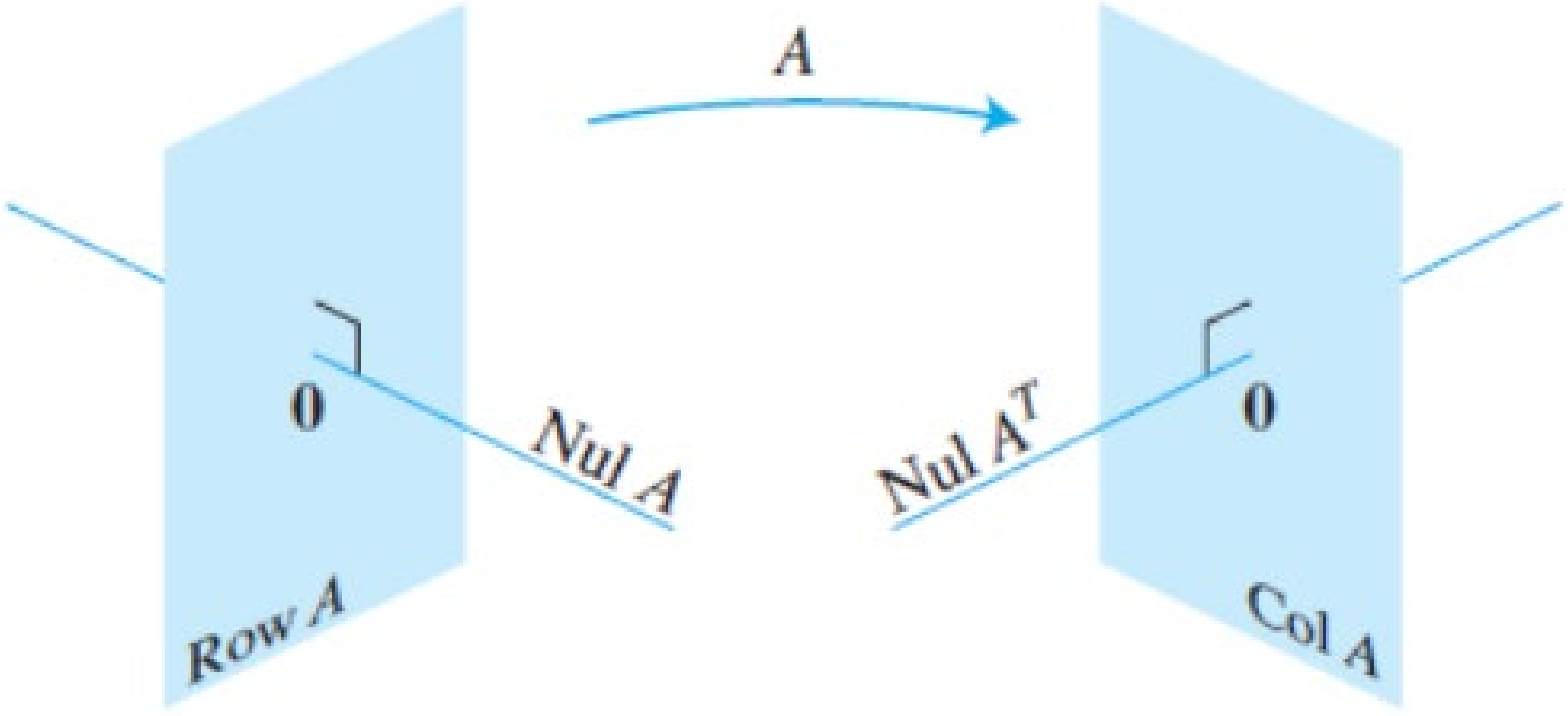


FIGURE 8 The fundamental subspaces determined by an $m \times n$ matrix A.

THEOREM 3

Let A be an $m \times n$ matrix. The orthogonal complement of the row space of A is the null space of A, and the orthogonal complement of the column space of A is the null space of A^T:

$$(\text{Row } A)^\perp = \text{Nul } A \quad \text{and} \quad (\text{Col } A)^\perp = \text{Nul } A^T$$

PROOF The row–column rule for computing $A\mathbf{x}$ shows that if $\mathbf{x}$ is in Nul A, then $\mathbf{x}$ is orthogonal to each row of A (with the rows treated as vectors in $\mathbb{R}^n$). Since the rows of A span the row space, $\mathbf{x}$ is orthogonal to Row A. Conversely, if $\mathbf{x}$ is orthogonal to Row A, then $\mathbf{x}$ is certainly orthogonal to each row of A, and hence $A\mathbf{x} = \mathbf{0}$. This proves the first statement of the theorem. Since this statement is true for any matrix, it is true for A^T. That is, the orthogonal complement of the row space of A^T is the null space of A^T. This proves the second statement, because Row A^T = Col A. ■

Remark: A common way to prove that two sets, say S and T, are equal is to show that S is a subset of T and T is a subset of S. The proof of the next theorem that Nul $A = (\text{Row } A)^\perp$ is established by showing that Nul A is a subset of $(\text{Row } A)^\perp$ and $(\text{Row } A)^\perp$ is a subset of Nul A. That is, an arbitrary element $\mathbf{x}$ in Nul A is shown to be in $(\text{Row } A)^\perp$, and then an arbitrary element $\mathbf{x}$ in $(\text{Row } A)^\perp$ is shown to be in Nul A.

Ref: Lay 5e, pg 337

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}}_{b}$$

CX1104: **Linear Algebra for Computing**

Chap. No : **6.2.2**

Lecture : **Orthogonality**

Topic : **Orthogonality**

Concept : **Orthogonal Projections**

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su, Vishal Choudhari**

Rev: 26th June 2020

Orthogonal Projections

Decomposition in R^2 space

In many applications it is necessary to "decompose" a vector **u** into a sum of two terms, one term being a scalar multiple of a specified nonzero vector **a** and the other term being orthogonal to **a**. For example, if **u** and **a** are vectors in R^2 that are positioned so their initial points coincide at a point Q, then we can create such a decomposition as follows (Figure 3.3.2):

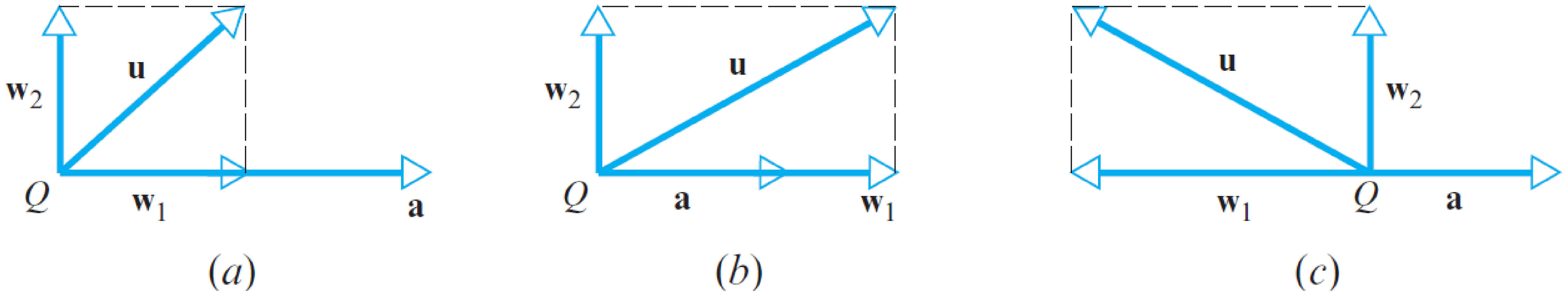


Figure 3.3.2 Three possible cases.

- Drop a perpendicular from the tip of **u** to the line through **a**.
- Construct the vector $\mathbf{w}_1$ from Q to the foot of the perpendicular.
- Construct the vector $\mathbf{w}_2 = \mathbf{u} - \mathbf{w}_1$.

Since

$$\mathbf{w}_1 + \mathbf{w}_2 = \mathbf{w}_1 + (\mathbf{u} - \mathbf{w}_1) = \mathbf{u}$$

we have decomposed **u** into a sum of two orthogonal vectors, the first term being a scalar multiple of **a** and the second being orthogonal to **a**.

Decomposing vectors into orthogonal basis has many advantages!
The standard basis is an orthogonal basis.
Orthogonal basis is explained in 6.2.3.

EXAMPLE 1 The Standard Basis for R^n

Recall from Example 11 of Section 4.2 that the standard unit vectors

$$\mathbf{e}_1 = (1, 0, 0, \dots, 0), \quad \mathbf{e}_2 = (0, 1, 0, \dots, 0), \dots, \quad \mathbf{e}_n = (0, 0, 0, \dots, 1)$$

span R^n and from Example 1 of Section 4.3 that they are linearly independent. Thus, they form a basis for R^n that we call the ***standard basis for R^n***. In particular,

$$\mathbf{i} = (1, 0, 0), \quad \mathbf{j} = (0, 1, 0), \quad \mathbf{k} = (0, 0, 1)$$

is the standard basis for R^3.

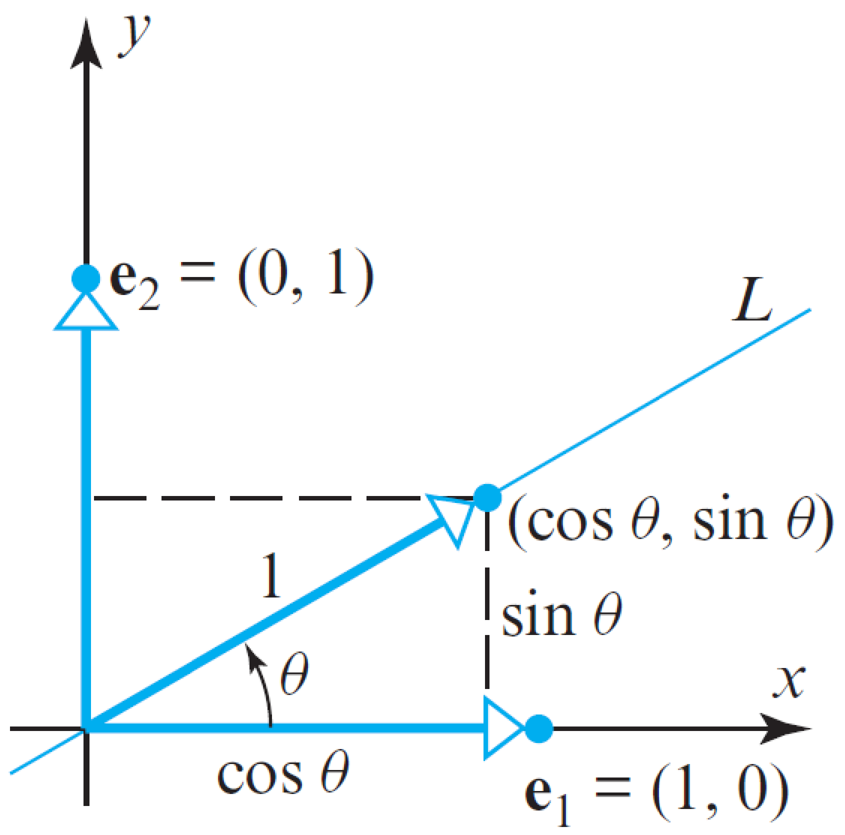
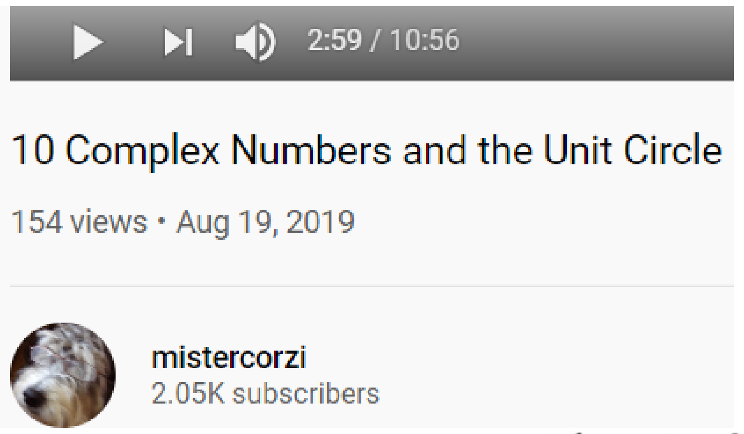
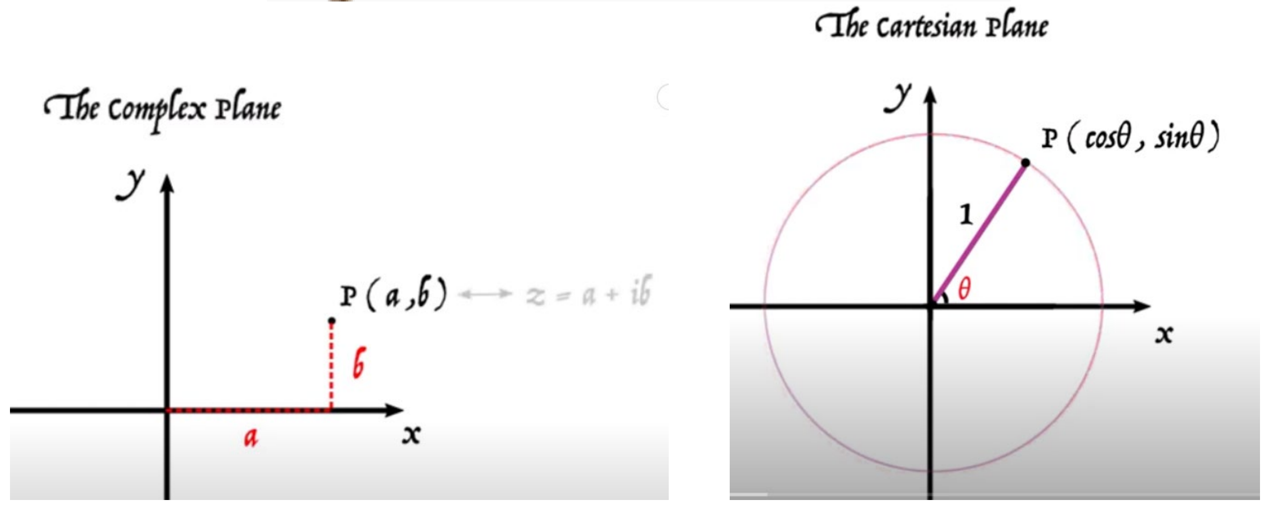


Figure 3.3.3





Mister Corzi: https://www.youtube.com/watch?v=UF172DZOiWo

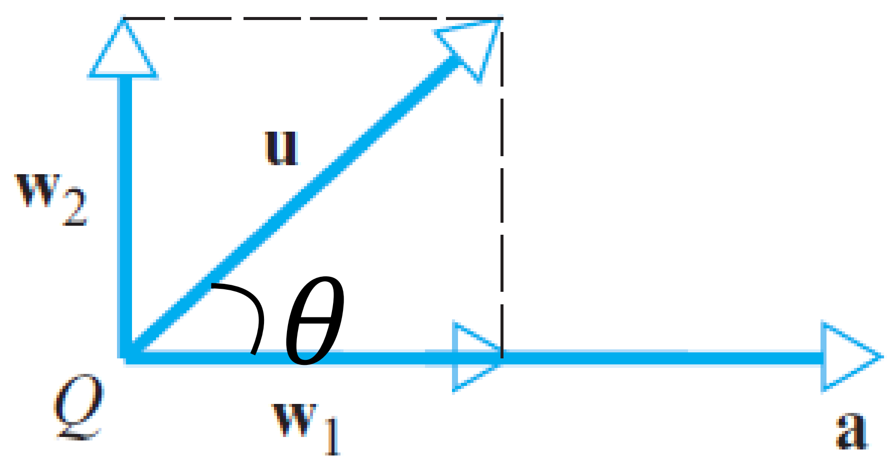
Example above:
Commonly found in complex number of unit circle.
Where e_1 is the real line, and e_2 is the imaginary line

Khan Academy:
https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:complex/x9e81a4f98389efdf:complex-polar/v/polar-form-complex-number
https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:complex/x9e81a4f98389efdf:complex-mul-div-polar/a/complex-number-polar-form-review

Orthogonal Projections

THEOREM 3.3.2 Projection Theorem

If $\mathbf{u}$ *and* $\mathbf{a}$ *are vectors in* R^n, *and if* $\mathbf{a} \neq 0$, *then* $\mathbf{u}$ *can be expressed in exactly one way in the form* $\mathbf{u} = \mathbf{w}_1 + \mathbf{w}_2$, *where* $\mathbf{w}_1$ *is a scalar multiple of* $\mathbf{a}$ *and* $\mathbf{w}_2$ *is orthogonal to* $\mathbf{a}$.



(*a*)

The vectors $\mathbf{w}_1$ and $\mathbf{w}_2$ in the Projection Theorem have associated names—the vector $\mathbf{w}_1$ is called the ***orthogonal projection of*** $\mathbf{u}$ ***on*** $\mathbf{a}$ or sometimes ***the vector component of*** $\mathbf{u}$ ***along*** $\mathbf{a}$, and the vector $\mathbf{w}_2$ is called the vector ***component of*** $\mathbf{u}$ ***orthogonal to*** $\mathbf{a}$. The vector $\mathbf{w}_1$ is commonly denoted by the symbol $\text{proj}_{\mathbf{a}}\mathbf{u}$, in which case it follows from (8) that $\mathbf{w}_2 = \mathbf{u} - \text{proj}_{\mathbf{a}}\mathbf{u}$. In summary,

$$w_1 = \text{proj}_{\mathbf{a}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{a}}{\|\mathbf{a}\|^2}\mathbf{a} \quad (\textit{vector component of } \mathbf{u} \textit{ along } \mathbf{a}) = ||u||cos\theta \tag{10}$$

$$w_2 = \mathbf{u} - \text{proj}_{\mathbf{a}}\mathbf{u} = \mathbf{u} - \frac{\mathbf{u}\cdot\mathbf{a}}{\|\mathbf{a}\|^2}\mathbf{a} \quad (\textit{vector component of } \mathbf{u} \textit{ orthogonal to } \mathbf{a}) \tag{11}$$

EXAMPLE 5 Vector Component of u Along a

Let $\mathbf{u} = (2, -1, 3)$ and $\mathbf{a} = (4, -1, 2)$. Find the vector component of $\mathbf{u}$ along $\mathbf{a}$ and the vector component of $\mathbf{u}$ orthogonal to $\mathbf{a}$.

Solution

$$\mathbf{u}\cdot\mathbf{a} = (2)(4) + (-1)(-1) + (3)(2) = 15$$
$$\|\mathbf{a}\|^2 = 4^2 + (-1)^2 + 2^2 = 21$$

Thus the vector component of $\mathbf{u}$ along $\mathbf{a}$ is

$$\text{proj}_{\mathbf{a}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{a}}{\|\mathbf{a}\|^2}\mathbf{a} = \tfrac{15}{21}(4, -1, 2) = \left(\tfrac{20}{7}, -\tfrac{5}{7}, \tfrac{10}{7}\right)$$

and the vector component of $\mathbf{u}$ orthogonal to $\mathbf{a}$ is

$$\mathbf{u} - \text{proj}_{\mathbf{a}}\mathbf{u} = (2, -1, 3) - \left(\tfrac{20}{7}, -\tfrac{5}{7}, \tfrac{10}{7}\right) = \left(-\tfrac{6}{7}, -\tfrac{2}{7}, \tfrac{11}{7}\right)$$

As a check, you may wish to verify that the vectors $\mathbf{u} - \text{proj}_{\mathbf{a}}\mathbf{u}$ and $\mathbf{a}$ are perpendicular by showing that their dot product is zero. ◀

Linear Algebra: Projection onto a Subspace

12,031 views • Jun 28, 2014

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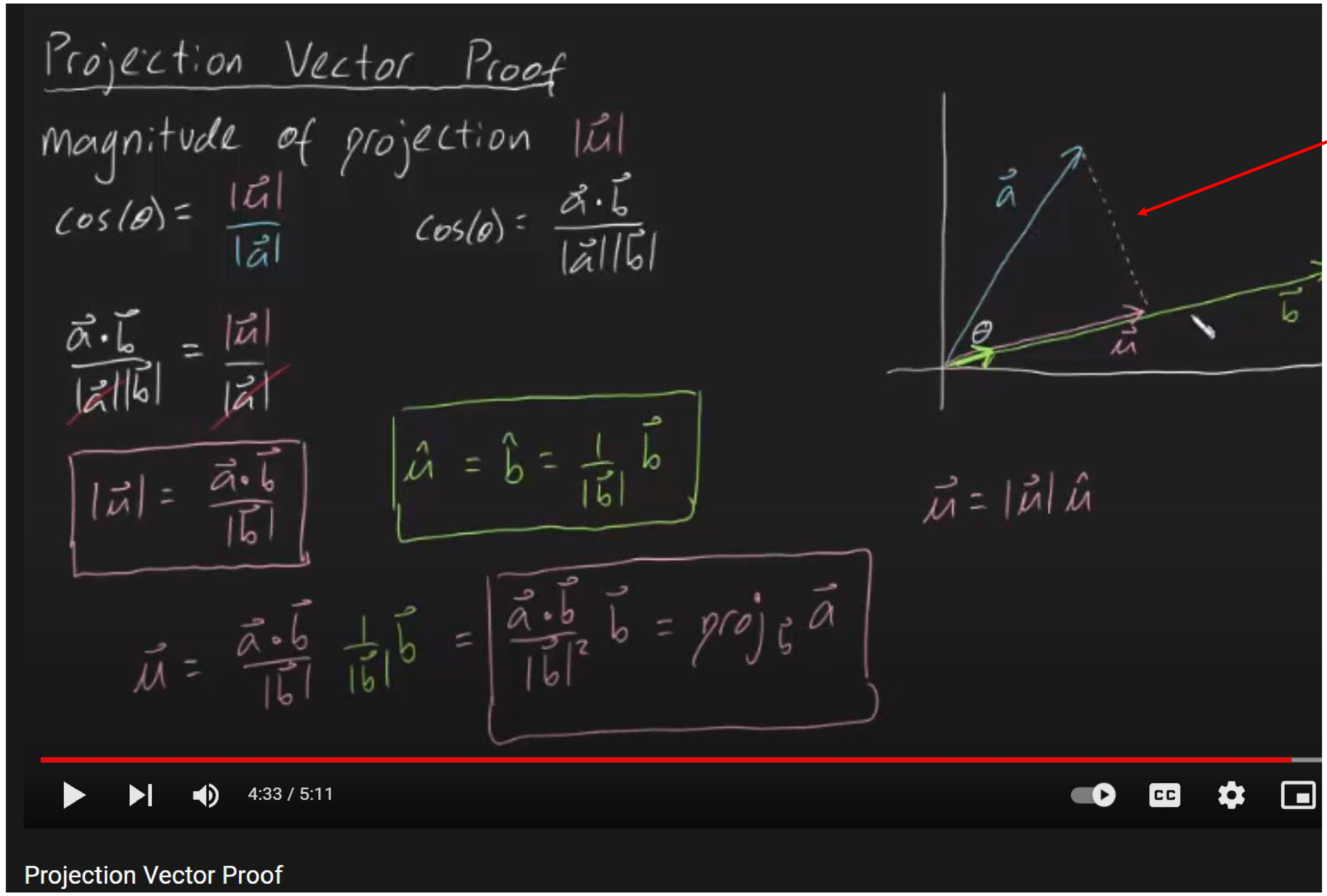
Watch these:
Projection onto Line: https://www.youtube.com/watch?v=GnvYEbaSBoY
Projection onto Subspace: https://www.youtube.com/watch?v=zZW6JV4yA54

Proof: $u = Proj_b a = \left(\frac{a.b}{||b||^2}\right) b$

Method 1:

In this example, ***u*** is the projection of **a onto b**



Ref:

1) https://www.youtube.com/watch?v=aTBtgW7U-Y8

Method 2

Let $\boldsymbol{e} = \boldsymbol{a} - \boldsymbol{u}$ /* the residual */

The residual $\boldsymbol{e}$ is orthogonal to $\boldsymbol{b}$

Therefore,

$$\boldsymbol{e}.\boldsymbol{b} = 0$$

$$(\boldsymbol{a} - \boldsymbol{u}).\boldsymbol{b} = 0$$

$$\boldsymbol{a}.\boldsymbol{b} - \boldsymbol{u}.\boldsymbol{b} = 0$$

$$\boldsymbol{a}.\boldsymbol{b} = \boldsymbol{u}.\boldsymbol{b}$$

multiply with $\boldsymbol{b}$ $\boldsymbol{on\ rhs}$

$$(\boldsymbol{a}.\boldsymbol{b})\boldsymbol{b} = (\boldsymbol{u}.\boldsymbol{b})\boldsymbol{b}$$

$$(\boldsymbol{a}.\boldsymbol{b})\boldsymbol{b} = \boldsymbol{u}(\boldsymbol{b}.\boldsymbol{b})$$

$$\boldsymbol{u} = \frac{(\boldsymbol{a}.\boldsymbol{b})}{\boldsymbol{b}.\boldsymbol{b}}\boldsymbol{b} = \frac{(\boldsymbol{a}.\boldsymbol{b})}{||\boldsymbol{b}||^2}\boldsymbol{b}$$

Orthogonal Projections

Fig. 2.5 shows two vectors $\mathbf{u}, \mathbf{v}$ in R^2 space. The orthogonal projection of $\mathbf{u}$ on $\mathbf{v}$, i.e. $Proj_{\mathbf{v}}\mathbf{u}$, can be thought of as the shadow formed by vector $\mathbf{u}$ onto $\mathbf{v}$ when an imaginary light is directed to $\mathbf{u}$ along the normal of $\mathbf{v}$; The vector $Proj_{\mathbf{v}}\mathbf{u}$ is the best approximation of $\mathbf{u}$ using $\mathbf{v}$. The following are the definitions of orthogonal projection and the equation to find the orthogonal residual.

$$Proj_{\mathbf{v}}\mathbf{u} = \mathbf{v}\left(\frac{\mathbf{u}^T\mathbf{v}}{||\mathbf{v}||^2}\right), \quad (2.34)$$

$$\mathbf{w}_{\mathbf{v}} = \mathbf{u} - Proj_{\mathbf{v}}\mathbf{u}, \quad (2.35)$$

where $||\mathbf{v}|| = \sqrt{\mathbf{v}^T\mathbf{v}}$ is the Euclidean norm of $\mathbf{v}$ and the vector $\mathbf{w}_{\mathbf{v}}$ is the residual when $\mathbf{v}$ is used to approximate $\mathbf{u}$.

Why is the denominator $||v||^2$? (eqn 2.34).

Ans: it is to normalize vector v to have unit length.
Notice that v occurs twice, to get a vector to have unit length,
We need to divide v by $||v||$

$$\left(\frac{v}{||v||}\right) * u^T\left(\frac{v}{||v||}\right) = v * \frac{u^T v}{(||v||^2)}$$

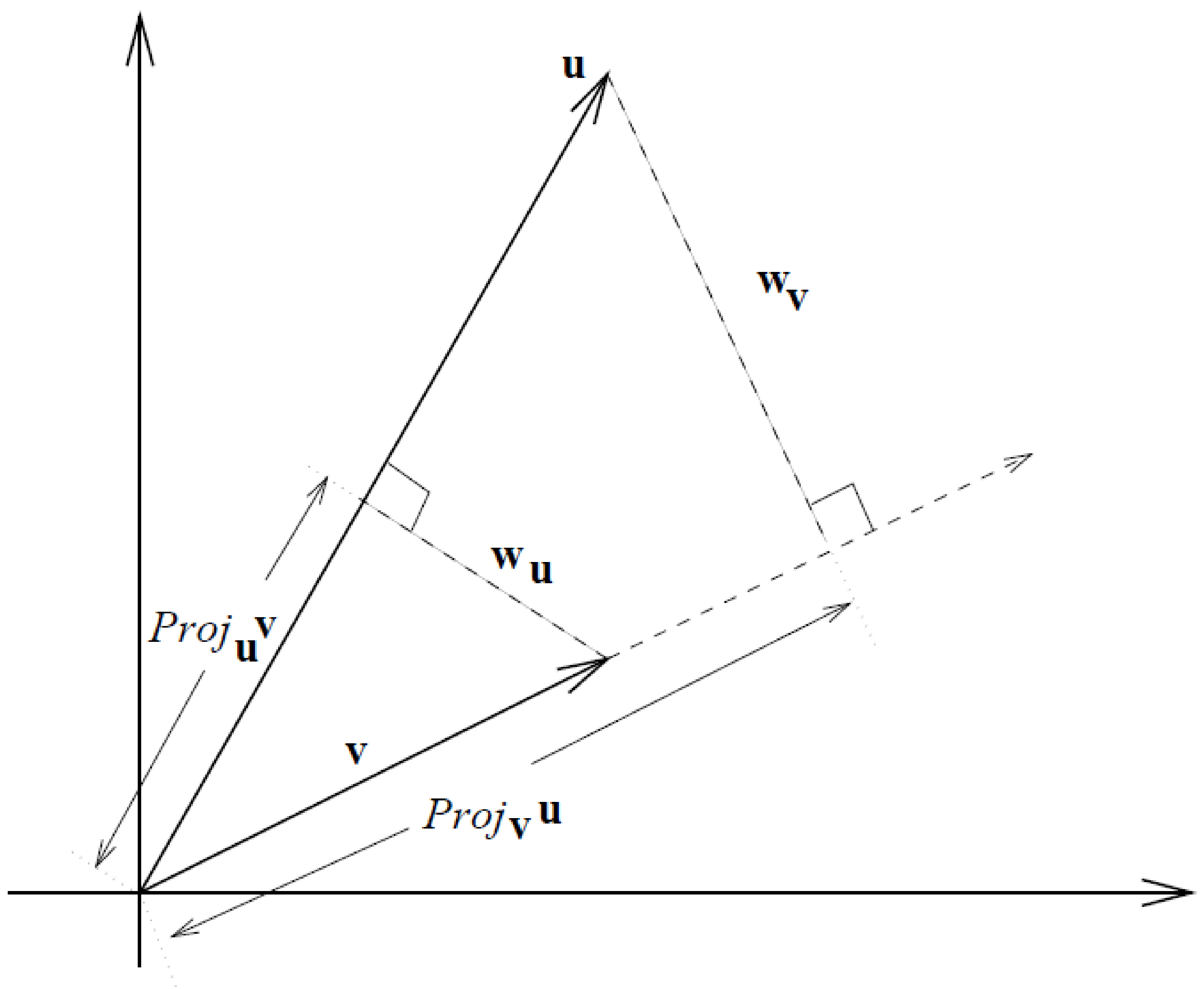
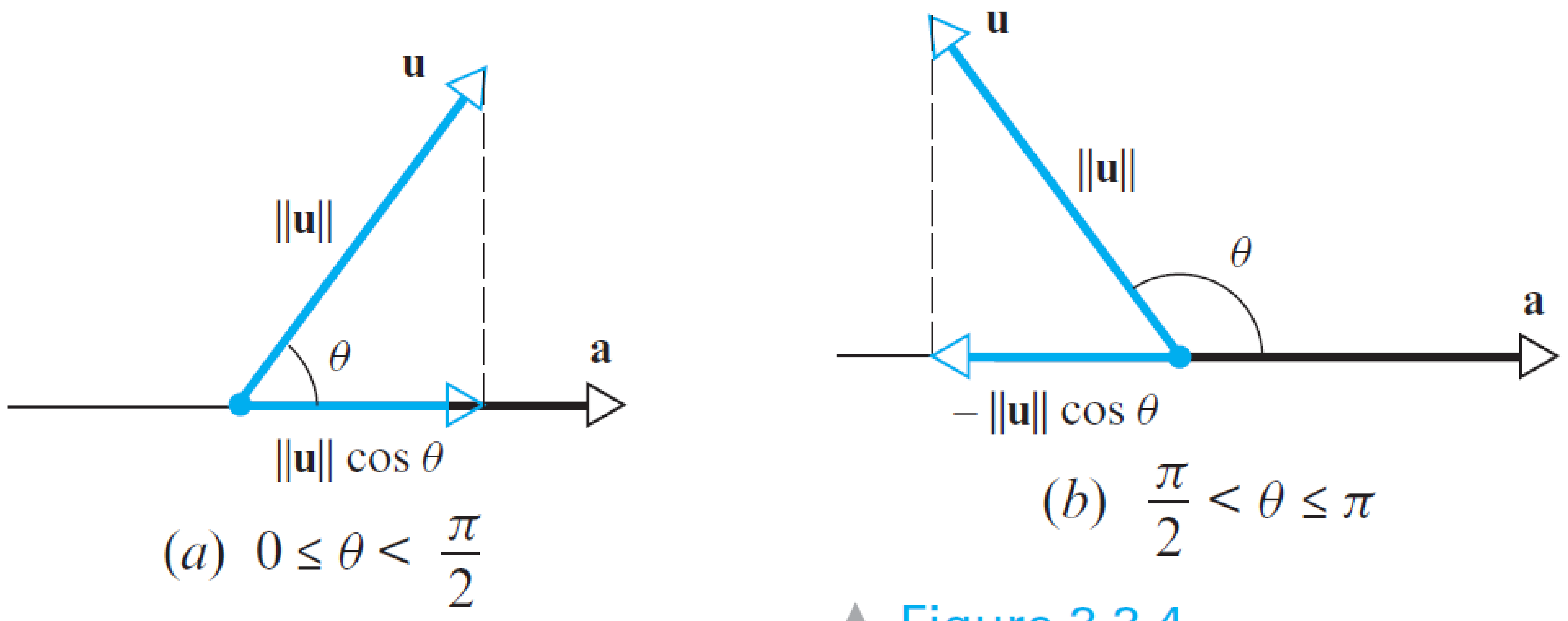


Figure 2.5: Orthogonal projections

Orthogonal Projections

Finding the norm of the projected component.



▲ Figure 3.3.4

Sometimes we will be more interested in the *norm* of the vector component of **u** along **a** than in the vector component itself. A formula for this norm can be derived as follows:

$$\|\text{proj}_{\mathbf{a}}\mathbf{u}\| = \left\|\frac{\mathbf{u}\cdot\mathbf{a}}{\|\mathbf{a}\|^2}\mathbf{a}\right\| = \left|\frac{\mathbf{u}\cdot\mathbf{a}}{\|\mathbf{a}\|^2}\right|\|\mathbf{a}\| = \frac{|\mathbf{u}\cdot\mathbf{a}|}{\|\mathbf{a}\|^2}\|\mathbf{a}\|$$

where the second equality follows from part (c) of Theorem 3.2.1 and the third from the fact that $\|\mathbf{a}\|^2 > 0$. Thus,

$$\|\text{proj}_{\mathbf{a}}\mathbf{u}\| = \frac{|\mathbf{u}\cdot\mathbf{a}|}{\|\mathbf{a}\|} \qquad (12)$$

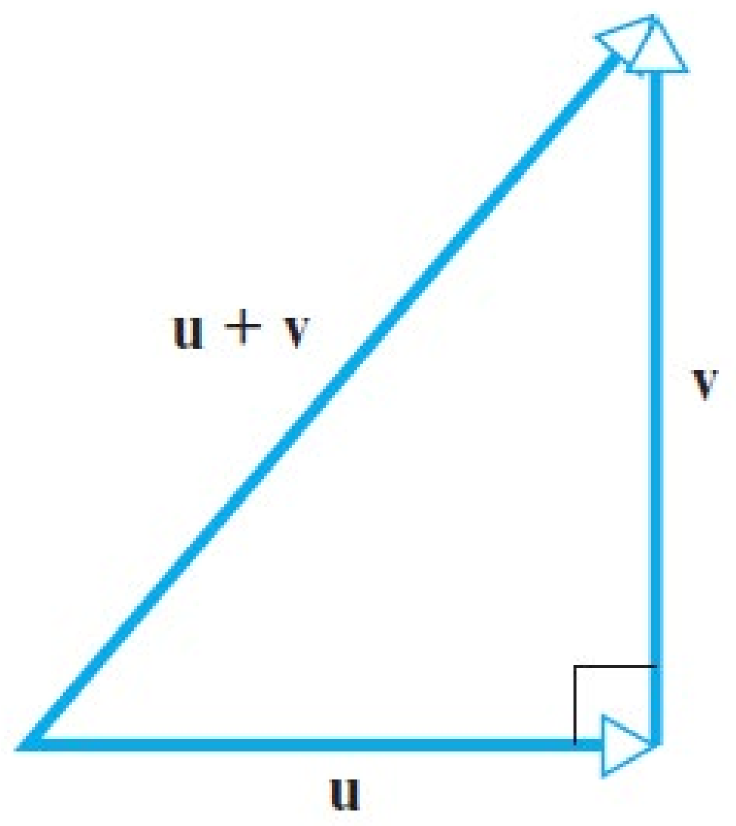
If θ denotes the angle between **u** and **a**, then $\mathbf{u}\cdot\mathbf{a} = \|\mathbf{u}\|\|\mathbf{a}\|\cos\theta$, so (12) can also be written as

$$\|\text{proj}_{\mathbf{a}}\mathbf{u}\| = \|\mathbf{u}\||\cos\theta| \qquad (13)$$

(Verify.) A geometric interpretation of this result is given in Figure 3.3.4.

Pythagoras Theorem

The Theorem of Pythagoras



▲ Figure 3.3.5

THEOREM 3.3.3 Theorem of Pythagoras in R^n

If **u** *and* **v** *are orthogonal vectors in* R^n *with the Euclidean inner product, then*

$$\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \tag{14}$$

Proof Since **u** and **v** are orthogonal, we have $\mathbf{u}\cdot\mathbf{v}=0$, from which it follows that

$$\|\mathbf{u}+\mathbf{v}\|^2 = (\mathbf{u}+\mathbf{v})\cdot(\mathbf{u}+\mathbf{v}) = \|\mathbf{u}\|^2 + 2(\mathbf{u}\cdot\mathbf{v}) + \|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$ ◄

► **EXAMPLE 6 Theorem of Pythagoras in R^4**

We showed in Example 1 that the vectors

$$\mathbf{u} = (-2, 3, 1, 4) \quad \text{and} \quad \mathbf{v} = (1, 2, 0, -1)$$

are orthogonal. Verify the Theorem of Pythagoras for these vectors.

Solution We leave it for you to confirm that

$$\begin{aligned} \mathbf{u}+\mathbf{v} &= (-1, 5, 1, 3) \\ \|\mathbf{u}+\mathbf{v}\|^2 &= 36 \\ \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 &= 30 + 6 \end{aligned}$$

Thus, $\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ ◄

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}}_{b}$$

CX1104: **Linear Algebra for Computing**

Chap. No : **6.2.3**

Lecture : **Orthogonality**

Topic : **Orthogonality**

Concept : **Orthogonal Sets & Orthogonal Basis**

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su, Vishal Choudhari**

Rev: 26th June 2020

Orthogonal Sets

A set of vectors $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ in $\mathbb{R}^n$ is said to be an **orthogonal set** if each pair of distinct vectors from the set is orthogonal, that is, if $\mathbf{u}_i \cdot \mathbf{u}_j = 0$ whenever $i \neq j$.

EXAMPLE 1 Show that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal set, where

$$\mathbf{u}_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}$$

SOLUTION Consider the three possible pairs of distinct vectors, namely, $\{\mathbf{u}_1, \mathbf{u}_2\}$, $\{\mathbf{u}_1, \mathbf{u}_3\}$, and $\{\mathbf{u}_2, \mathbf{u}_3\}$.

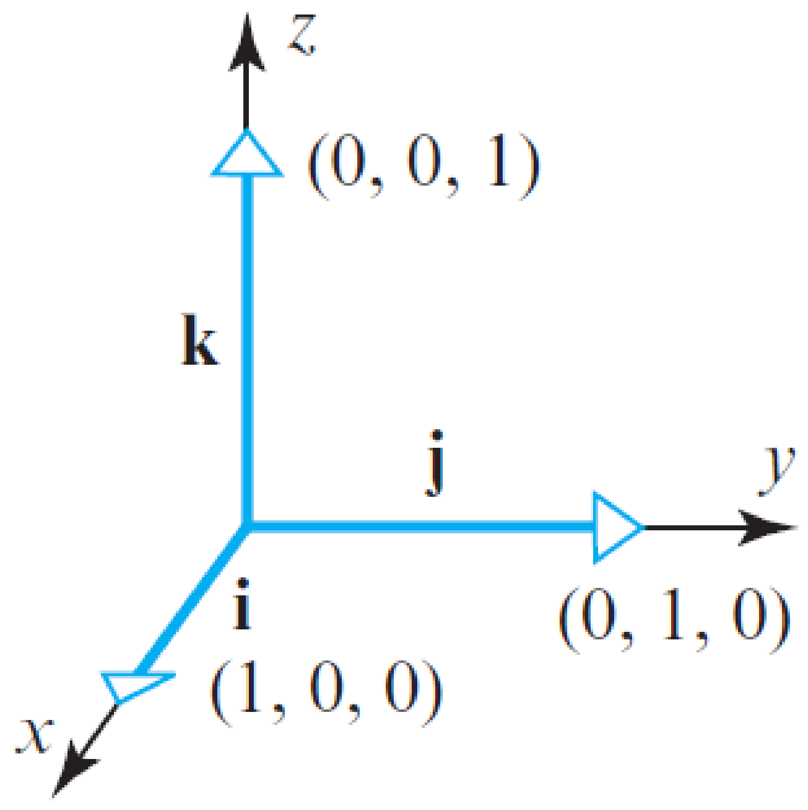
$$\mathbf{u}_1 \cdot \mathbf{u}_2 = 3(-1) + 1(2) + 1(1) = 0$$
$$\mathbf{u}_1 \cdot \mathbf{u}_3 = 3\left(-\tfrac{1}{2}\right) + 1(-2) + 1\left(\tfrac{7}{2}\right) = 0$$
$$\mathbf{u}_2 \cdot \mathbf{u}_3 = -1\left(-\tfrac{1}{2}\right) + 2(-2) + 1\left(\tfrac{7}{2}\right) = 0$$

Each pair of distinct vectors is orthogonal, and so $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal set. See Fig. 1; the three line segments there are mutually perpendicular. ■

Standard basis for R^n is an orthogonal set

Standard Basis

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



(b)

▲ Figure 3.2.2

Ref: https://mathworld.wolfram.com/StandardBasis.html

Orthogonal Sets and Orthogonal Basis

THEOREM 4 $Note: p \le n$

If $S = \{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^n$, then S is linearly independent and hence is a basis for the subspace spanned by S.

PROOF If $\mathbf{0} = c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p$ for some scalars $c_1, \dots, c_p$, then

$$\begin{aligned} 0 = \mathbf{0}\cdot\mathbf{u}_1 &= (c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_p\mathbf{u}_p)\cdot\mathbf{u}_1 \\ &= (c_1\mathbf{u}_1)\cdot\mathbf{u}_1 + (c_2\mathbf{u}_2)\cdot\mathbf{u}_1 + \cdots + (c_p\mathbf{u}_p)\cdot\mathbf{u}_1 \\ &= c_1(\mathbf{u}_1\cdot\mathbf{u}_1) + c_2(\mathbf{u}_2\cdot\mathbf{u}_1) + \cdots + c_p(\mathbf{u}_p\cdot\mathbf{u}_1) \\ &= c_1(\mathbf{u}_1\cdot\mathbf{u}_1) \end{aligned}$$

because $\mathbf{u}_1$ is orthogonal to $\mathbf{u}_2, \dots, \mathbf{u}_p$. Since $\mathbf{u}_1$ is nonzero, $\mathbf{u}_1\cdot\mathbf{u}_1$ is not zero and so $c_1 = 0$. Similarly, $c_2, \dots, c_p$ must be zero. Thus S is linearly independent. ■

DEFINITION An **orthogonal basis** for a subspace W of $\mathbb{R}^n$ is a basis for W that is also an orthogonal set.

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Examples of orthogonal set of vectors in R^3

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 3 & -1 & -1/2 \\ 1 & 2 & -2 \\ 1 & 1 & 7/2 \end{bmatrix}$$

```
y = -8/6;
S = [ 1 0 ;
      2 4 ;
      6 y ];
S
checkOrthognality = S'*S
```

$\mathbf{u}_1 \; \mathbf{u}_2$

```
S =

    1.0000         0
    2.0000    4.0000
    6.0000   -1.3333


checkOrthognality =

   41.0000    0.0000
    0.0000   17.7778
```

$$S = \begin{bmatrix} \uparrow & \uparrow \\ u_1 & u_2 \\ \downarrow & \downarrow \end{bmatrix}_{3\times2} \quad S^T = \begin{bmatrix} \leftarrow & u_1^T & \rightarrow \\ \leftarrow & u_2^T & \rightarrow \end{bmatrix}_{2\times3} \quad S^T \times S = \begin{bmatrix} ||u_1||^2 & u_1^T u_2 \\ u_2^T u_1 & ||u_2||^2 \end{bmatrix}_{2\times2}$$

Dot product between vectors u_1 and u_2

Projecting a vector onto a subspace (span by an orthogonal basis)

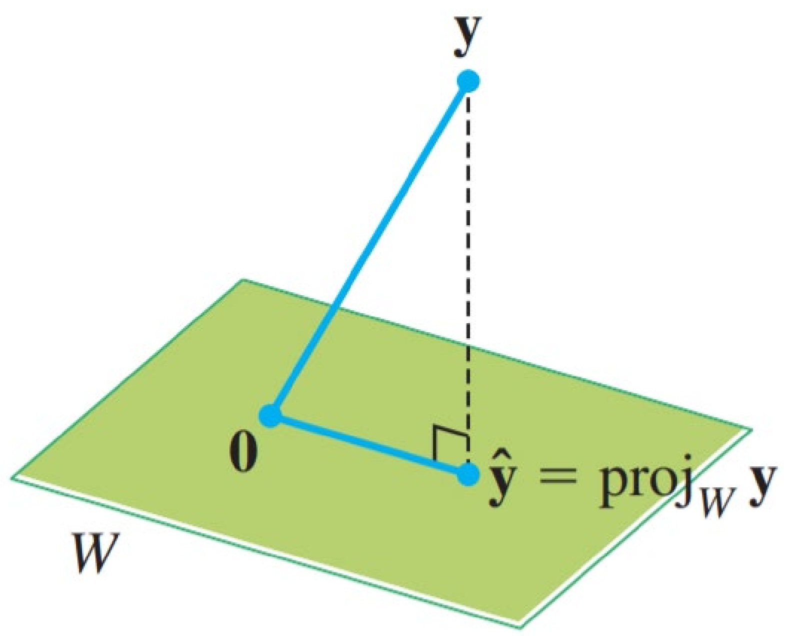

FIGURE 1

Given a vector $\mathbf{y}$ and a subspace W in $\mathbb{R}^n$, there is a vector $\hat{\mathbf{y}}$ in W such that (1) $\hat{\mathbf{y}}$ is the unique vector in W for which $\mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to W, and (2) $\hat{\mathbf{y}}$ is the unique vector in W closest to $\mathbf{y}$. See Figure 1. These two properties of $\hat{\mathbf{y}}$ provide the key to finding least-squares solutions of linear systems, mentioned in the introductory example for this chapter.

To prepare for the first theorem, observe that whenever a vector $\mathbf{y}$ is written as a linear combination of vectors $\mathbf{u}_1, \dots, \mathbf{u}_n$ in $\mathbb{R}^n$, the terms in the sum for $\mathbf{y}$ can be grouped into two parts so that $\mathbf{y}$ can be written as

$$\mathbf{y} = \mathbf{z}_1 + \mathbf{z}_2$$

where $\mathbf{z}_1$ is a linear combination of some of the $\mathbf{u}_i$ and $\mathbf{z}_2$ is a linear combination of the rest of the $\mathbf{u}_i$. This idea is particularly useful when $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ is an orthogonal basis.

Lay 5e, pg 349

EXAMPLE 1 Let $\{\mathbf{u}_1, \dots, \mathbf{u}_5\}$ be an orthogonal basis for $\mathbb{R}^5$ and let

$$\mathbf{y} = c_1\mathbf{u}_1 + \cdots + c_5\mathbf{u}_5$$

Consider the subspace $W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$, and write $\mathbf{y}$ as the sum of a vector $\mathbf{z}_1$ in W and a vector $\mathbf{z}_2$ in $W^\perp$.

SOLUTION Write

$$\mathbf{y} = \underbrace{c_1\mathbf{u}_1 + c_2\mathbf{u}_2}_{\mathbf{z}_1} + \underbrace{c_3\mathbf{u}_3 + c_4\mathbf{u}_4 + c_5\mathbf{u}_5}_{\mathbf{z}_2}$$

where $\quad \mathbf{z}_1 = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 \quad$ is in $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$

and $\quad \mathbf{z}_2 = c_3\mathbf{u}_3 + c_4\mathbf{u}_4 + c_5\mathbf{u}_5 \quad$ is in $\text{Span}\{\mathbf{u}_3, \mathbf{u}_4, \mathbf{u}_5\}$.

To show that $\mathbf{z}_2$ is in $W^\perp$, it suffices to show that $\mathbf{z}_2$ is orthogonal to the vectors in the basis $\{\mathbf{u}_1, \mathbf{u}_2\}$ for W. (See Section 6.1.) Using properties of the inner product, compute

$$\begin{aligned}
\mathbf{z}_2 \cdot \mathbf{u}_1 &= (c_3\mathbf{u}_3 + c_4\mathbf{u}_4 + c_5\mathbf{u}_5) \cdot \mathbf{u}_1 \\
&= c_3\mathbf{u}_3 \cdot \mathbf{u}_1 + c_4\mathbf{u}_4 \cdot \mathbf{u}_1 + c_5\mathbf{u}_5 \cdot \mathbf{u}_1 \\
&= 0
\end{aligned}$$

because $\mathbf{u}_1$ is orthogonal to $\mathbf{u}_3$, $\mathbf{u}_4$, and $\mathbf{u}_5$. A similar calculation shows that $\mathbf{z}_2 \cdot \mathbf{u}_2 = 0$. Thus $\mathbf{z}_2$ is in $W^\perp$. ∎

Orthogonal Sets and Orthogonal Basis

The next theorem suggests why an orthogonal basis is much nicer than other bases. The weights in a linear combination can be computed easily.

THEOREM 5

Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $\mathbf{y}$ in W, the weights in the linear combination

$$\mathbf{y} = c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p$$

are given by

$$c_j = \frac{\mathbf{y}\cdot\mathbf{u}_j}{\mathbf{u}_j\cdot\mathbf{u}_j} \qquad (j = 1, \ldots, p)$$

PROOF As in the preceding proof, the orthogonality of $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ shows that

$$\mathbf{y}\cdot\mathbf{u}_1 = (c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_p\mathbf{u}_p)\cdot\mathbf{u}_1 = c_1(\mathbf{u}_1\cdot\mathbf{u}_1)$$

Since $\mathbf{u}_1\cdot\mathbf{u}_1$ is not zero, the equation above can be solved for c_1. To find c_j for $j = 2, \ldots, p$, compute $\mathbf{y}\cdot\mathbf{u}_j$ and solve for c_j. ■

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THEOREM 8

The Orthogonal Decomposition Theorem

Let W be a subspace of $\mathbb{R}^n$. Then each $\mathbf{y}$ in $\mathbb{R}^n$ can be written uniquely in the form

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z} \qquad (1)$$

where $\hat{\mathbf{y}}$ is in W and $\mathbf{z}$ is in $W^\perp$. In fact, if $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is any orthogonal basis of W, then

$$\hat{\mathbf{y}} = \frac{\mathbf{y}\cdot\mathbf{u}_1}{\mathbf{u}_1\cdot\mathbf{u}_1}\mathbf{u}_1 + \cdots + \frac{\mathbf{y}\cdot\mathbf{u}_p}{\mathbf{u}_p\cdot\mathbf{u}_p}\mathbf{u}_p \qquad (2)$$

and $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$.

The vector $\hat{\mathbf{y}}$ in (1) is called the **orthogonal projection of y onto** W and often is written as $\text{proj}_W\,\mathbf{y}$. See Figure 2. When W is a one-dimensional subspace, the formula for $\hat{\mathbf{y}}$ matches the formula given in Section 6.2.

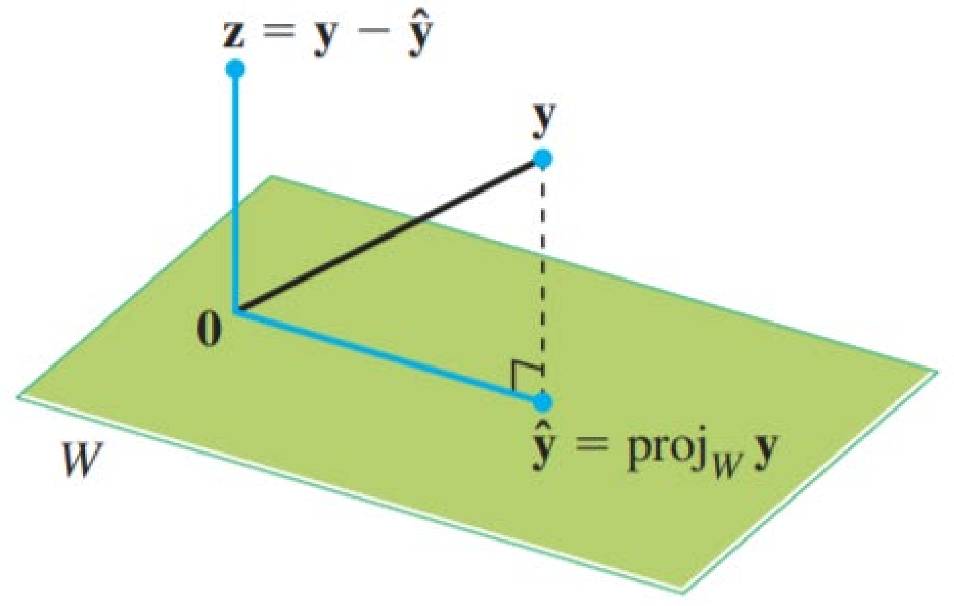

FIGURE 2 The orthogonal projection of $\mathbf{y}$ onto W.

Proof later: ch 6.2.5

Example

$$S = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Decompose $y = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix}$ using the standard basis.

$$\begin{bmatrix} 6 \\ 1 \\ 8 \end{bmatrix} = 6 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + (-8) \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

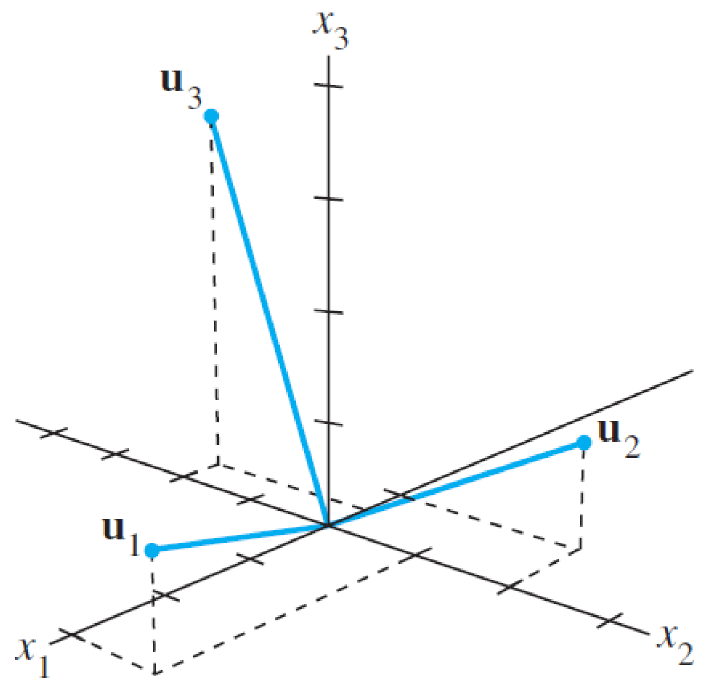

FIGURE 1

EXAMPLE 1 Show that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal set, where

$$\mathbf{u}_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}$$

SOLUTION Consider the three possible pairs of distinct vectors, namely, $\{\mathbf{u}_1, \mathbf{u}_2\}$, $\{\mathbf{u}_1, \mathbf{u}_3\}$, and $\{\mathbf{u}_2, \mathbf{u}_3\}$.

$$\mathbf{u}_1 \cdot \mathbf{u}_2 = 3(-1) + 1(2) + 1(1) = 0$$
$$\mathbf{u}_1 \cdot \mathbf{u}_3 = 3\left(-\tfrac{1}{2}\right) + 1(-2) + 1\left(\tfrac{7}{2}\right) = 0$$
$$\mathbf{u}_2 \cdot \mathbf{u}_3 = -1\left(-\tfrac{1}{2}\right) + 2(-2) + 1\left(\tfrac{7}{2}\right) = 0$$

Each pair of distinct vectors is orthogonal, and so $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal set. See Figure 1; the three line segments there are mutually perpendicular. ■

EXAMPLE 2 The set $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ in Example 1 is an orthogonal basis for $\mathbb{R}^3$. Express the vector $\mathbf{y} = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix}$ as a linear combination of the vectors in S.

SOLUTION Compute

$$\mathbf{y} \cdot \mathbf{u}_1 = 11, \qquad \mathbf{y} \cdot \mathbf{u}_2 = -12, \qquad \mathbf{y} \cdot \mathbf{u}_3 = -33$$
$$\mathbf{u}_1 \cdot \mathbf{u}_1 = 11, \qquad \mathbf{u}_2 \cdot \mathbf{u}_2 = 6, \qquad \mathbf{u}_3 \cdot \mathbf{u}_3 = 33/2$$

By Theorem 5,

$$\begin{aligned} \mathbf{y} &= \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2}\mathbf{u}_2 + \frac{\mathbf{y} \cdot \mathbf{u}_3}{\mathbf{u}_3 \cdot \mathbf{u}_3}\mathbf{u}_3 \\ &= \frac{11}{11}\mathbf{u}_1 + \frac{-12}{6}\mathbf{u}_2 + \frac{-33}{33/2}\mathbf{u}_3 \\ &= \mathbf{u}_1 - 2\mathbf{u}_2 - 2\mathbf{u}_3 \end{aligned}$$ ■

Notice how easy it is to compute the weights needed to build $\mathbf{y}$ from an orthogonal basis. If the basis were not orthogonal, it would be necessary to solve a system of linear equations in order to find the weights, as in Chapter 1.

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$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}}_{b}$$

CX1104: **Linear Algebra for Computing**

Chap. No : **6.2.4**

Lecture : **Orthogonality**

Topic : **Orthogonality**

Concept : **Orthonormal Sets & Orthogonal Matrices**

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su, Vishal Choudhari**

Rev: 26th June 2020

Orthonormal Set and Orthonormal basis

Orthonormal Sets

A set $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is an **orthonormal set** if it is an orthogonal set of unit vectors. If W is the subspace spanned by such a set, then $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is an **orthonormal basis** for W, since the set is automatically linearly independent, by Theorem 4.

The simplest example of an orthonormal set is the standard basis $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ for $\mathbb{R}^n$. Any nonempty subset of $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ is orthonormal, too. Here is a more complicated example.

THEOREM 4 If $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^n$, then S is linearly independent and hence is a basis for the subspace spanned by S.

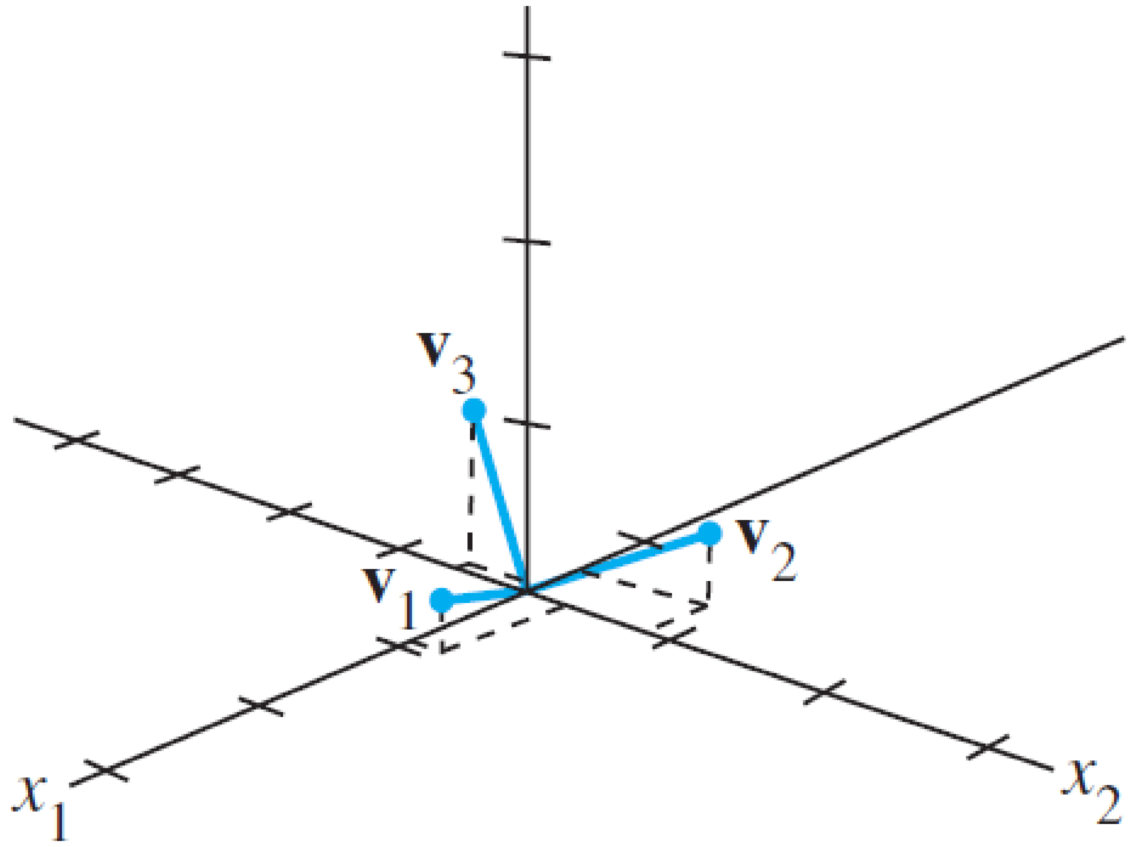

FIGURE 6

Ref: https://www.youtube.com/watch?v=ZJu26chXEiw

Linear Algebra: Orthonormal Basis
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EXAMPLE 5 Show that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthonormal basis of $\mathbb{R}^3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 3/\sqrt{11} \\ 1/\sqrt{11} \\ 1/\sqrt{11} \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1/\sqrt{66} \\ -4/\sqrt{66} \\ 7/\sqrt{66} \end{bmatrix}$$

SOLUTION Compute

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = -3/\sqrt{66} + 2/\sqrt{66} + 1/\sqrt{66} = 0$$
$$\mathbf{v}_1 \cdot \mathbf{v}_3 = -3/\sqrt{726} - 4/\sqrt{726} + 7/\sqrt{726} = 0$$
$$\mathbf{v}_2 \cdot \mathbf{v}_3 = 1/\sqrt{396} - 8/\sqrt{396} + 7/\sqrt{396} = 0$$

Thus $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthogonal set. Also,

$$\mathbf{v}_1 \cdot \mathbf{v}_1 = 9/11 + 1/11 + 1/11 = 1$$
$$\mathbf{v}_2 \cdot \mathbf{v}_2 = 1/6 + 4/6 + 1/6 = 1$$
$$\mathbf{v}_3 \cdot \mathbf{v}_3 = 1/66 + 16/66 + 49/66 = 1$$

which shows that $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ are unit vectors. Thus $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthonormal set. Since the set is linearly independent, its three vectors form a basis for $\mathbb{R}^3$. See Fig. 6. ■

```
S3 = [ 3/sqrt(11)   -1/sqrt(6)   -1/sqrt(66);
       1/sqrt(11)   2/sqrt(6)    -4/sqrt(66);
       1/sqrt(11)   1/sqrt(6)     7/sqrt(66)];

S3
checkOrthognality = S3'*S3
```

```
S3 =

    0.9045   -0.4082   -0.1231
    0.3015    0.8165   -0.4924
    0.3015    0.4082    0.8616


checkOrthognality =

    1.0000    0.0000    0.0000
    0.0000    1.0000    0.0000
    0.0000    0.0000    1.0000
```

2

NOTE: The 0's correspond to dot products of orthogonal vectors. See next slide for explanation of result!

Orthonormal Sets and U^TU

THEOREM 6

An $m \times n$ matrix U has orthonormal columns if and only if $U^TU = I$.

PROOF To simplify notation, we suppose that U has only three columns, each a vector in $\mathbb{R}^m$. The proof of the general case is essentially the same. Let $U = [\mathbf{u}_1 \ \ \mathbf{u}_2 \ \ \mathbf{u}_3]$ and compute

$$U^TU = \begin{bmatrix} \mathbf{u}_1^T \\ \mathbf{u}_2^T \\ \mathbf{u}_3^T \end{bmatrix} [\mathbf{u}_1 \ \ \mathbf{u}_2 \ \ \mathbf{u}_3] = \begin{bmatrix} \mathbf{u}_1^T\mathbf{u}_1 & \mathbf{u}_1^T\mathbf{u}_2 & \mathbf{u}_1^T\mathbf{u}_3 \\ \mathbf{u}_2^T\mathbf{u}_1 & \mathbf{u}_2^T\mathbf{u}_2 & \mathbf{u}_2^T\mathbf{u}_3 \\ \mathbf{u}_3^T\mathbf{u}_1 & \mathbf{u}_3^T\mathbf{u}_2 & \mathbf{u}_3^T\mathbf{u}_3 \end{bmatrix} \tag{4}$$

The entries in the matrix at the right are inner products, using transpose notation. The columns of U are orthogonal if and only if

$$\mathbf{u}_1^T\mathbf{u}_2 = \mathbf{u}_2^T\mathbf{u}_1 = 0, \quad \mathbf{u}_1^T\mathbf{u}_3 = \mathbf{u}_3^T\mathbf{u}_1 = 0, \quad \mathbf{u}_2^T\mathbf{u}_3 = \mathbf{u}_3^T\mathbf{u}_2 = 0 \tag{5}$$

The columns of U all have unit length if and only if

$$\mathbf{u}_1^T\mathbf{u}_1 = 1, \quad \mathbf{u}_2^T\mathbf{u}_2 = 1, \quad \mathbf{u}_3^T\mathbf{u}_3 = 1 \tag{6}$$

The theorem follows immediately from (4)–(6). ■

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6.2 Orthogonal Sets 343

Matlab Example: When U is square (3x3)

```
>> U = [3/sqrt(11) -1/sqrt(6) -1/sqrt(66);
1/sqrt(11) 2/sqrt(6) -4/sqrt(66);
1/sqrt(11) 1/sqrt(6) 7/sqrt(66)]

U =

    0.9045   -0.4082   -0.1231
    0.3015    0.8165   -0.4924
    0.3015    0.4082    0.8616

>> U'*U

ans =

    1.0000    0.0000    0.0000
    0.0000    1.0000    0.0000
    0.0000    0.0000    1.0000

>> U*U'

ans =

    1.0000   -0.0000    0.0000
   -0.0000    1.0000    0.0000
    0.0000    0.0000    1.0000
```

vs U is rectangle (3x2)

```
>> U2=[U(:,1) U(:,2)]

U2 =

    0.9045   -0.4082
    0.3015    0.8165
    0.3015    0.4082

>> U2'*U2

ans =

    1.0000    0.0000
    0.0000    1.0000

>> U2*U2'

ans =

    0.9848   -0.0606    0.1061
   -0.0606    0.7576    0.4242
    0.1061    0.4242    0.2576
```

```
>> P=U2*U2'

P =

    0.9848   -0.0606    0.1061
   -0.0606    0.7576    0.4242
    0.1061    0.4242    0.2576

>> P*P

ans =

    0.9848   -0.0606    0.1061
   -0.0606    0.7576    0.4242
    0.1061    0.4242    0.2576

>> rank(P*P)

ans =

     2
```

Question: If U (a mx2 matrix) ONLY has 2 **orthonormal** columns, (m>2), what is the characteristics of matrixes: U^TU and UU^T?

Ans: U^TU = identity (2x2) matrix

$P = UU^T$ = (mxm)matrix but not identity (only rank 2)

It spans the column space of U, lets call it *W*.

(later we show) it is a projection matrix P

P can be used to projecting a given vector y onto W by

$\hat{y} = UU^Ty = Py$ (see Ch 6.2.5, pg 7 Theorem 10)

Orthonormal Sets and Orthogonal matrix

THEOREM 7

Let U be an $m \times n$ matrix with orthonormal columns, and let $\mathbf{x}$ and $\mathbf{y}$ be in $\mathbb{R}^n$. Then

a. $\|U\mathbf{x}\| = \|\mathbf{x}\|$

b. $(U\mathbf{x})\cdot(U\mathbf{y}) = \mathbf{x}\cdot\mathbf{y}$

c. $(U\mathbf{x})\cdot(U\mathbf{y}) = 0$ if and only if $\mathbf{x}\cdot\mathbf{y} = 0$

Properties (a) and (c) say that the linear mapping $\mathbf{x} \mapsto U\mathbf{x}$ preserves lengths and orthogonality. These properties are crucial for many computer algorithms.

EXAMPLE 6 Let $U = \begin{bmatrix} 1/\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -2/3 \\ 0 & 1/3 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} \sqrt{2} \\ 3 \end{bmatrix}$. Notice that U has orthonormal columns and

$$U^TU = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 2/3 & -2/3 & 1/3 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -2/3 \\ 0 & 1/3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Verify that $\|U\mathbf{x}\| = \|\mathbf{x}\|$.

SOLUTION

$$U\mathbf{x} = \begin{bmatrix} 1/\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -2/3 \\ 0 & 1/3 \end{bmatrix} \begin{bmatrix} \sqrt{2} \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$$

$$\|U\mathbf{x}\| = \sqrt{9+1+1} = \sqrt{11}$$

$$\|\mathbf{x}\| = \sqrt{2+9} = \sqrt{11}$$

Theorems 6 and 7 are particularly useful when applied to *square* matrices. An **orthogonal matrix** is a square invertible matrix U such that $U^{-1} = U^T$. By Theorem 6, such a matrix has orthonormal columns.[1] It is easy to see that any *square* matrix with orthonormal columns is an orthogonal matrix. Surprisingly, such a matrix must have orthonormal *rows*, too.

Orthogonal Matrix definitions

Ref: https://en.wikipedia.org/wiki/Orthogonal_matrix **Important!**

In linear algebra, an **orthogonal matrix**, or **orthonormal matrix**, is a real square matrix whose columns and rows are orthonormal vectors.

One way to express this is

$$Q^{\mathrm{T}}Q = QQ^{\mathrm{T}} = I,$$

where Q^{T} is the transpose of Q and I is the identity matrix.

This leads to the equivalent characterization: a matrix Q is orthogonal if its transpose is equal to its inverse:

$$Q^{\mathrm{T}} = Q^{-1},$$

where Q^{-1} is the inverse of Q.

Orthogonal Matrix
MUST be SQUARE!

Note: confusion
If you have a mxn matrix called U *with* its column ortho-normal, and m>n (tall matrix)

1) IT IS NOT an orthogonal matrix since it satisfy ONLY $U^TU = I$ (nxn)
2) its UU^T is mxm matrix BUT it is not equals to I) .Instead UU^T is a projection matrix and has rank n.

Note: There is no standard name for "rectangular matrix with orthonormal columns"

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}}_{b}$$

CX1104: **Linear Algebra for Computing**

Chap. No : **6.2.5**

Lecture : **Orthogonality**

Topic : **Orthogonality**

Concept : **Orthogonal Decomposition**

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su, Vishal Choudhari**

Rev: 26th June 2020

Orthogonal Decomposition

THEOREM 8

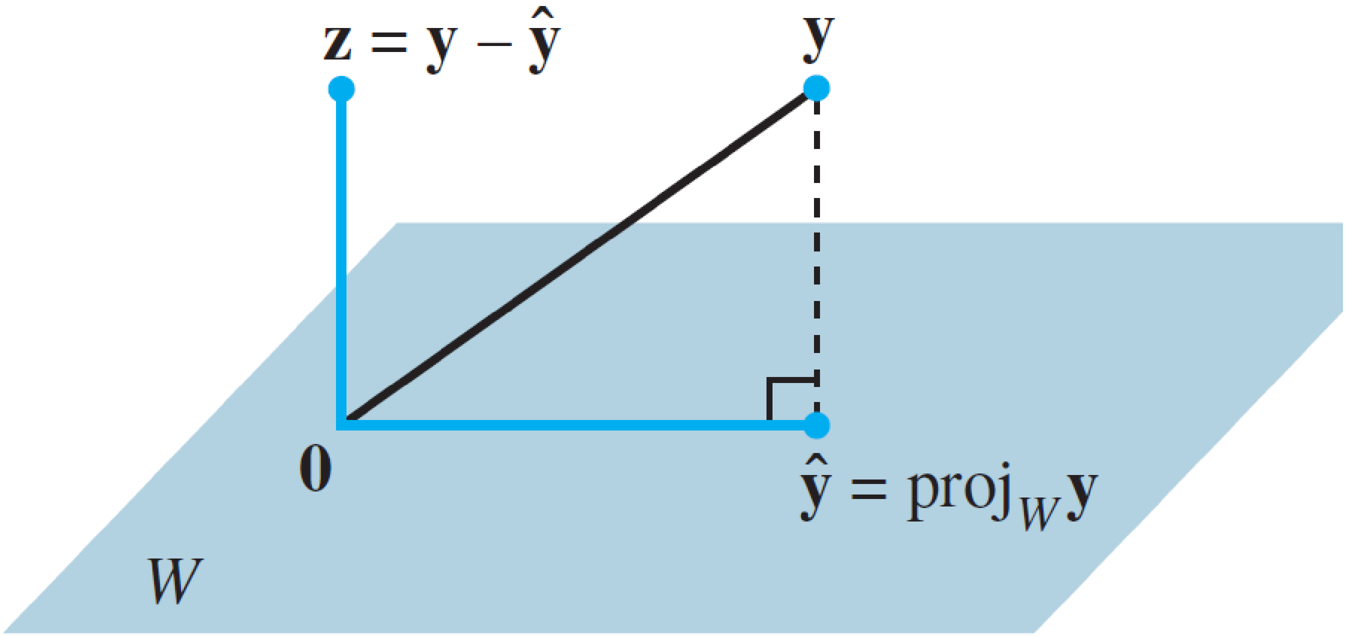

FIGURE 2 The orthogonal projection of $\mathbf{y}$ onto W.

The vector $\hat{\mathbf{y}}$ in (1) is called the **orthogonal projection of y onto** W and often is written as $\text{proj}_W \mathbf{y}$. See Fig. 2.

The Orthogonal Decomposition Theorem

Let W be a subspace of $\mathbb{R}^n$. Then each $\mathbf{y}$ in $\mathbb{R}^n$ can be written uniquely in the form

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z} \tag{1}$$

where $\hat{\mathbf{y}}$ is in W and $\mathbf{z}$ is in $W^\perp$. In fact, if $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is any orthogonal basis of W, then

$$\hat{\mathbf{y}} = \frac{\mathbf{y}\cdot\mathbf{u}_1}{\mathbf{u}_1\cdot\mathbf{u}_1}\mathbf{u}_1 + \cdots + \frac{\mathbf{y}\cdot\mathbf{u}_p}{\mathbf{u}_p\cdot\mathbf{u}_p}\mathbf{u}_p \tag{2}$$

and $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$.

Note: $W^\perp$ is the set of all vectors orthogonal to the subspace W.

Lay's Linear Algebra and Applications

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PROOF Let $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ be any orthogonal basis for W, and define $\hat{\mathbf{y}}$ by (2).[1] Then $\hat{\mathbf{y}}$ is in W because $\hat{\mathbf{y}}$ is a linear combination of the basis $\mathbf{u}_1, \dots, \mathbf{u}_p$. Let $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$. Since $\mathbf{u}_1$ is orthogonal to $\mathbf{u}_2, \dots, \mathbf{u}_p$, it follows from (2) that

$$\begin{aligned}\mathbf{z}\cdot\mathbf{u}_1 &= (\mathbf{y} - \hat{\mathbf{y}})\cdot\mathbf{u}_1 = \mathbf{y}\cdot\mathbf{u}_1 - \left(\frac{\mathbf{y}\cdot\mathbf{u}_1}{\mathbf{u}_1\cdot\mathbf{u}_1}\right)\mathbf{u}_1\cdot\mathbf{u}_1 - 0 - \cdots - 0 \\ &= \mathbf{y}\cdot\mathbf{u}_1 - \mathbf{y}\cdot\mathbf{u}_1 = 0\end{aligned}$$

Thus $\mathbf{z}$ is orthogonal to $\mathbf{u}_1$. Similarly, $\mathbf{z}$ is orthogonal to each $\mathbf{u}_j$ in the basis for W. Hence $\mathbf{z}$ is orthogonal to every vector in W. That is, $\mathbf{z}$ is in $W^\perp$.

To show that the decomposition in (1) is unique, suppose $\mathbf{y}$ can also be written as $\mathbf{y} = \hat{\mathbf{y}}_1 + \mathbf{z}_1$, with $\hat{\mathbf{y}}_1$ in W and $\mathbf{z}_1$ in $W^\perp$. Then $\hat{\mathbf{y}} + \mathbf{z} = \hat{\mathbf{y}}_1 + \mathbf{z}_1$ (since both sides equal $\mathbf{y}$), and so

$$\hat{\mathbf{y}} - \hat{\mathbf{y}}_1 = \mathbf{z}_1 - \mathbf{z}$$

This equality shows that the vector $\mathbf{v} = \hat{\mathbf{y}} - \hat{\mathbf{y}}_1$ is in W and in $W^\perp$ (because $\mathbf{z}_1$ and $\mathbf{z}$ are both in $W^\perp$, and $W^\perp$ is a subspace). Hence $\mathbf{v}\cdot\mathbf{v} = 0$, which shows that $\mathbf{v} = \mathbf{0}$. This proves that $\hat{\mathbf{y}} = \hat{\mathbf{y}}_1$ and also $\mathbf{z}_1 = \mathbf{z}$. ■

[1] We may assume that W is not the zero subspace, for otherwise $W^\perp = \mathbb{R}^n$ and (1) is simply $\mathbf{y} = \mathbf{0} + \mathbf{y}$.

Example

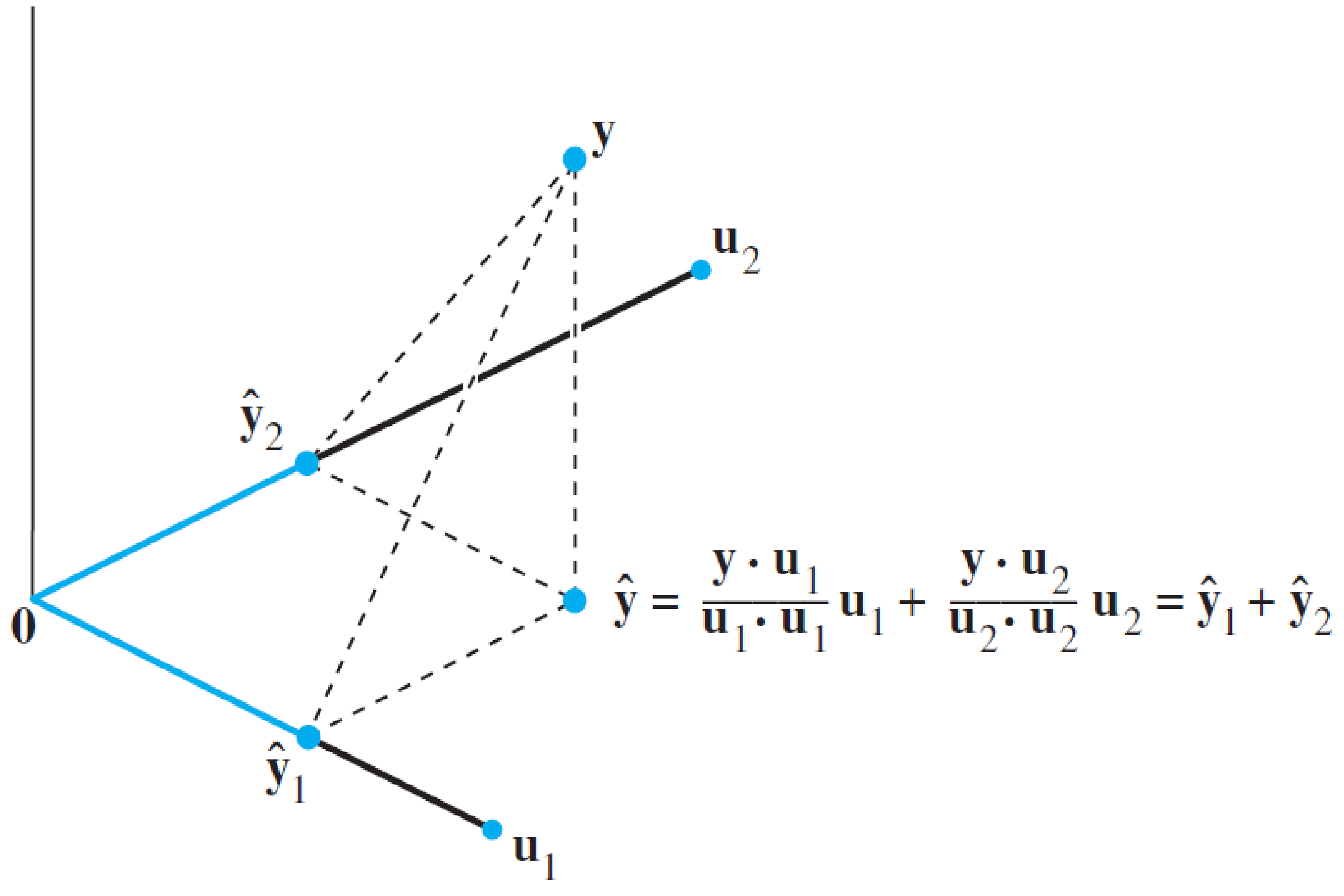

FIGURE 3 The orthogonal projection of $\mathbf{y}$ is the sum of its projections onto one-dimensional subspaces that are mutually orthogonal.

Figure 3 W is a subspace of $\mathbb{R}^3$ spanned by $\mathbf{u}_1$ and $\mathbf{u}_2$. Here $\hat{\mathbf{y}}_1$ and $\hat{\mathbf{y}}_2$ denote the projections of $\mathbf{y}$ onto the lines spanned by $\mathbf{u}_1$ and $\mathbf{u}_2$, respectively. The orthogonal projection $\hat{\mathbf{y}}$ of $\mathbf{y}$ onto W is the sum of the projections of $\mathbf{y}$ onto one-dimensional subspaces that are orthogonal to each other.

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EXAMPLE 2 Let $\mathbf{u}_1 = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$, and $\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Observe that $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal basis for $W = \text{Span}\,\{\mathbf{u}_1, \mathbf{u}_2\}$. Write $\mathbf{y}$ as the sum of a vector in W and a vector orthogonal to W.

SOLUTION The orthogonal projection of $\mathbf{y}$ onto W is

$$\begin{aligned}\hat{\mathbf{y}} &= \frac{\mathbf{y}\cdot\mathbf{u}_1}{\mathbf{u}_1\cdot\mathbf{u}_1}\mathbf{u}_1 + \frac{\mathbf{y}\cdot\mathbf{u}_2}{\mathbf{u}_2\cdot\mathbf{u}_2}\mathbf{u}_2 \\ &= \frac{9}{30}\begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} + \frac{3}{6}\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = \frac{9}{30}\begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} + \frac{15}{30}\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix}\end{aligned}$$

Also

$$\mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix} = \begin{bmatrix} 7/5 \\ 0 \\ 14/5 \end{bmatrix}$$

Theorem 8 ensures that $\mathbf{y} - \hat{\mathbf{y}}$ is in $W^\perp$. To check the calculations, however, it is a good idea to verify that $\mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to both $\mathbf{u}_1$ and $\mathbf{u}_2$ and hence to all of W. The desired decomposition of $\mathbf{y}$ is

$$\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix} + \begin{bmatrix} 7/5 \\ 0 \\ 14/5 \end{bmatrix}$$

Best Approximation Theorem

THEOREM 9

The Best Approximation Theorem

Let W be a subspace of $\mathbb{R}^n$, let $\mathbf{y}$ be any vector in $\mathbb{R}^n$, and let $\hat{\mathbf{y}}$ be the orthogonal projection of $\mathbf{y}$ onto W. Then $\hat{\mathbf{y}}$ is the closest point in W to $\mathbf{y}$, in the sense that

$$\|\mathbf{y} - \hat{\mathbf{y}}\| < \|\mathbf{y} - \mathbf{v}\| \tag{3}$$

for all $\mathbf{v}$ in W distinct from $\hat{\mathbf{y}}$.

Properties of Orthogonal Projections

If $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthogonal basis for W and if $\mathbf{y}$ happens to be in W, then the formula for $\text{proj}_W\, \mathbf{y}$ is exactly the same as the representation of $\mathbf{y}$ given in Theorem 5 in Section 6.2. In this case, $\text{proj}_W\, \mathbf{y} = \mathbf{y}$.

If $\mathbf{y}$ is in $W = \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$, then $\text{proj}_W\, \mathbf{y} = \mathbf{y}$.

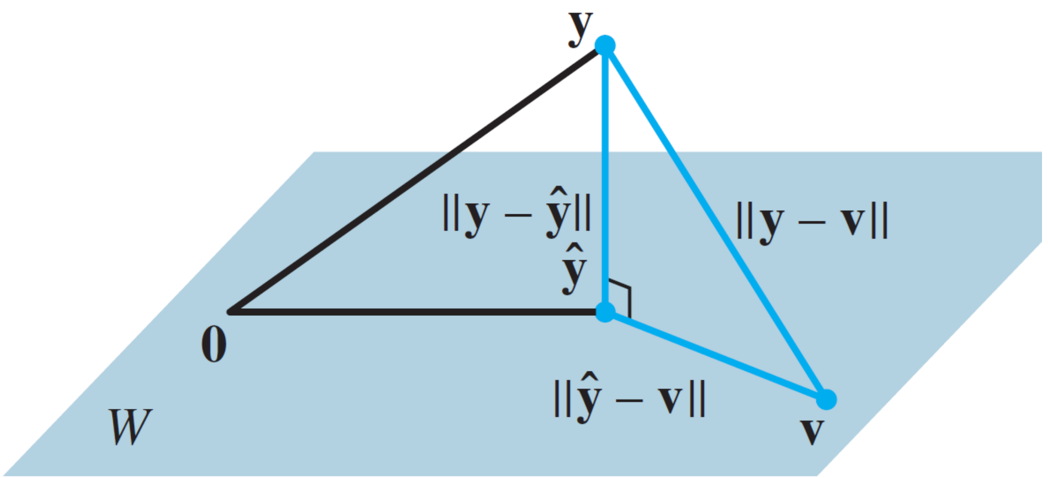

FIGURE 4 The orthogonal projection of $\mathbf{y}$ onto W is the closest point in W to $\mathbf{y}$.

The vector $\hat{\mathbf{y}}$ in Theorem 9 is called **the best approximation to y by elements of** W.

PROOF Take $\mathbf{v}$ in W distinct from $\hat{\mathbf{y}}$. See Fig. 4. Then $\hat{\mathbf{y}} - \mathbf{v}$ is in W. By the Orthogonal Decomposition Theorem, $\mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to W. In particular, $\mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to $\hat{\mathbf{y}} - \mathbf{v}$ (which is in W). Since

$$\mathbf{y} - \mathbf{v} = (\mathbf{y} - \hat{\mathbf{y}}) + (\hat{\mathbf{y}} - \mathbf{v})$$

the Pythagorean Theorem gives

$$\|\mathbf{y} - \mathbf{v}\|^2 = \|\mathbf{y} - \hat{\mathbf{y}}\|^2 + \|\hat{\mathbf{y}} - \mathbf{v}\|^2$$

(See the colored right triangle in Fig. 4. The length of each side is labeled.) Now $\|\hat{\mathbf{y}} - \mathbf{v}\|^2 > 0$ because $\hat{\mathbf{y}} - \mathbf{v} \neq \mathbf{0}$, and so inequality (3) follows immediately. ■

Conclusion: if y is in the space of W, then we can show $\hat{y}$ to be equal to y using theorem 8. Else, it's the best approximation according to theorem 9.

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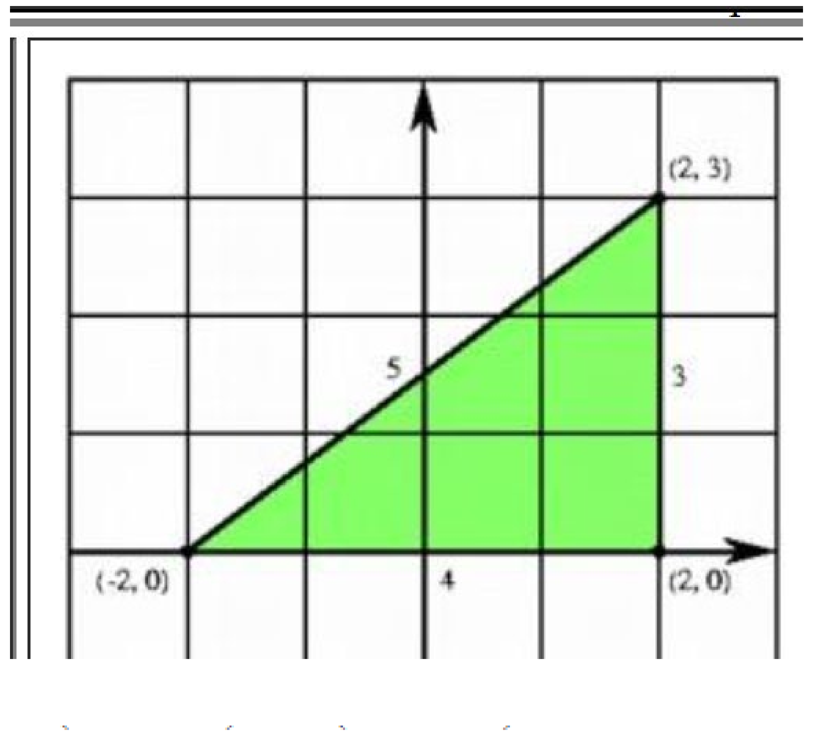
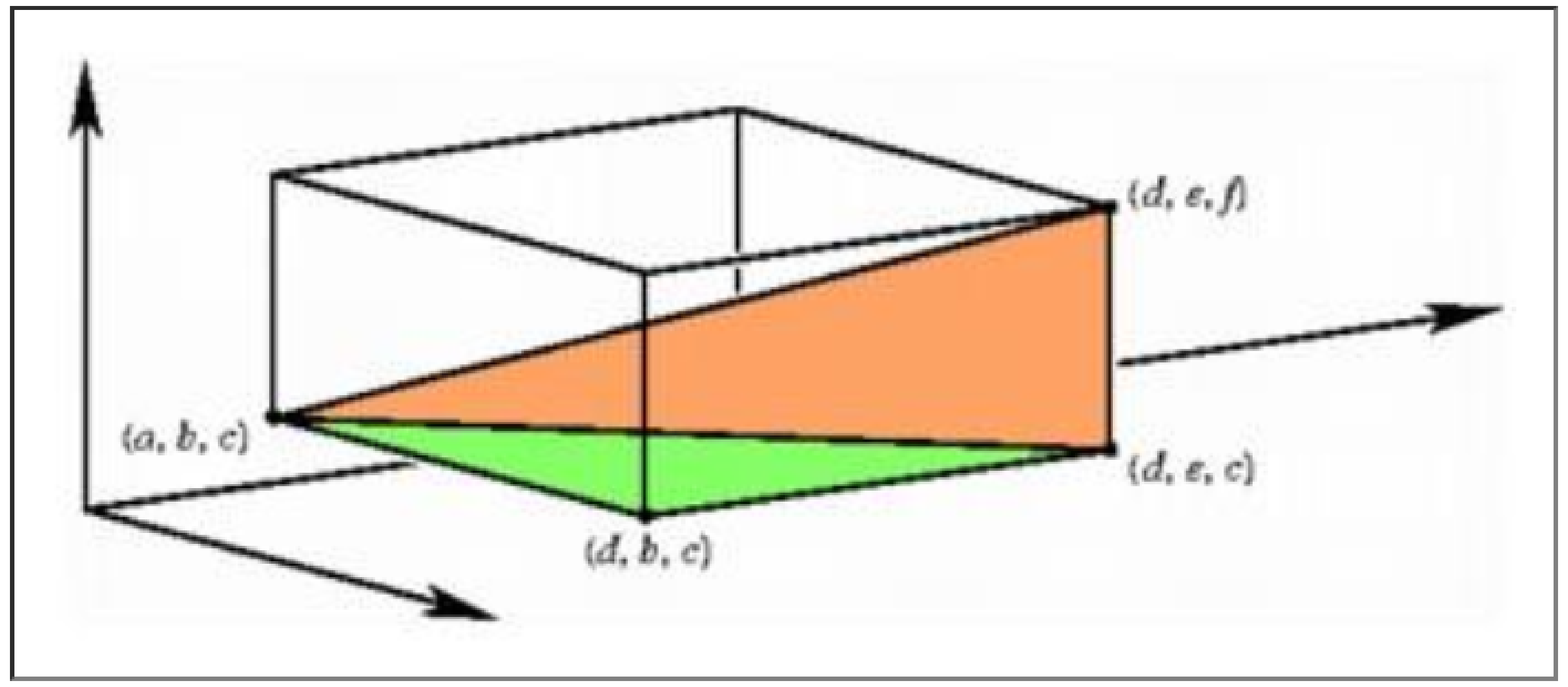
Using Pythagoras Theorem in Higher Dimensions

Ref:

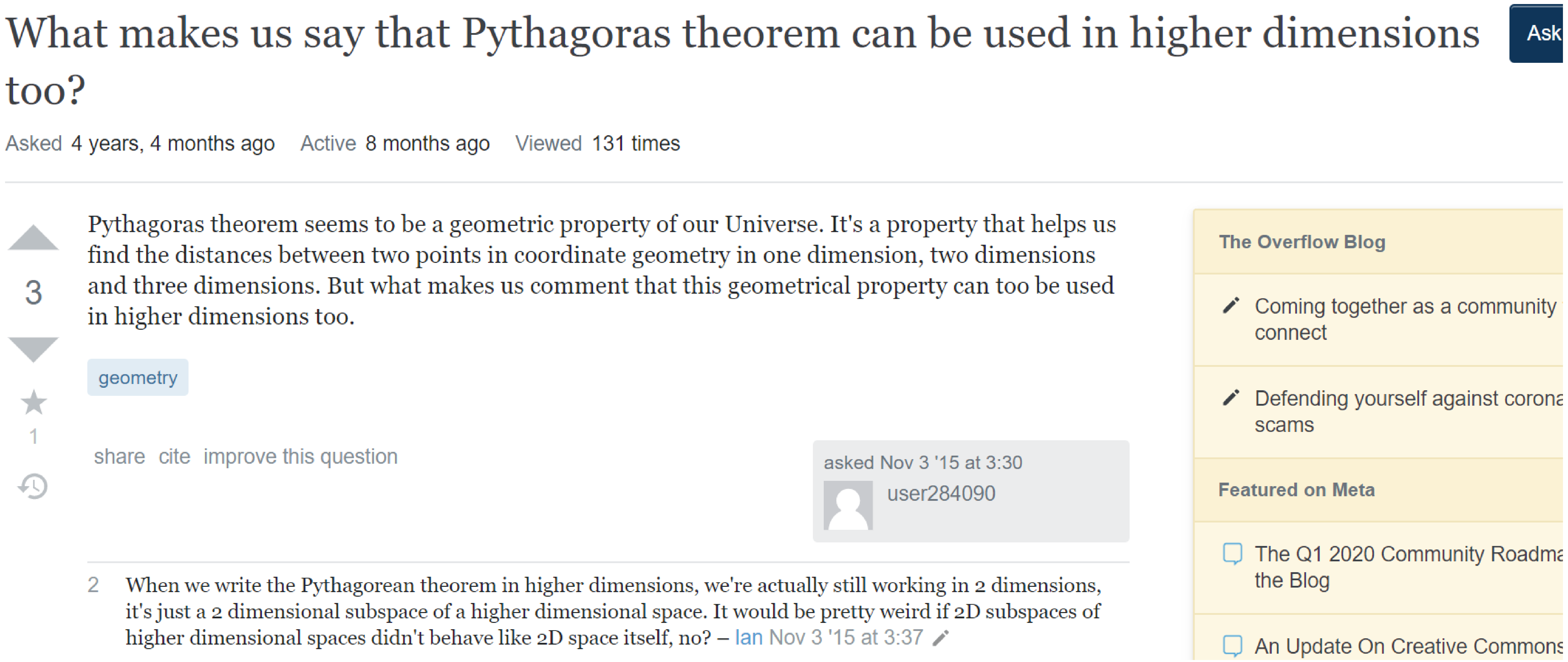
i) https://math.stackexchange.com/questions/1510549/what-makes-us-say-that-pythagoras-theorem-can-be-used-in-higher-dimensions-too

ii) http://www.math.brown.edu/~banchoff/Beyond3d/chapter8/section02.html





To find the distance formula in three-space, we apply the planar distance formula twice, once to find the length of the diagonal of one of the faces, and then to find the hypotenuse of a right triangle having this diagonal and an edge of the cube as its sides.



What makes us say that Pythagoras theorem can be used in higher dimensions too?

Asked 4 years, 4 months ago Active 8 months ago Viewed 131 times

3

Pythagoras theorem seems to be a geometric property of our Universe. It's a property that helps us find the distances between two points in coordinate geometry in one dimension, two dimensions and three dimensions. But what makes us comment that this geometrical property can too be used in higher dimensions too.

geometry

share cite improve this question

asked Nov 3 '15 at 3:30

user284090

2 When we write the Pythagorean theorem in higher dimensions, we're actually still working in 2 dimensions, it's just a 2 dimensional subspace of a higher dimensional space. It would be pretty weird if 2D subspaces of higher dimensional spaces didn't behave like 2D space itself, no? – Ian Nov 3 '15 at 3:37

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Example

EXAMPLE 3 If $\mathbf{u}_1 = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, and $W = \text{Span}\,\{\mathbf{u}_1, \mathbf{u}_2\}$, as in Example 2, then the closest point in W to $\mathbf{y}$ is

$$\hat{\mathbf{y}} = \frac{\mathbf{y}\cdot\mathbf{u}_1}{\mathbf{u}_1\cdot\mathbf{u}_1}\mathbf{u}_1 + \frac{\mathbf{y}\cdot\mathbf{u}_2}{\mathbf{u}_2\cdot\mathbf{u}_2}\mathbf{u}_2 = \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix}$$

■

EXAMPLE 4 The distance from a point $\mathbf{y}$ in $\mathbb{R}^n$ to a subspace W is defined as the distance from $\mathbf{y}$ to the nearest point in W. Find the distance from $\mathbf{y}$ to $W = \text{Span}\,\{\mathbf{u}_1, \mathbf{u}_2\}$, where

$$\mathbf{y} = \begin{bmatrix} -1 \\ -5 \\ 10 \end{bmatrix}, \quad \mathbf{u}_1 = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

SOLUTION By the Best Approximation Theorem, the distance from $\mathbf{y}$ to W is $\|\mathbf{y} - \hat{\mathbf{y}}\|$, where $\hat{\mathbf{y}} = \text{proj}_W\,\mathbf{y}$. Since $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal basis for W,

$$\hat{\mathbf{y}} = \frac{15}{30}\mathbf{u}_1 + \frac{-21}{6}\mathbf{u}_2 = \frac{1}{2}\begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix} - \frac{7}{2}\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ -8 \\ 4 \end{bmatrix}$$

$$\mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} -1 \\ -5 \\ 10 \end{bmatrix} - \begin{bmatrix} -1 \\ -8 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 6 \end{bmatrix}$$

$$\|\mathbf{y} - \hat{\mathbf{y}}\|^2 = 3^2 + 6^2 = 45$$

The distance from $\mathbf{y}$ to W is $\sqrt{45} = 3\sqrt{5}$. ■

Best Approximation Theorem

THEOREM 10

If $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthonormal basis for a subspace W of $\mathbb{R}^n$, then

$$\operatorname{proj}_W \mathbf{y} = (\mathbf{y}\cdot\mathbf{u}_1)\mathbf{u}_1 + (\mathbf{y}\cdot\mathbf{u}_2)\mathbf{u}_2 + \cdots + (\mathbf{y}\cdot\mathbf{u}_p)\mathbf{u}_p \tag{4}$$

If $U = [\,\mathbf{u}_1 \;\; \mathbf{u}_2 \;\; \cdots \;\; \mathbf{u}_p\,]$, then

$$\operatorname{proj}_W \mathbf{y} = UU^T\mathbf{y} \quad \text{for all } \mathbf{y} \text{ in } \mathbb{R}^n \tag{5}$$

$Note: \; p < n$

PROOF Formula (4) follows immediately from (2) in Theorem 8. Also, (4) shows that $\operatorname{proj}_W \mathbf{y}$ is a linear combination of the columns of U using the weights $\mathbf{y}\cdot\mathbf{u}_1, \mathbf{y}\cdot\mathbf{u}_2, \dots, \mathbf{y}\cdot\mathbf{u}_p$. The weights can be written as $\mathbf{u}_1^T\mathbf{y}, \mathbf{u}_2^T\mathbf{y}, \dots, \mathbf{u}_p^T\mathbf{y}$, showing that they are the entries in $U^T\mathbf{y}$ and justifying (5). ■

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Suppose U is an $n \times p$ matrix with orthonormal columns, and let W be the column space of U. Then

$$U^TU\mathbf{x} = I_p\mathbf{x} = \mathbf{x} \quad \text{for all } \mathbf{x} \text{ in } \mathbb{R}^p \qquad \text{Theorem 6}$$

$$UU^T\mathbf{y} = \operatorname{proj}_W \mathbf{y} \quad \text{for all } \mathbf{y} \text{ in } \mathbb{R}^n \qquad \text{Theorem 10}$$

If U is an $n \times n$ (square) matrix with orthonormal columns, then U is an *orthogonal* matrix, the column space W is all of $\mathbb{R}^n$, and $UU^T\mathbf{y} = I\mathbf{y} = \mathbf{y}$ for all $\mathbf{y}$ in $\mathbb{R}^n$.

Although formula (4) is important for theoretical purposes, in practice it usually involves calculations with square roots of numbers (in the entries of the $\mathbf{u}_i$). Formula (2) is recommended for hand calculations.

PRACTICE PROBLEM

Let $\mathbf{u}_1 = \begin{bmatrix} -7 \\ 1 \\ 4 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} -9 \\ 1 \\ 6 \end{bmatrix}$, and $W = \operatorname{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$. Use the fact that $\mathbf{u}_1$ and $\mathbf{u}_2$ are orthogonal to compute $\operatorname{proj}_W \mathbf{y}$.

Note: above u_1 ,and u_2 are orthogonal, but their norm is not 1.

To use

Theorem 10 (orthonormality is needed).

You need to convert u_1 and u_2 to normalized col

Else

Use theorem 8 (only orthogonality is needed).

Example

PRACTICE PROBLEM

Let $\mathbf{u}_1 = \begin{bmatrix} -7 \\ 1 \\ 4 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} -9 \\ 1 \\ 6 \end{bmatrix}$, and $W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$. Use the fact that $\mathbf{u}_1$ and $\mathbf{u}_2$ are orthogonal to compute $\text{proj}_W\, \mathbf{y}$.

example4_Walsh.m | example5_orthogonalSet.m | example6_Lay_pg352.m

```matlab
% Lay Pg 352, orthogonality, projecting y onto U
U = [- 7 -1; 1 1; 4 -2];
y = [-9 1 6]'

U  % sanity check
y  % to see the values

orthogonalityCheck = U'*U   % quick way to find transpose column of U
    % and dot producting its other column

u1 = U(:,1) % extracting u1 and u2 from the column of U
u2 = U(:,2)

u1'*u2   % checking u1 is orthogonal to u2 using dot product

Proj_y_onto_u1 = y'*u1/(u1'*u1)   % remember to normalize using denorminator
Proj_y_onto_u2 = y'*u2/(u2'*u2)   % to make it unit vector

est_y = Proj_y_onto_u1*u1 + Proj_y_onto_u2*u2
est_y - y

% Using projection matrix to find est_y2
U_norm(:,1) = u1/sqrt((u1'*u1))
U_norm(:,2) = u2/sqrt((u2'*u2))

est_y = U_norm*U_norm'*y


y2 = [-8.5, 1,6]'
est_y2 = U_norm*U_norm'*y2
error = y2 - est_y2
```


Screen shot showing the answer:

```
U =

    -7    -1
     1     1
     4    -2


y =

    -9
     1
     6


orthogonalityCheck =

    66     0
     0     6
```

```
Proj_y_onto_u1 =

    1.3333


Proj_y_onto_u2 =

   -0.3333


est_y =

   -9.0000
    1.0000
    6.0000

ans =

   1.0e-14 *

    0.1776
         0
         0
```

```
U_norm =

   -0.8616   -0.4082
    0.1231    0.4082
    0.4924   -0.8165


est_y =

   -9.0000
    1.0000
    6.0000
```

```
y2 =

   -8.5000
    1.0000
    6.0000


est_y2 =

   -8.5455
    0.8636
    5.9545


error =

    0.0455
    0.1364
    0.0455
```

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}}_{b}$$

CX1104: **Linear Algebra for Computing**

Chap. No : **6.3.1**

Lecture : **Orthogonality**

Topic : **Gram–Schmidt for QR**

Concept : **Motivation and Review of Concepts**

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su, Vishal Choudhari**

Rev: ces-22July2020

Rev: 29th June 2020

Introducing   A = QR

QR decomposition, given A (mxn) matrix, we can decompose it into the product of 2 matrixes,

$$A = QR$$

- Properties of Q:
  - C(Q) == C(A)
  - $Q^T Q = I$, i.e Q has orthonormal column (BUT not necessary square)
  - QQ^T = projection matrix into col (A)

- R is a square upper triangle matrix and Depending if
  - A has independent col, then R is invertible,
  - A has dependent col, then R is NOT-invertible.

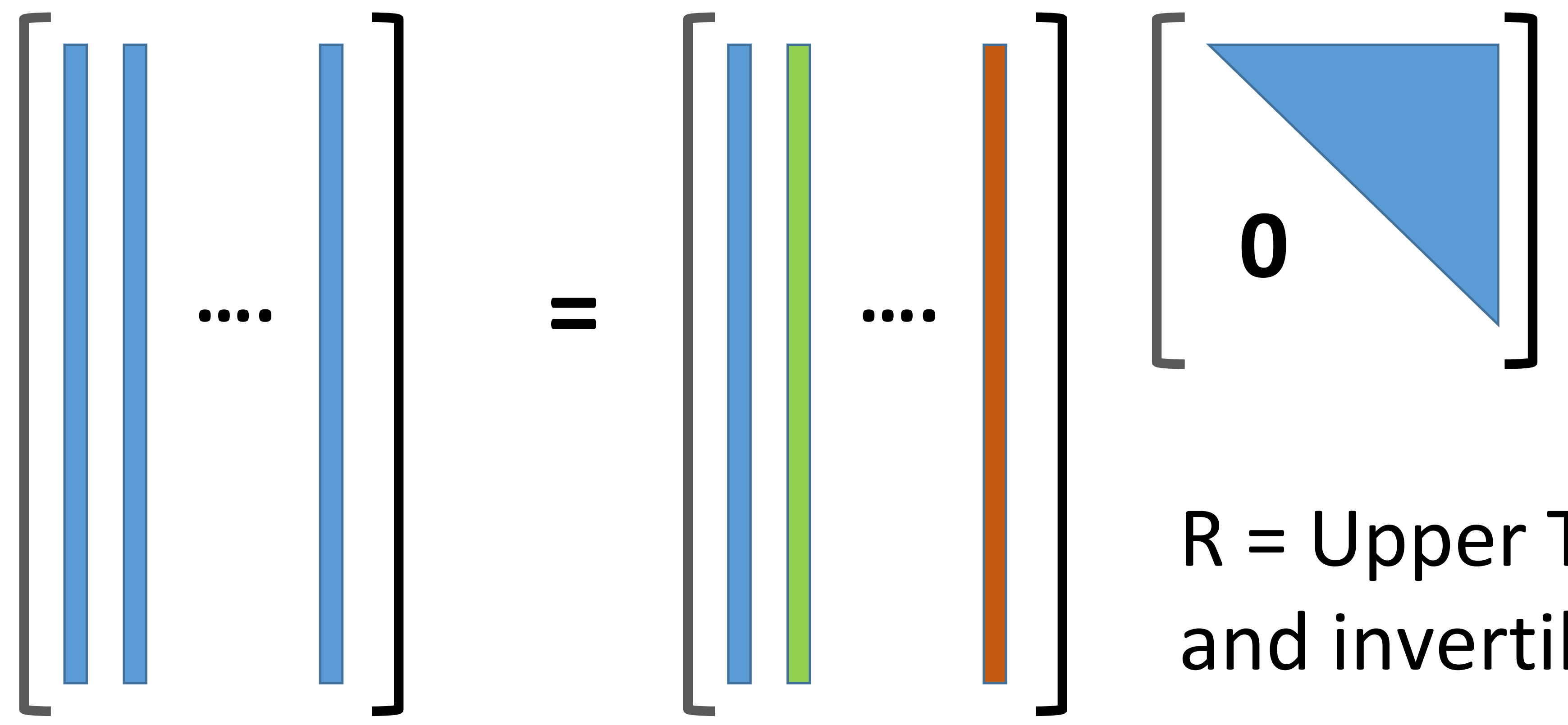
Warning: Theorem 12 is for A Having independent column.

THEOREM 12

The QR Factorization

If A is an $m \times n$ matrix with linearly independent columns, then A can be factored as $A = QR$, where Q is an $m \times n$ matrix whose columns form an orthonormal basis for Col A and R is an $n \times n$ upper triangular invertible matrix with positive entries on its diagonal.

$$A = [\,\mathbf{x}_1 \quad \cdots \quad \mathbf{x}_n\,] = [\,Q\mathbf{r}_1 \quad \cdots \quad Q\mathbf{r}_n\,] = QR$$



Ref: 1) https://en.wikipedia.org/wiki/QR_decomposition
2) http://ee263.stanford.edu/lectures/qr.pdf

Motivation for QR

It has many applications, e.g,
solving least squares:

$$Ax = y$$
$$QRx = y$$
$$Q^T QRx = Q^T y$$

$$Rx = Q^T y$$

Is can be easily solved because R is upper triangle
(If R is not invertible, it will be more involved, see least squares chapter)

There are at least 3 approaches to realise QR decomposition,
we will only introduce Gram-Schmidt orthogonalization to get Q first
https://www.math.ucla.edu/~yanovsky/Teaching/Math151B/handouts/GramSchmidt.pdf
https://towardsdatascience.com/can-qr-decomposition-be-actually-faster-schwarz-rutishauser-algorithm-a32c0cde8b9b

What does having same column space mean? C(Q) == C(A)

When we can convert

$$Ax = b$$

To

$$QRx = b$$

Then finding the solution
x is easier and more efficient especially
when A is large sized
(e.g thousands of rows and columns)

The found solution x will be the same.

Chapter 3. Vector Spaces and Subspaces

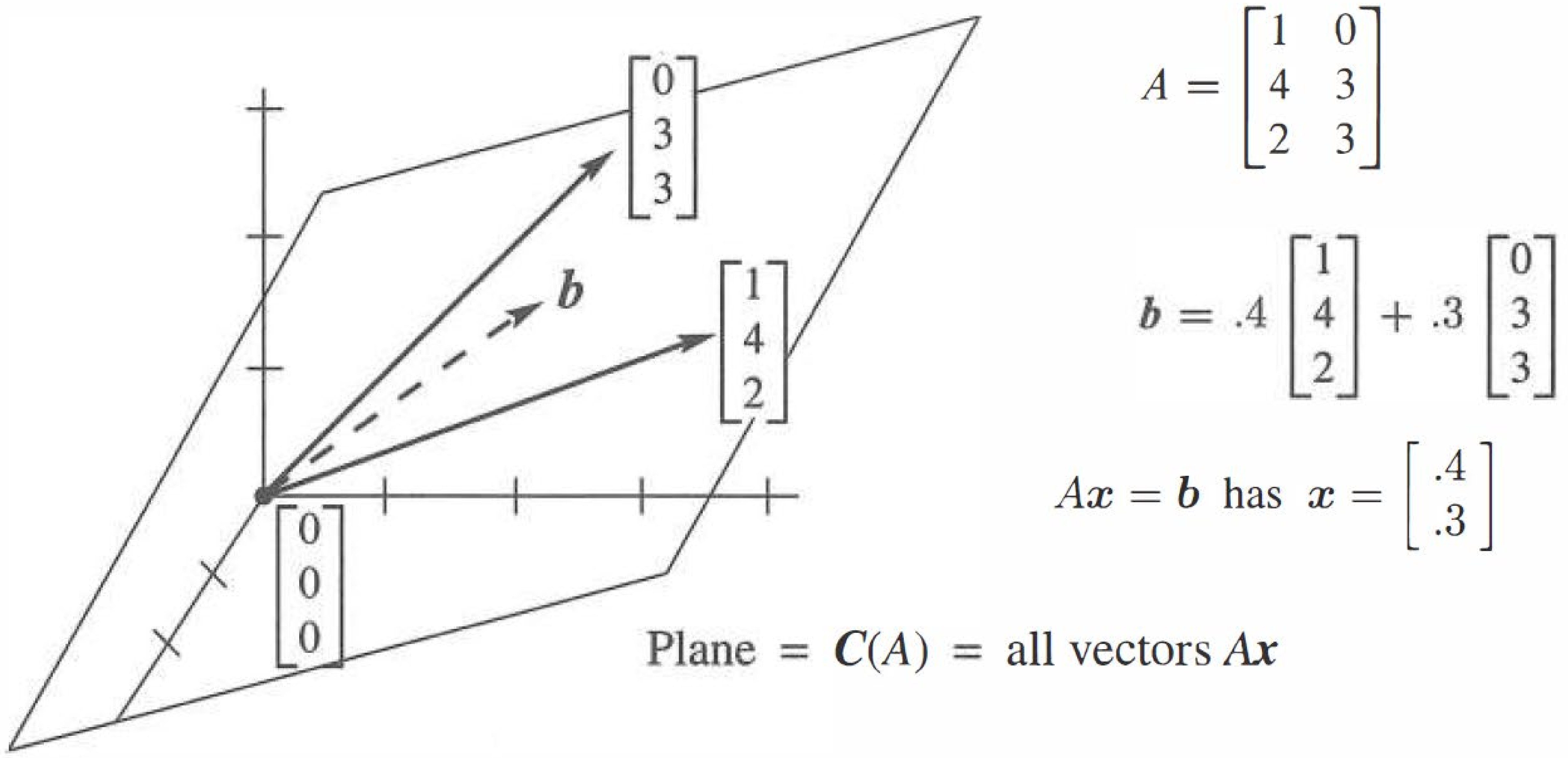


Figure 3.2: The column space $C(A)$ is a plane containing the two columns. $Ax = b$ is solvable when b is on that plane. Then b is a combination of the columns.

Interpretation: since C(Q) == C(A),
then Columns of Q are orthonormal basis for range(A), since $Q^T Q = I$

Revision: Projecting y onto an orthogonal vs orthonormal basis

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THEOREM 8 **The Orthogonal Decomposition Theorem**

Let W be a subspace of $\mathbb{R}^n$. Then each $\mathbf{y}$ in $\mathbb{R}^n$ can be written uniquely in the form

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z} \tag{1}$$

where $\hat{\mathbf{y}}$ is in W and $\mathbf{z}$ is in $W^\perp$. In fact, if $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is any orthogonal basis of W, then

$$\hat{\mathbf{y}} = \frac{\mathbf{y}\cdot\mathbf{u}_1}{\mathbf{u}_1\cdot\mathbf{u}_1}\mathbf{u}_1 + \cdots + \frac{\mathbf{y}\cdot\mathbf{u}_p}{\mathbf{u}_p\cdot\mathbf{u}_p}\mathbf{u}_p \tag{2}$$

and $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$.

The vector $\hat{\mathbf{y}}$ in (1) is called the **orthogonal projection of y onto** W and often is written as $\text{proj}_W\,\mathbf{y}$. See Figure 2. When W is a one-dimensional subspace, the formula for $\hat{\mathbf{y}}$ matches the formula given in Section 6.2.

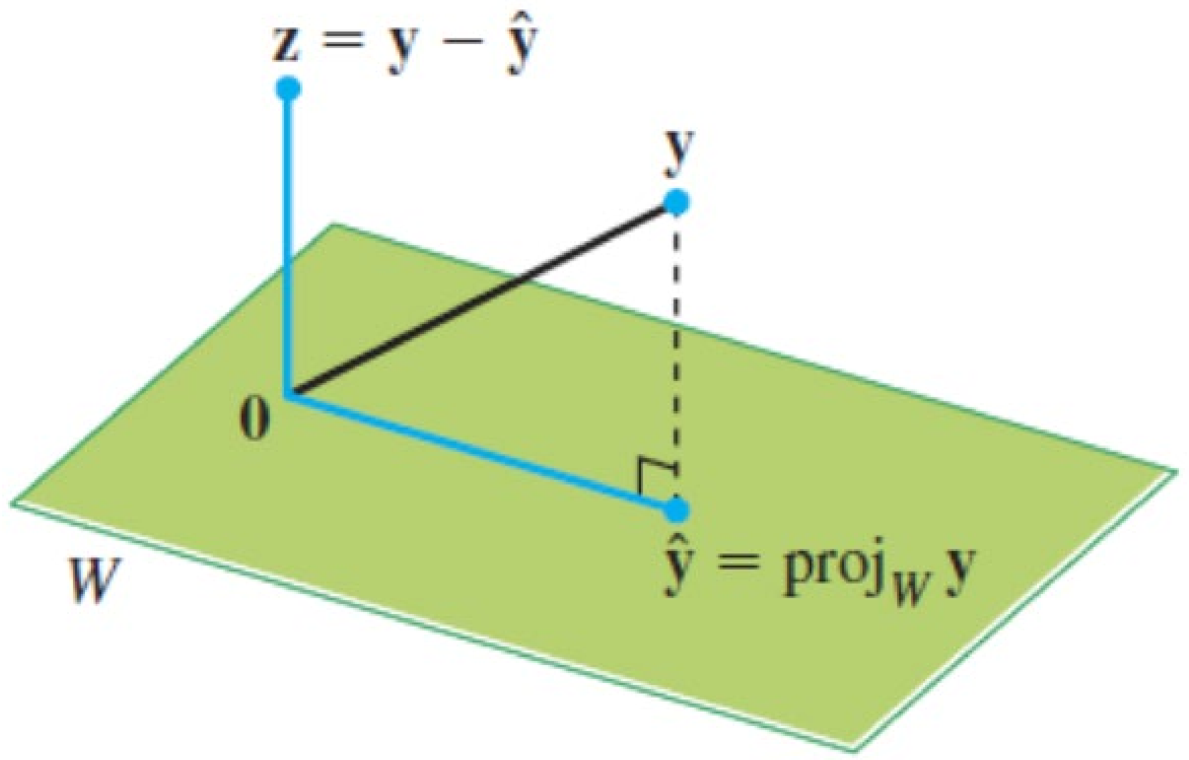

FIGURE 2 The orthogonal projection of $\mathbf{y}$ onto W.

Lay5e pg 353

The final theorem in this section shows how formula (2) for $\text{proj}_W\,\mathbf{y}$ is simplified when the basis for W is an orthonormal set.

THEOREM 10

If $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthonormal basis for a subspace W of $\mathbb{R}^n$, then

$$\text{proj}_W\,\mathbf{y} = (\mathbf{y}\cdot\mathbf{u}_1)\mathbf{u}_1 + (\mathbf{y}\cdot\mathbf{u}_2)\mathbf{u}_2 + \cdots + (\mathbf{y}\cdot\mathbf{u}_p)\mathbf{u}_p \tag{4}$$

If $U = [\,\mathbf{u}_1 \;\; \mathbf{u}_2 \;\; \cdots \;\; \mathbf{u}_p\,]$, then

$$\text{proj}_W\,\mathbf{y} = UU^T\mathbf{y} \quad \text{for all } \mathbf{y} \text{ in } \mathbb{R}^n \tag{5}$$

PROOF Formula (4) follows immediately from (2) in Theorem 8. Also, (4) shows that $\text{proj}_W\,\mathbf{y}$ is a linear combination of the columns of U using the weights $\mathbf{y}\cdot\mathbf{u}_1, \mathbf{y}\cdot\mathbf{u}_2, \dots, \mathbf{y}\cdot\mathbf{u}_p$. The weights can be written as $\mathbf{u}_1^T\mathbf{y}, \mathbf{u}_2^T\mathbf{y}, \dots, \mathbf{u}_p^T\mathbf{y}$, showing that they are the entries in $U^T\mathbf{y}$ and justifying (5). ■

How to find Q from A?

What does Gram-Schmidt Process Do?

It orthogonalises a set of vectors!

- A is a mxn matrix,
  assume columns are Not orthogonal
- $\boldsymbol{v_i} \in \boldsymbol{R^m}, i = 1..\boldsymbol{n}$

- The columns of Q are Orthonormal u_i
  $i = 1..n$
- **And spans the same space as A**

$$A = \{v_1, v_2, v_3, \cdots, v_n\} \rightarrow \boxed{\text{Gram-Schmidt Process}} \rightarrow Q = \{u_1, u_2, u_3, \cdots, u_n\}$$

$$Span(A) = Span(Q)$$

Note:Q spans the same m-dimensional subspace of R^m as that of A

Ref: http://www.seas.ucla.edu/~vandenbe/133A/lectures/qr.pdf Slide 6.7
See Matlab: https://www.mathworks.com/help/matlab/ref/qr.html
(economy QR factorization vs full QR decomposition)
More explanations of full vs economy qr : http://www.ece.northwestern.edu/local-apps/matlabhelp/techdoc/math_anal/mat_li23.html

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}}_{b}$$

CX1104: **Linear Algebra for Computing**

Chap. No : **6.3.2**

Lecture : **Orthogonality**

Topic : **Gram–Schmidt Process**

Concept : **Gram-Schmidt Process for QR decomposition**

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QR Factorisation revisited

The QR decomposition can be performed by Gram–Schmidt. Given a matrix A (mxn sized),

$$A = QR$$

$$A = \begin{bmatrix} q_1 & q_2 & \cdots & q_n \end{bmatrix} \begin{bmatrix} R_{11} & R_{12} & \cdots & R_{1n} \\ 0 & R_{22} & \cdots & R_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R_{nn} \end{bmatrix}$$

The Q Factor (economy qr):

- Q is $m \times n$ with orthonormal columns and $Q^TQ = I$ dimension nxn
- If A is square $(m = n)$, then Q is orthogonal, i.e, $Q^TQ = QQ^T = I$
- If A is tall (m>n), then $QQ^T \neq I$,
  The matrix QQ^T is a projection matrix
  of dimension mxm, and it will project a vector R^m onto
  the columns space of A. In other words,
  $QQ^Ty = \hat{y}$, where $\hat{y}$ is the least squares error approximation of y in the
  column space of A (see sec 7.1.4)

The R Factor:

- R is $n \times n$ upper triangular,
- If A has independent column, then R is invertible,
  else R is singular (not-invertible)

- Vectors $q_1, q_2, \dots, q_n$ are orthonormal m-dimensional vectors: $||q_i|| = 1 \;\; and \;\; q_i^T q_j = 0 \; if \;\; i \neq j$

NOTE:

Q is obtained by performing GS Process on A

The Gram Schmidt Process

What does Gram-Schmidt Process Do?

It orthogonalises a set of vectors!

$W = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_n\}$
$\boldsymbol{x}_i \in R^m$

Gram-Schmidt Process

$W' = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n\}$
$\boldsymbol{v}_i \in R^m$ and
$\boldsymbol{v}_i . \boldsymbol{v}_j = 0$ when $i \neq j$

$$Span(W) = Span(W')$$

Typically, we are give a matrix A, and these x_i are columns of A.

The Gram Schmidt Process

In basic Gram-Schmidt, we assume that $\{x_1, x_2, ..\}$ are independent columns,

THEOREM 11

The Gram–Schmidt Process

Given a basis $\{\mathbf{x}_1, \dots, \mathbf{x}_p\}$ for a nonzero subspace W of $\mathbb{R}^n$, define

$$\begin{aligned}
\mathbf{v}_1 &= \mathbf{x}_1 \\
\mathbf{v}_2 &= \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 \\
\mathbf{v}_3 &= \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2 \\
&\vdots \\
\mathbf{v}_p &= \mathbf{x}_p - \frac{\mathbf{x}_p \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{x}_p \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2 - \cdots - \frac{\mathbf{x}_p \cdot \mathbf{v}_{p-1}}{\mathbf{v}_{p-1} \cdot \mathbf{v}_{p-1}}\mathbf{v}_{p-1}
\end{aligned}$$

Then $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is an orthogonal basis for W. In addition

$$\operatorname{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \operatorname{Span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\} \qquad \text{for } 1 \le k \le p \tag{1}$$

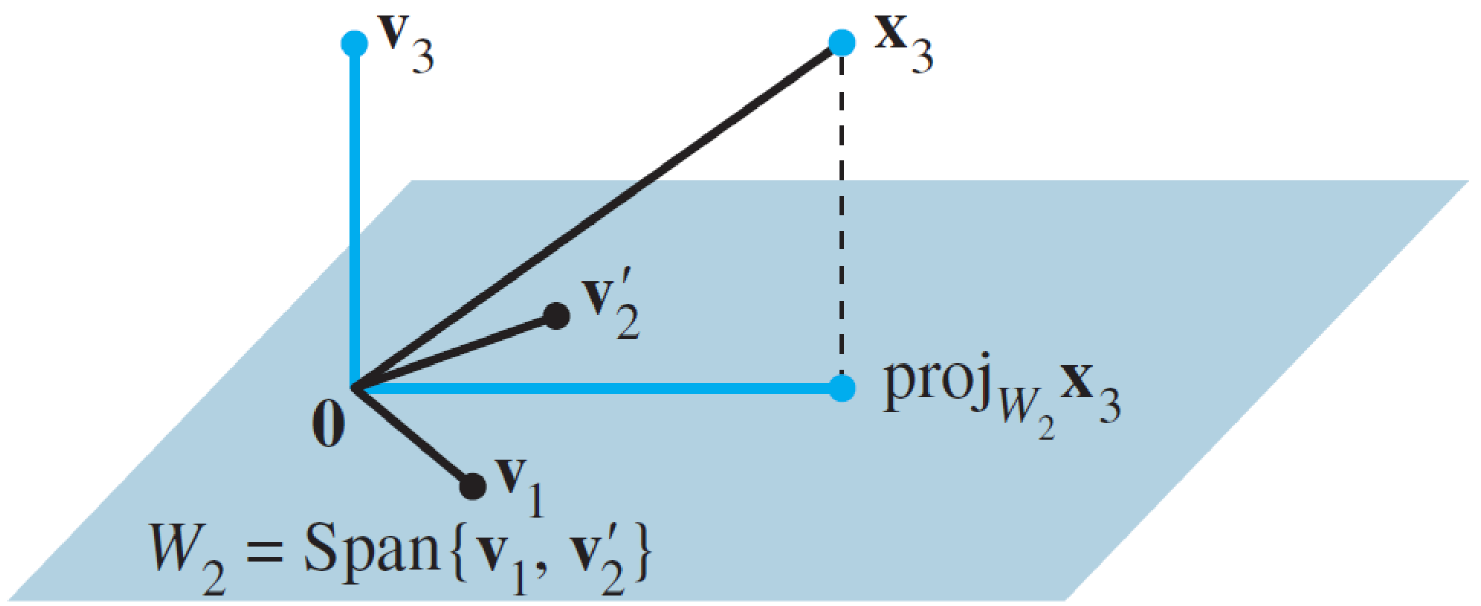

FIGURE 2 The construction of $\mathbf{v}_3$ from $\mathbf{x}_3$ and W_2.

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Gram-Schmidt Orthogonalization
Instructor: Ana Rita Pires
View the complete course: http://ocw.mit.edu/18-06SCF11

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Watch these worked out examples:

1. GramSchmidt: https://www.youtube.com/watch?v=Aslf3KGq2UE
2. QR: https://www.youtube.com/watch?v=6DybLNNkWyE
3. MIT Gram Schmidt: https://www.youtube.com/watch?v=TRktLuAktBQ&t=17s

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Proof Theorem 11: Span of vectors generated by GS is same as original set of vectors

PROOF For $1 \leq k \leq p$, let $W_k = \text{Span}\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$. Set $\mathbf{v}_1 = \mathbf{x}_1$, so that $\text{Span}\{\mathbf{v}_1\} = \text{Span}\{\mathbf{x}_1\}$. Suppose, for some $k < p$, we have constructed $\mathbf{v}_1, \ldots, \mathbf{v}_k$ so that $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is an orthogonal basis for W_k. Define

$$\mathbf{v}_{k+1} = \mathbf{x}_{k+1} - \text{proj}_{W_k} \mathbf{x}_{k+1} \tag{2}$$

By the Orthogonal Decomposition Theorem, $\mathbf{v}_{k+1}$ is orthogonal to W_k. Note that $\text{proj}_{W_k} \mathbf{x}_{k+1}$ is in W_k and hence also in W_{k+1}. Since $\mathbf{x}_{k+1}$ is in W_{k+1}, so is $\mathbf{v}_{k+1}$ (because W_{k+1} is a subspace and is closed under subtraction). Furthermore, $\mathbf{v}_{k+1} \neq \mathbf{0}$ because $\mathbf{x}_{k+1}$ is not in $W_k = \text{Span}\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$. Hence $\{\mathbf{v}_1, \ldots, \mathbf{v}_{k+1}\}$ is an orthogonal set of nonzero vectors in the $(k+1)$-dimensional space W_{k+1}. By the Basis Theorem in Section 4.5, this set is an orthogonal basis for W_{k+1}. Hence $W_{k+1} = \text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_{k+1}\}$. When $k + 1 = p$, the process stops. ∎

Theorem 11 shows that any nonzero subspace W of $\mathbb{R}^n$ has an orthogonal basis, because an ordinary basis $\{\mathbf{x}_1, \ldots, \mathbf{x}_p\}$ is always available (by Theorem 11 in Section 4.5), and the Gram–Schmidt process depends only on the existence of orthogonal projections onto subspaces of W that already have orthogonal bases.

Example:

EXAMPLE 2 Let $\mathbf{x}_1 = \begin{bmatrix}1\\1\\1\\1\end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix}0\\1\\1\\1\end{bmatrix}$, and $\mathbf{x}_3 = \begin{bmatrix}0\\0\\1\\1\end{bmatrix}$. Then $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ is clearly linearly independent and thus is a basis for a subspace W of $\mathbb{R}^4$. Construct an orthogonal basis for W.

SOLUTION

Step 1. Let $\mathbf{v}_1 = \mathbf{x}_1$ and $W_1 = \text{Span}\{\mathbf{x}_1\} = \text{Span}\{\mathbf{v}_1\}$.

Step 2. Let $\mathbf{v}_2$ be the vector produced by subtracting from $\mathbf{x}_2$ its projection onto the subspace W_1. That is, let

$$\begin{aligned}\mathbf{v}_2 &= \mathbf{x}_2 - \text{proj}_{W_1}\mathbf{x}_2 \\ &= \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 \qquad \text{Since } \mathbf{v}_1 = \mathbf{x}_1 \\ &= \begin{bmatrix}0\\1\\1\\1\end{bmatrix} - \frac{3}{4}\begin{bmatrix}1\\1\\1\\1\end{bmatrix} = \begin{bmatrix}-3/4\\1/4\\1/4\\1/4\end{bmatrix}\end{aligned}$$

As in Example 1, $\mathbf{v}_2$ is the component of $\mathbf{x}_2$ orthogonal to $\mathbf{x}_1$, and $\{\mathbf{v}_1, \mathbf{v}_2\}$ is an orthogonal basis for the subspace W_2 spanned by $\mathbf{x}_1$ and $\mathbf{x}_2$.

Note: $v_2' = v_2 * 4$ to get rid of denominator in v_2

Step 3. Let $\mathbf{v}_3$ be the vector produced by subtracting from $\mathbf{x}_3$ its projection onto the subspace W_2. Use the orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2'\}$ to compute this projection onto W_2:

$$\text{proj}_{W_2}\mathbf{x}_3 = \underbrace{\frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1}_{\text{Projection of } \mathbf{x}_3 \text{ onto } \mathbf{v}_1} + \underbrace{\frac{\mathbf{x}_3 \cdot \mathbf{v}_2'}{\mathbf{v}_2' \cdot \mathbf{v}_2'}\mathbf{v}_2'}_{\text{Projection of } \mathbf{x}_3 \text{ onto } \mathbf{v}_2'} = \frac{2}{4}\begin{bmatrix}1\\1\\1\\1\end{bmatrix} + \frac{2}{12}\begin{bmatrix}-3\\1\\1\\1\end{bmatrix} = \begin{bmatrix}0\\2/3\\2/3\\2/3\end{bmatrix}$$

Then $\mathbf{v}_3$ is the component of $\mathbf{x}_3$ orthogonal to W_2, namely,

$$\mathbf{v}_3 = \mathbf{x}_3 - \text{proj}_{W_2}\mathbf{x}_3 = \begin{bmatrix}0\\0\\1\\1\end{bmatrix} - \begin{bmatrix}0\\2/3\\2/3\\2/3\end{bmatrix} = \begin{bmatrix}0\\-2/3\\1/3\\1/3\end{bmatrix}$$

See Slide 3 of Chapter 6.2.5 for explanation.

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Example:

$$\mathbf{v}_3 = \mathbf{x}_3 - \text{proj}_{W_2}\,\mathbf{x}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 2/3 \\ 2/3 \\ 2/3 \end{bmatrix} = \begin{bmatrix} 0 \\ -2/3 \\ 1/3 \\ 1/3 \end{bmatrix}$$

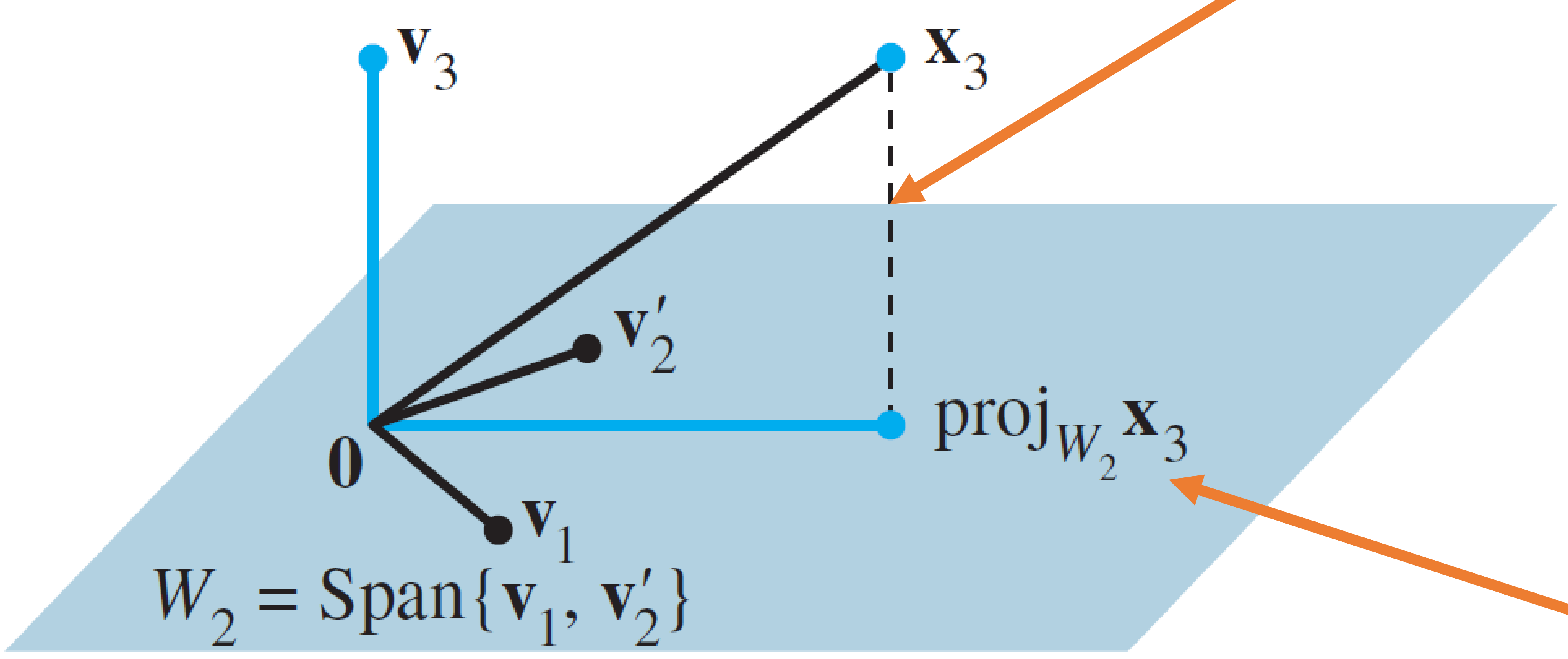

FIGURE 2 The construction of $\mathbf{v}_3$ from $\mathbf{x}_3$ and W_2.

See Fig. 2 for a diagram of this construction. Observe that $\mathbf{v}_3$ is in W, because $\mathbf{x}_3$ and $\text{proj}_{W_2}\mathbf{x}_3$ are both in W. Thus $\{\mathbf{v}_1, \mathbf{v}_2', \mathbf{v}_3\}$ is an orthogonal set of nonzero vectors and hence a linearly independent set in W. Note that W is three-dimensional since it was defined by a basis of three vectors. Hence, by the Basis Theorem in Section 4.5, $\{\mathbf{v}_1, \mathbf{v}_2', \mathbf{v}_3\}$ is an orthogonal basis for W. ■

$$\text{proj}_{W_2}\,\mathbf{x}_3 = \underbrace{\frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1}_{\substack{\text{Projection of} \\ \mathbf{x}_3 \text{ onto } \mathbf{v}_1}} + \underbrace{\frac{\mathbf{x}_3 \cdot \mathbf{v}_2'}{\mathbf{v}_2' \cdot \mathbf{v}_2'}\mathbf{v}_2'}_{\substack{\text{Projection of} \\ \mathbf{x}_3 \text{ onto } \mathbf{v}_2'}} = \frac{2}{4}\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \frac{2}{12}\begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2/3 \\ 2/3 \\ 2/3 \end{bmatrix}$$

The Basis Theorem

Let V be a p-dimensional vector space, $p \geq 1$. Any linearly independent set of exactly p elements in V is automatically a basis for V. Any set of exactly p elements that spans V is automatically a basis for V.

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Orthogonal basis vs Orthonormal basis

The columns of Q
are orthogonal,
as well as orthonormal!
$Q^T Q = I$

Orthonormal == orthogonal +
(length of vector ==1)

Orthonormal Bases

An orthonormal basis is constructed easily from an orthogonal basis $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$: simply normalize (i.e., "scale") all the $\mathbf{v}_k$. When working problems by hand, this is easier than normalizing each $\mathbf{v}_k$ as soon as it is found (because it avoids unnecessary writing of square roots).

EXAMPLE 3 Example 1 constructed the orthogonal basis

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

An orthonormal basis is

$$\mathbf{u}_1 = \frac{1}{\|\mathbf{v}_1\|}\mathbf{v}_1 = \frac{1}{\sqrt{45}}\begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \\ 0 \end{bmatrix}$$

$$\mathbf{u}_2 = \frac{1}{\|\mathbf{v}_2\|}\mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \qquad \blacksquare$$

The entries of R matrix during GS

Consider A has independent col (mxn matrix)

$$A = [\, x_1, x_2, x_3, \dots, x_n]$$

Gram–Schmidt Process

$$\mathrm{Proj}_v x = \frac{v.x}{v.v}\, v$$

$$v_1 = x_1$$

$$v_2 = x_2 - \mathrm{Proj}_{v_1} x_2$$

$$v_3 = x_3 - \mathrm{Proj}_{v_1} x_3 - \mathrm{Proj}_{v_2} x_3$$

$$v_i = x_i - \sum_{j=1}^{i-1} \mathrm{Proj}_{v_j} x_i$$

Ortho-normalization

$$u_1 = \frac{v_1}{||v_1||}$$

$$u_2 = \frac{v_2}{||v_2||}$$

$$u_i = \frac{v_i}{||v_i||}$$

We can now express the x_i over our newly computed orthonormal basis: e_j

$$x_1 = u_1.x_1\ u_1$$

$$x_2 = u_1.x_2\ u_1 + u_2.x_2\ u_2$$

$$x_3 = u_1.x_3\ u_1 + u_2.x_3\ u_2 + u_3.x_3\ u_3$$

$$\vdots$$

$$x_n = \sum_{j=1}^{n} u_j . x_n \ \ u_j$$

This can be written in matrix form:

$$A = QR$$

where:

$$Q = [\, u_1, u_2, u_3, \dots, u_n]$$

and

$$R = \begin{pmatrix} u_1.x_1 & u_1.x_2 & u_1.x_3 & \cdots \\ 0 & u_2.x_2 & u_2.x_3 & \cdots \\ 0 & 0 & u_3.x_3 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

https://www.math.ucla.edu/~yanovsky/Teaching/Math151B/handouts/GramSchmidt.pdf

Example: Gram–Schmidt on a 3x3 matrix

Example [edit]

Consider the decomposition of

$$A = \begin{pmatrix} 12 & -51 & 4 \\ 6 & 167 & -68 \\ -4 & 24 & -41 \end{pmatrix}.$$

Recall that an orthonormal matrix Q has the property

$$Q^\mathsf{T} Q = I.$$

Then, we can calculate Q by means of Gram–Schmidt as follows:

$$U = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{pmatrix} = \begin{pmatrix} 12 & -69 & -58/5 \\ 6 & 158 & 6/5 \\ -4 & 30 & -33 \end{pmatrix};$$

$$Q = \begin{pmatrix} \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|} & \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|} & \frac{\mathbf{u}_3}{\|\mathbf{u}_3\|} \end{pmatrix} = \begin{pmatrix} 6/7 & -69/175 & -58/175 \\ 3/7 & 158/175 & 6/175 \\ -2/7 & 6/35 & -33/35 \end{pmatrix}$$

Thus, we have

$$Q^\mathsf{T} A = Q^\mathsf{T} Q\, R = R;$$

$$R = Q^\mathsf{T} A = \begin{pmatrix} 14 & 21 & -14 \\ 0 & 175 & -70 \\ 0 & 0 & 35 \end{pmatrix}$$

Hence,

$$A = \begin{pmatrix} 12 & -51 & 4 \\ 6 & 167 & -68 \\ -4 & 24 & -41 \end{pmatrix} =$$

$$\begin{pmatrix} 6/7 & -69/175 & -58/175 \\ 3/7 & 158/175 & 6/175 \\ -2/7 & 6/35 & -33/35 \end{pmatrix} \times \begin{pmatrix} 14 & 21 & -14 \\ 0 & 175 & -70 \\ 0 & 0 & 35 \end{pmatrix}$$

Q: Orthogonal Matrix $\quad$ R: Upper Triangular Matrix

Ref: https://en.wikipedia.org/wiki/QR_decomposition#Using_the_Gram–Schmidt_process

Warning: modify QR when A has dependent column!

From: pg 4-6 http://ee263.stanford.edu/lectures/qr.pdf

Note: here the columns of Q are denoted as q_i

QR decomposition

written in matrix form: $A = QR$, where $A \in \mathbb{R}^{m\times n}$, $Q \in \mathbb{R}^{m\times n}$, $R \in \mathbb{R}^{n\times n}$:

$$\underbrace{\begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} q_1 & q_2 & \cdots & q_n \end{bmatrix}}_{Q} \underbrace{\begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ 0 & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & r_{nn} \end{bmatrix}}_{R}$$

- in basic G-S we assume $a_1, \ldots, a_n \in \mathbb{R}^m$ are independent
- if $a_1, \ldots, a_n$ are dependent, we find $\tilde{q}_j = 0$ for some j, which means a_j is linearly dependent on $a_1, \ldots, a_{j-1}$
- modified algorithm: when we encounter $\tilde{q}_j = 0$, skip to next vector a_{j+1} and continue:

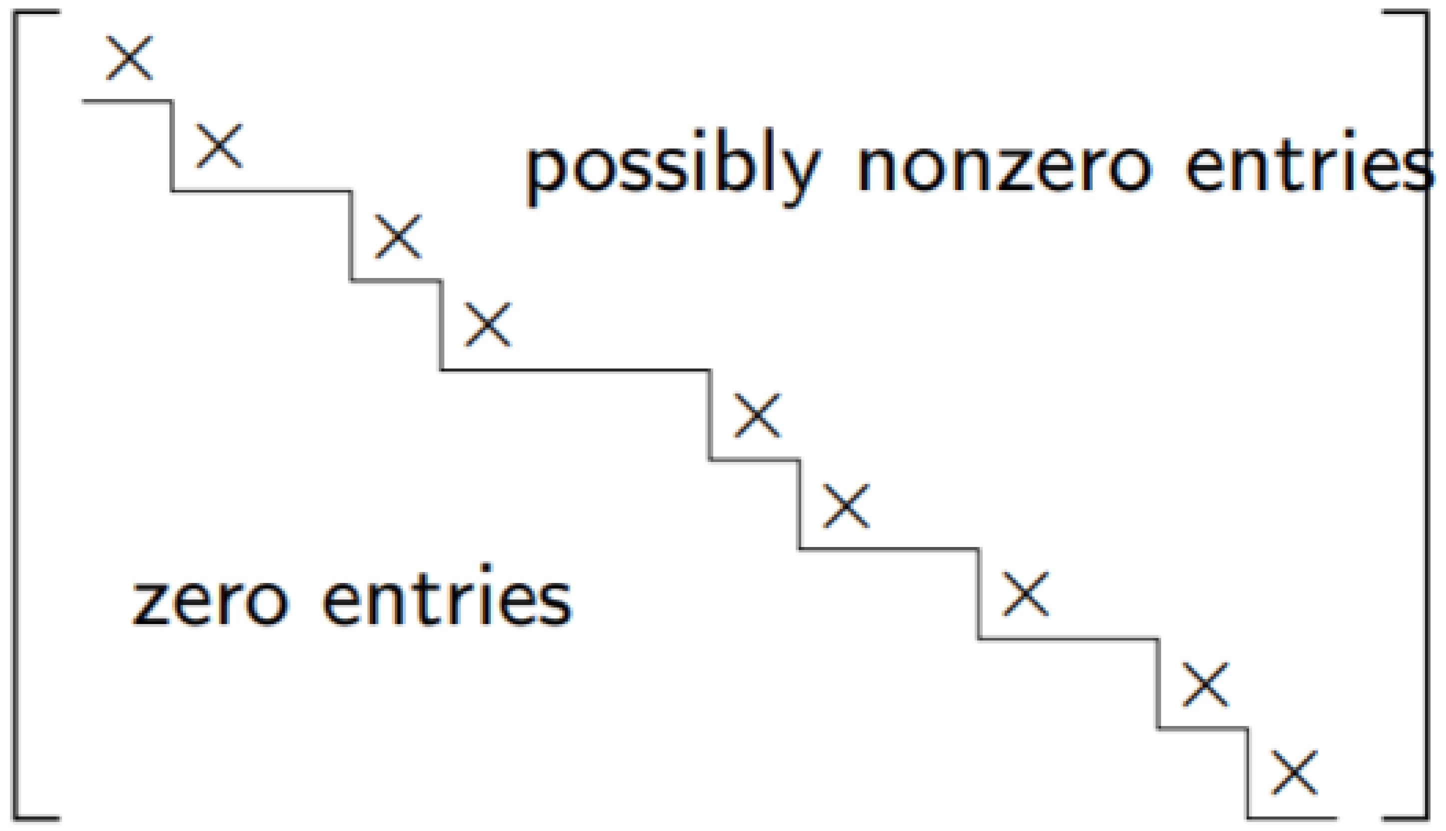
$r = 0$
for $i = 1, \ldots, n$
  $\tilde{a} = a_i - \sum_{j=1}^{r} q_j q_j^\mathsf{T} a_i$
  if $\tilde{a} \neq 0$
    $r = r + 1$
    $q_r = \tilde{a}/\|\tilde{a}\|$

Warning: modify QR when A has dependent column!

on exit,

- $q_1, \ldots, q_r$ is an orthonormal basis for **range**(A) (hence $r =$ **Rank**(A))
- each a_i is linear combination of previously generated q_j's

in matrix notation we have $A = QR$ with $Q^\mathsf{T} Q = I$ and $R \in \mathbb{R}^{r \times n}$ in *upper staircase form*



'corner' entries (shown as $\times$) are nonzero

From: pg 4-6 http://ee263.stanford.edu/lectures/qr.pdf

How to get full QR decomposition when A is tall and skinny?

'Full' QR factorization

with $A = Q_1 R_1$ the QR factorization as above, write

$$A = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix} \begin{bmatrix} R_1 \\ 0 \end{bmatrix}$$

where $\begin{bmatrix} Q_1 & Q_2 \end{bmatrix}$ is orthogonal, $i.e.$, columns of $Q_2 \in \mathbb{R}^{m\times(m-r)}$ are orthonormal, orthogonal to Q_1

to find Q_2:

- find any matrix $\tilde{A}$ s.t. $\begin{bmatrix} A & \tilde{A} \end{bmatrix}$ has rank m ($e.g.$, $\tilde{A} = I$)
- apply general Gram-Schmidt to $\begin{bmatrix} A & \tilde{A} \end{bmatrix}$
- Q_1 are orthonormal vectors obtained from columns of A
- Q_2 are orthonormal vectors obtained from extra columns ($\tilde{A}$)

$i.e.$, any set of orthonormal vectors can be *extended* to an orthonormal basis for $\mathbb{R}^m$

look ahead: QQ^T relationship to Least Squares

Ref: relating QQ^T to least squares solution (see Sec 7.1.4)
and Boyd's lecture:
https://see.stanford.edu/materials/lsoeldsee263/05-ls.pdf Pg 5-8

Least-squares via QR factorization

- $A \in \mathbf{R}^{m\times n}$ skinny, full rank
- factor as $A = QR$ with $Q^TQ = I_n$, $R \in \mathbf{R}^{n\times n}$ upper triangular, invertible
- pseudo-inverse is

$$(A^TA)^{-1}A^T = (R^TQ^TQR)^{-1}R^TQ^T = R^{-1}Q^T$$

so $x_{\rm ls} = R^{-1}Q^Ty$

- projection on $\mathcal{R}(A)$ given by matrix

$$A(A^TA)^{-1}A^T = AR^{-1}Q^T = QQ^T$$

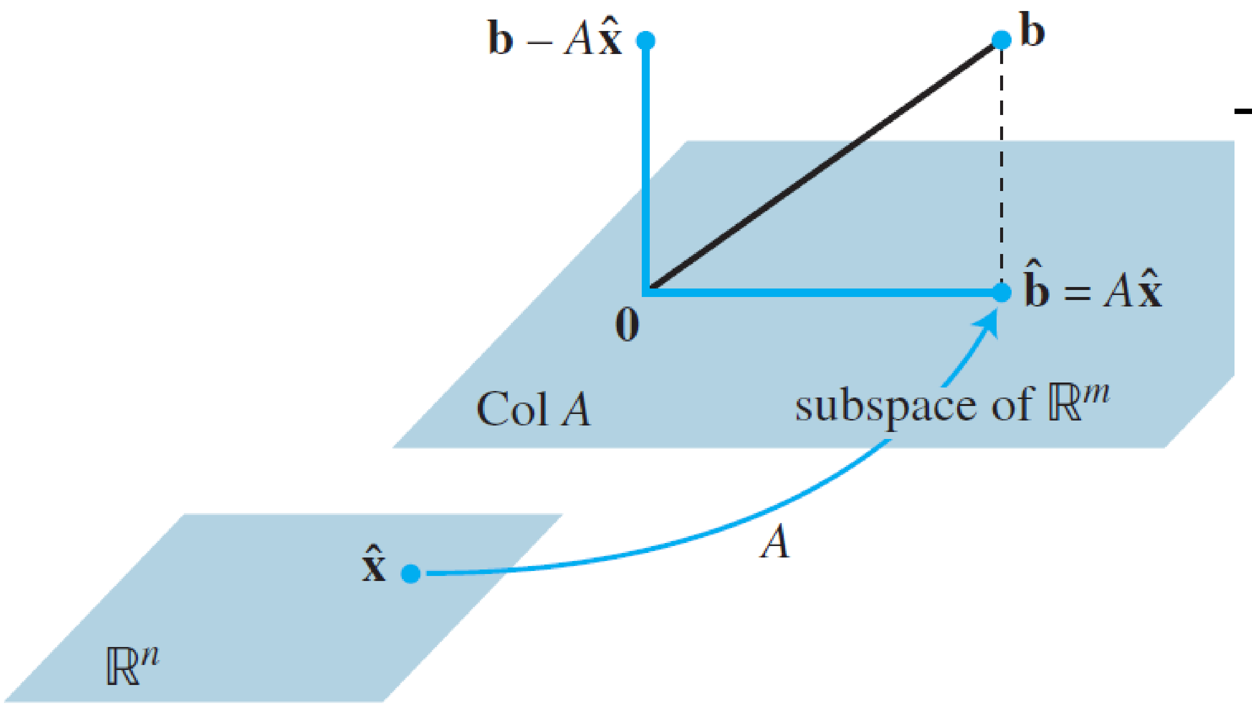


FIGURE 2 The least-squares solution $\hat{\mathbf{x}}$ is in $\mathbb{R}^n$.

THEOREM 14

Let A be an $m \times n$ matrix. The following statements are logically equivalent:

a. The equation $A\mathbf{x} = \mathbf{b}$ has a unique least-squares solution for each $\mathbf{b}$ in $\mathbb{R}^m$.

b. The columns of A are linearly indpendent.

c. The matrix A^TA is invertible.

When these statements are true, the least-squares solution $\hat{\mathbf{x}}$ is given by

$$\hat{\mathbf{x}} = (A^TA)^{-1}A^T\mathbf{b} \tag{4}$$

Therefore if $\hat{\mathrm{x}} = (A^TA)^{-1}A^Tb$, then

$\hat{b} = A\hat{x}$, means

$\hat{b} = A(A^TA)^{-1}A^T\ b$

And this matrix $A(A^TA)^{-1}A^T$ is a projection matrix, projecting b into the column space of A.

See 7.1.3 (pg 4) : projection matrix of least squares

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}}_{b}$$

CX1104: **Linear Algebra for Computing**

Chap. No : **6.3.3**

Lecture : **Orthogonality**

Topic : **Gram–Schmidt Process**

Concept : **Using Matlab to get QR and 4 cases of A**

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Using Matlab for QR

We will consider only the two common cases:

i) A is a square matrix

ii) A has dimension mxn (m>n)

Using Matlab, there are 2 options, complete (full) vs economy decomposition.

Matlab

```
qr     Orthogonal-triangular decomposition.
   [Q,R] = qr(A), where A is m-by-n, produces an m-by-n upper triangular
   matrix R and an m-by-m unitary matrix Q so that A = Q*R.

   [Q,R] = qr(A,0) produces the "economy size" decomposition.
   If m>n, only the first n columns of Q and the first n rows of R are
   computed. If m<=n, this is the same as [Q,R] = qr(A).
```

numpy.linalg.qr

`linalg.qr(a, mode='reduced')`

Compute the qr factorization of a matrix.

Factor the matrix *a* as *qr*, where *q* is orthonormal and *r* is upper-triangular.

Parameters: **a** : *array_like, shape (M, N)*

Matrix to be factored.

mode : *{'reduced', 'complete', 'r', 'raw'}, optional*

If K = min(M, N), then

- 'reduced' : returns q, r with dimensions (M, K), (K, N) (default)
- 'complete' : returns q, r with dimensions (M, M), (M, N)

Note: option for numpy 'reduced' == matlab 'economy' when performing QR

Ref: 1) https://en.wikipedia.org/wiki/QR_decomposition
2) http://ee263.stanford.edu/lectures/qr.pdf

QR using Matlab and the 4 cases

We explore QR of A for the following 4 cases

1) Square A matrix (size 3x3)
   - Ex1) where A has 3 independent col
   - Ex2) where A has 2 independent col, and 1 dependent col

2) Tall A matrix (size 3x2)
   - Ex3) where A has 2 independent col
   - Ex4) where A col are dependent (col2 == 2xcol1)

Study the effect decomposition has on Q and R, as well
as selection of Matlab economy vs complete (full) QR decomposition.

Introducing A = QR for Square A

Example 1: A is 3x3 square and has independent column

Square matrix [edit]

Any real square matrix A may be decomposed as

$$A = QR,$$

where Q is an orthogonal matrix (its columns are orthogonal unit vectors meaning $Q^\mathsf{T} = Q^{-1}$) and R is an upper triangular matrix (also called right triangular matrix). If A is invertible, then the factorization is unique if we require the diagonal elements of R to be positive.

If instead A is a complex square matrix, then there is a decomposition $A = QR$ where Q is a unitary matrix (so $Q^* = Q^{-1}$).

If A has n linearly independent columns, then the first n columns of Q form an orthonormal basis for the column space of A. More generally, the first k columns of Q form an orthonormal basis for the span of the first k columns of A for any $1 \le k \le n$.[1] The fact that any column k of A only depends on the first k columns of Q is responsible for the triangular form of R.[1]

```
>> A = [1 2 3; 1 2 1; 1 1 3]

A =

     1     2     3
     1     2     1
     1     1     3

>> rank(A)

ans =

     3
```

```
>> [Q,R] = qr(A)

Q =

   -0.5774   -0.4082   -0.7071
   -0.5774   -0.4082    0.7071
   -0.5774    0.8165   -0.0000


R =

   -1.7321   -2.8868   -4.0415
         0   -0.8165    0.8165
         0         0   -1.4142
```

```
>> rank(R)

ans =

     3
```

```
>> Q*R

ans =

    1.0000    2.0000    3.0000
    1.0000    2.0000    1.0000
    1.0000    1.0000    3.0000
```

```
>> Q'*Q

ans =

    1.0000    0.0000    0.0000
    0.0000    1.0000   -0.0000
    0.0000   -0.0000    1.0000

>> Q*Q'

ans =

    1.0000   -0.0000   -0.0000
   -0.0000    1.0000   -0.0000
   -0.0000   -0.0000    1.0000
```

Matlab Notation:

$$Q' == Q^T$$

When A is square and has independent column, factorizing A to QR, then
Q'*Q == Q*Q' == identity matrix
And R which is square is invertible!

Square A with dependent col

Example 2: A is 3x3 square and
A has 1 dependent column

```
>> A = [1 2 3; 1 2 3; 1 1 2]

A =

     1     2     3
     1     2     3
     1     1     2

>> rank(A)

ans =

     2
```

```
>> [Q,R] = qr(A)

Q =

   -0.5774   -0.4082   -0.7071
   -0.5774   -0.4082    0.7071
   -0.5774    0.8165   -0.0000


R =

   -1.7321   -2.8868   -4.6188
         0   -0.8165   -0.8165
         0         0   -0.0000
```

```
>> Q*R

ans =

    1.0000    2.0000    3.0000
    1.0000    2.0000    3.0000
    1.0000    1.0000    2.0000
```

```
>> rank(R)

ans =

     2
```

```
>> Q'*Q

ans =

    1.0000    0.0000    0.0000
    0.0000    1.0000   -0.0000
    0.0000   -0.0000    1.0000

>> Q*Q'

ans =

    1.0000   -0.0000   -0.0000
   -0.0000    1.0000   -0.0000
   -0.0000   -0.0000    1.0000
```

When A is square and has dependent column,
It still can be decomposed into QR, and
Q'*Q == Q*Q' == identity matrix
BUT now, R which is square is NOT invertible!
R has rank 2 bcos A has 2 independent col.

Ref: https://en.wikipedia.org/wiki/QR_decomposition

Introducing $A = QR$ for Tall A (m>n)

Rectangular matrix [edit]

More generally, we can factor a complex $m \times n$ matrix A, with $m \geq n$, as the product of an $m \times m$ unitary matrix Q and an $m \times n$ upper triangular matrix R. As the bottom $(m-n)$ rows of an $m \times n$ upper triangular matrix consist entirely of zeroes, it is often useful to partition R, or both R and Q:

$$A = QR = Q\begin{bmatrix} R_1 \\ 0 \end{bmatrix} = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix}\begin{bmatrix} R_1 \\ 0 \end{bmatrix} = Q_1 R_1,$$

where R_1 is an $n \times n$ upper triangular matrix, 0 is an $(m - n) \times n$ zero matrix, Q_1 is $m \times n$, Q_2 is $m \times (m - n)$, and Q_1 and Q_2 both have orthogonal columns.

Golub & Van Loan (1996, §5.2) call Q_1R_1 the *thin QR factorization* of A; Trefethen and Bau call this the *reduced QR factorization*.[1] If A is of full rank n and we require that the diagonal elements of R_1 are positive then R_1 and Q_1 are unique, but in general Q_2 is not. R_1 is then equal to the upper triangular factor of the Cholesky decomposition of $A^* A$ (= A^TA if A is real).

Note:

col(Q_1)== col(A),

while

col(Q_2) = orthogonal complement of col(A)

see example 3 and 4 (in next pages)

Tall A

$$A = QR = Q\begin{bmatrix} R_1 \\ 0 \end{bmatrix} = [Q_1 \quad Q_2]\begin{bmatrix} R_1 \\ 0 \end{bmatrix} = Q_1 R_1,$$

Example 3: A is tall (3x2) and A has 2 independent column

```
A =

     1     2
     1     2
     1     1

>> [Q,R] = qr(A)

Q =

   -0.5774   -0.4082   -0.7071
   -0.5774   -0.4082    0.7071
   -0.5774    0.8165   -0.0000


R =

   -1.7321   -2.8868
         0   -0.8165
         0         0
```

```
>> Q'*Q

ans =

    1.0000    0.0000    0.0000
    0.0000    1.0000   -0.0000
    0.0000   -0.0000    1.0000

>> Q*Q'

ans =

    1.0000   -0.0000   -0.0000
   -0.0000    1.0000   -0.0000
   -0.0000   -0.0000    1.0000
```

[Q,R] = qr(A) (complete decomposition)

- Complete Q has size == 3x3 (and it an orthogonal matrix even though A is 3x2.
- R is 3x2 (row 1 and 2 of R are non-zero bcos A has 2 independent col)

Q_1 is the first 2 columns of Q (bcos A has 2 independent col)
$\text{col}(Q_1) == \text{col}(\text{A})$
Q_2 is the last column of Q

```
A =

     1     2
     1     2
     1     1

>> [Q,R] = qr(A,0)

Q =

   -0.5774   -0.4082
   -0.5774   -0.4082
   -0.5774    0.8165


R =

   -1.7321   -2.8868
         0   -0.8165
```

```
>> Q'*Q

ans =

    1.0000    0.0000
    0.0000    1.0000

>> Q*Q'

ans =

    0.5000    0.5000   -0.0000
    0.5000    0.5000    0.0000
   -0.0000    0.0000    1.0000
```

[Q,R] = qr(A,0) (using economy decomposition)
reduced Q has size == 3x2.
Note that Q'*Q = I (size (2x2))
While Q*Q' = P
a projection matrix size 3x3.

A projection matrix: see Strang lecture 16
https://www.youtube.com/watch?v=osh80YCg_GM

Tall A

$$A = QR = Q\begin{bmatrix} R_1 \\ 0 \end{bmatrix} = [Q_1 \quad Q_2]\begin{bmatrix} R_1 \\ 0 \end{bmatrix} = Q_1 R_1,$$

Example 4: A is tall (3x2) and A has dependent column

```
A =

     1     2
     1     2
     1     2

>> [Q,R] = qr(A)

Q =

   -0.5774    0.8165   -0.0000
   -0.5774   -0.4082   -0.7071
   -0.5774   -0.4082    0.7071


R =

   -1.7321   -3.4641
         0   -0.0000
         0         0
```

```
>> Q*R

ans =

    1.0000    2.0000
    1.0000    2.0000
    1.0000    2.0000
```

[Q,R] = qr(A) (complete decomposition)
Complete Q has size == 3x3 even though A is 3x2
And R is 3x2 (Only row 1 of R is non-zero, bcos there is ONLY 1 independent col in A)

Q_1 is ONLY the first column of Q (bcos A has ONLY 1 independent col), $col(Q_1) == col(A)$
Q_2 is the last 2 columns of Q

```
A =

     1     2
     1     2
     1     2

>> [Q,R] = qr(A,0)

Q =

   -0.5774    0.8165
   -0.5774   -0.4082
   -0.5774   -0.4082


R =

   -1.7321   -3.4641
         0   -0.0000
```

```
>> Q*R

ans =

    1.0000    2.0000
    1.0000    2.0000
    1.0000    2.0000
```

[Q,R] = qr(A,0) (economy decomposition)
reduced Q has size == 3x2

Maths/LA/Tut6
Orthogonality

18 October 2020

CES

Last updated: 27 Oct2021
Last updated: 18 Mar 2024

Tutorial 6 Help links

Youtube link: playlist

https://www.youtube.com/playlist?list=PLki3aFwg-9exsbmLQXdb7jvhlTOtyu9f_

PDF

Q1-6: https://www.dropbox.com/s/mj0hrsw4vqqix55/Tut6_Q1_6_ces.pdf?dl=0

Q7-11: https://www.dropbox.com/s/6lpmr2z8afckir5/Tut6_Q7_11_ces.pdf?dl=0

Q1,5) Orthogonal vs Orthonormal set of vectors

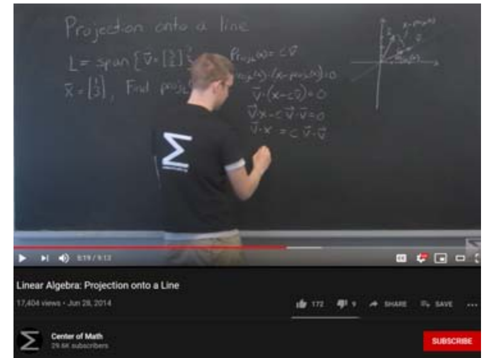
Ref:

1) Read 6.3 of UCL's writeup https://www.ucl.ac.uk/~ucahmdl/LessonPlans/Lesson10.pdf

2) Youtube (Prof Dave Explains): "Orthogonality vs Orthonormality" https://www.youtube.com/watch?v=6nqMegdbxik

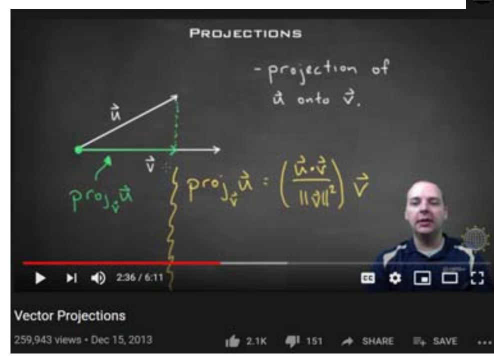
3) Center of Math, "Orthonormal basis": https://www.youtube.com/watch?v=ZJu26chXEiw

Q3,4) Examples: Videos of Projection onto a line



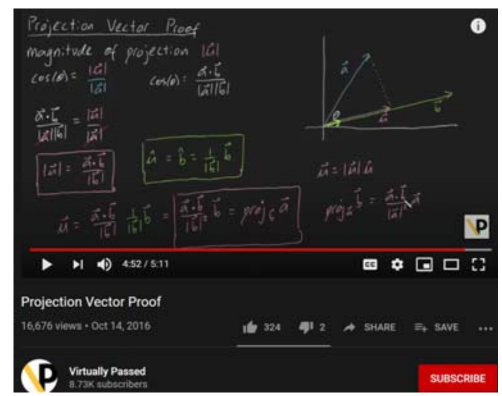
Ref

1) Projection of a vector onto a line
https://www.youtube.com/watch?v=GnvYEbaSBoY

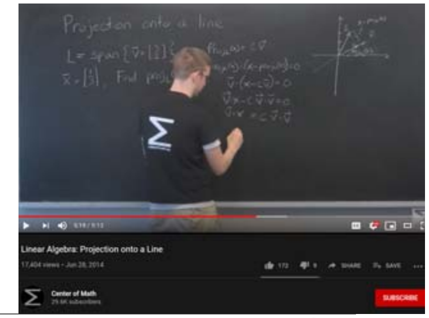


2) Firefly (Vector Projection)
https://www.youtube.com/watch?v=fqPiDICPkj8

3) Virtually Passed (Projection Vector proof)
https://www.youtube.com/watch?v=aTBtgW7U-Y8



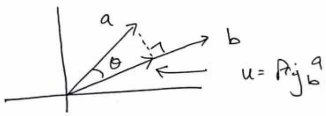
Detail Proof: Proj vector onto line



1

Proof: $Proj_b^a = \frac{a \cdot b}{|b||b|} b$

Method 1: projection vector proof by Virtually Passed - YouTube



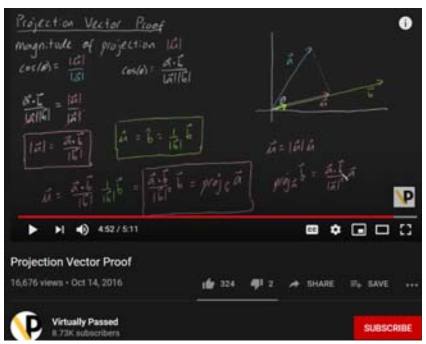
$|u|$ = magnitude of projection a onto b.

eq (1) $\cos(\theta) = \frac{|u|}{|a|}$; cosine rule for $\perp$ $\triangle$

and by dot product

$a \cdot b = |a||b| \cos\theta$

$\Rightarrow \cos(\theta) = \frac{a \cdot b}{|a||b|}$. — eq (2)



∴ equating eq (1) & eq (2)

$\frac{|u|}{|a|} = \frac{a \cdot b}{|a||b|}$; we need $|u|$

$\Rightarrow |u| = \frac{a \cdot b}{|b|}$.

now $Proj_b^a$ is a vector in direction of u, of b.

$\underbrace{\hat{u}}_{\text{normal length}} = \hat{b} = \frac{b}{|b|}$

∴ $u = |u| \hat{b} = \frac{a \cdot b}{|b|} \cdot \frac{b}{|b|}$

$u = Proj_b^a = \underbrace{\left(\frac{a \cdot b}{|b||b|}\right)}_{\alpha \text{ scaler}} b$ ☒

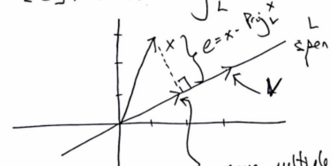
2

Proof: $Proj_L^x$

Method 2: projection onto a line by "Center of Math" YouTube.

$L = span\{\bar{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}\}$.

$x = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$. Find $Proj_L^x$



some multiple of v

$= Proj_L^x = Proj_v^x$

$= cv$

$e = x - Proj_L^x$; the error.

$x = Proj_L^x + e$

Also $e \perp v$, ∴

$\Rightarrow \underbrace{(x - Proj_L^x)}_{e} \cdot v = 0$, remember $Proj_L^x = cv$

$= v \cdot x - v \cdot (c v) = 0$.

$\Rightarrow v \cdot x = c(v \cdot v) = 0$.

$\Rightarrow c = \frac{v \cdot x}{v \cdot v} = \frac{v \cdot x}{|v|^2}$ ☒

$= \frac{v \cdot x}{\|v\|^2}$ ☒

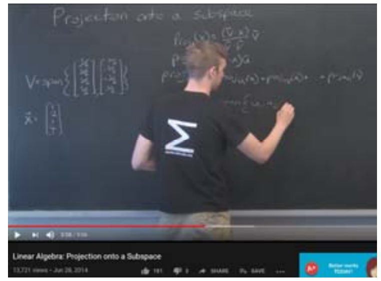
∴ $Proj_v^x = cv = \frac{v \cdot x}{\|v\|^2} v$ ☒

$Proj_v^x = \begin{bmatrix} 27/13 \\ 18/13 \end{bmatrix}$ ☒

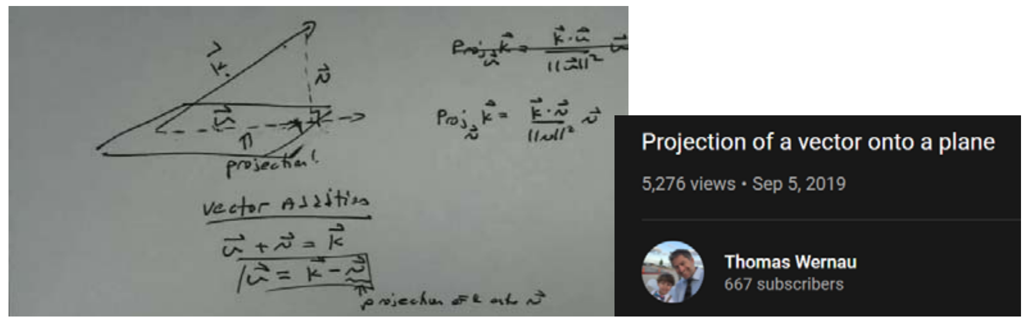
Q3,4,8) Examples: Projecting a vector onto a subspace

Ref:

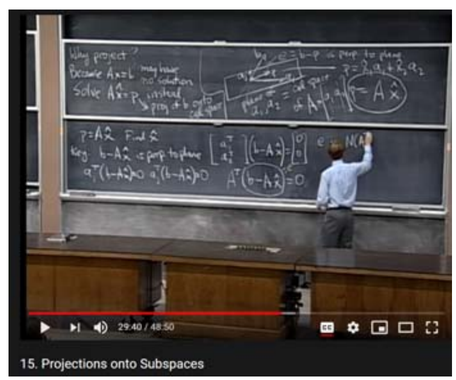
1. Center of Math: Projection onto a subspace (a direct way): https://www.youtube.com/watch?v=zZW6JV4yA54



2. Thomas Wernau: Projection of a vector onto a plane (a roundabout way – using the normal vector to the plane) : https://www.youtube.com/watch?v=qz3Q3v84k9Y



3. MIT Strang explains in L15 (This will lead to chapter 7 – least squares) https://www.youtube.com/watch?v=Y_Ac6KiQ1t0



FAQ) Confusion with the name "Orthogonal Matrix"

a) Orthogonal matrix are square and its columns form an orthonormal set, hence its inverse is simply its transpose: https://en.wikipedia.org/wiki/Orthogonal_matrix
We like orthogonal matrix because its inverse is its transpose, and it simplify orthogonal decomposition.

b) When we have Ax = b, where A is orthogonal,

- then the number of element (column length) of b MUST be the same as column length of A
- Then A^{-1} exist and is A^T and pre-multiply Ax = b by A^{-1} to solve for x , we have:
  $A^{-1}Ax \quad = \quad A^T Ax$

=> $x = A^T b$

c) A better name for orthogonal matrix is SQUARE Orthonormal matrix
Because it is a square matrix with a set of orthonormal columns.
This is the confusion of the word orthogonal matrix:

- See: pg 5.3 and 5.6 of
- https://www.seas.ucla.edu/~vandenbe/133A/lectures/orthogonal.pdf

Q11) Help in QR –
confusion in the word orthogonal for Q

QR decomposition decomposes a rectangle (or square) matrix into two matrixes Q,R, where

- Q has orthonormal columns,

  Note: if matrix Q is square , then Q is an orthogonal matrix bcos its columns are also orthonormal.

  If matrix Q is not square, then Q'*Q = Identity Matrix

  BUT Q*Q' is not equal to Identity Matrix (its col are only orthonormal) – it is a projection matrix on colSpace(A)
- R is upper triangle

QR may be confusing because

- Size of Q matrix :
  - Depending on size of matrix A and variant in implementing QR decomposition (complete vs reduced), the matrix Q has different sizes
- Invertibility of R matrix :
  - if columns of A are linearly or not linearly independent, it will affect invertibility of R,
  - also depends on variant of implementing QR (complete vs reduced)

See code : test_QR.m

Q11)
[Q,R] = qr(A) vs [Q,R] = qr(A,0) in Matlab

https://www.mathworks.com/help/matlab/ref/qr.html

Given A = mxn matrix where m> n,
Matlab's QR factorization will decompose
- Q into full square matrix for Q of dimension (mxm), and
- R into rectangle matrix mxn, the if the so-called 'full' or 'complete' (numpy-notation) qr decomposition.

Hence, some books will immediately say that Q is an orthogonal matrix, since its columns form an orthonormal set for full decoposition.

However: Economy qr –
[Q,R] = qr(A,0) ; the 0 indicates economy selection.
The Q will be dimesion mxn, and R will be square (nxn)

My Matlab code
https://www.dropbox.com/s/1xbzb7rpjty2y4u/test_QR.m?dl=0

```
>> A = [1 2; 3 4; 5 6]

A =

     1     2
     3     4
     5     6
```

```
>> [Q,R] = qr(A)

Q =

   -0.1690    0.8971    0.4082
   -0.5071    0.2760   -0.8165
   -0.8452   -0.3450    0.4082


R =

   -5.9161   -7.4374
         0    0.8281
         0         0
```

```
>> Q'*Q

ans =

    1.0000   -0.0000   -0.0000
   -0.0000    1.0000    0.0000
   -0.0000    0.0000    1.0000

>> Q*Q'

ans =

    1.0000    0.0000    0.0000
    0.0000    1.0000    0.0000
    0.0000    0.0000    1.0000
```

```
>> [Qecon,Recon] = qr(A,0)

Qecon =

   -0.1690    0.8971
   -0.5071    0.2760
   -0.8452   -0.3450


Recon =

   -5.9161   -7.4374
         0    0.8281
```

```
>> Qecon'*Qecon

ans =

    1.0000   -0.0000
   -0.0000    1.0000

>> Qecon*Qecon'

ans =

    0.8333    0.3333   -0.1667
    0.3333    0.3333    0.3333
   -0.1667    0.3333    0.8333
```

Q11)
[Q,R] = numpy.linalg.qr(A,'reduced' vs 'complete')

https://numpy.org/doc/stable/reference/generated/numpy.linalg.qr.html

In numpy, the choice of full vs economy uses the parameter as 'complete' vs 'reduced'

numpy.linalg.qr¶

numpy.linalg.qr(*a*, *mode='reduced'*)

Compute the qr factorization of a matrix.

Factor the matrix *a* as *qr*, where *q* is orthonormal and *r* is upper-triangular.

Parameters:

a : *array_like, shape (M, N)*

Matrix to be factored.

mode : *{'reduced', 'complete', 'r', 'raw'}, optional*

If K = min(M, N), then

- 'reduced' : returns q, r with dimensions (M, K), (K, N) (default)
- 'complete' : returns q, r with dimensions (M, M), (M, N)

In [1]:
```
import numpy as np
A = np.array([[1,2], [3,4],[5,6]])
print(A)
```

```
[[1 2]
 [3 4]
 [5 6]]
```

In [2]:
```
[Qfull,Rfull] = np.linalg.qr(A,'complete')
print(Qfull)
print(Rfull)
```

```
[[-0.16903085  0.89708523  0.40824829]
 [-0.50709255  0.27602622 -0.81649658]
 [-0.84515425 -0.34503278  0.40824829]]
[[-5.91607978 -7.43735744]
 [ 0.          0.82807867]
 [ 0.          0.        ]]
```

In [3]:
```
[Qreduced,Rreduced] = np.linalg.qr(A,'reduced')
print(Qreduced)
print(Rreduced)
```

```
[[-0.16903085  0.89708523]
 [-0.50709255  0.27602622]
 [-0.84515425 -0.34503278]]
[[-5.91607978 -7.43735744]
 [ 0.          0.82807867]]
```

myPythonCode =
https://www.dropbox.com/s/apmy59m8kn5hoau/test_QR_python.ipynb?dl=0

Q11) implementing your own QR decomposition (full)

If you wish to implement your own QR decomposition on A a tall and thin matrix of mxn dimension,

- note that using GS will stop after n columns.

- hence to stop GS from terminating, augment your A matrix with identity and perform GS. See discussion (right slide)

'Full' QR factorization

with $A = Q_1 R_1$ the QR factorization as above, write

$$A = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix} \begin{bmatrix} R_1 \\ 0 \end{bmatrix}$$

where $[Q_1 \; Q_2]$ is orthogonal, *i.e.*, columns of $Q_2 \in \mathbf{R}^{n \times (n-r)}$ are orthonormal, orthogonal to Q_1

to find Q_2:

- find any matrix $\tilde{A}$ s.t. $[A \; \tilde{A}]$ is full rank (*e.g.*, $\tilde{A} = I$)
- apply general Gram-Schmidt to $[A \; \tilde{A}]$
- Q_1 are orthonormal vectors obtained from columns of A
- Q_2 are orthonormal vectors obtained from extra columns ($\tilde{A}$)

Orthonormal sets of vectors and QR factorization 4–20

in pg 4-20
https://see.stanford.edu/materials/lsoeldsee263/04-qr.pdf

Some questions relating to rank and nullspace

1) Why is $A^T A$ invertible when A has full column rank?

https://www.youtube.com/watch?v=ESSMQH6Y5OA

2) Null space of AA^T is the same as $N(A)$

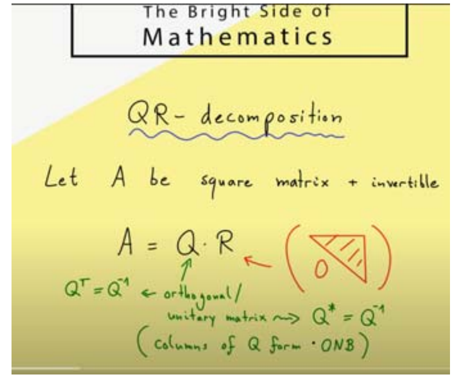
https://math.stackexchange.com/questions/66560/null-space-for-aat-is-the-same-as-null-space-for-at

3) Sum of 2 rank 1 matrix with certain properties in == rank 2

https://math.stackexchange.com/questions/2623005/sum-of-two-rank-1-matrices-with-some-property-gives-rank-2-matrix

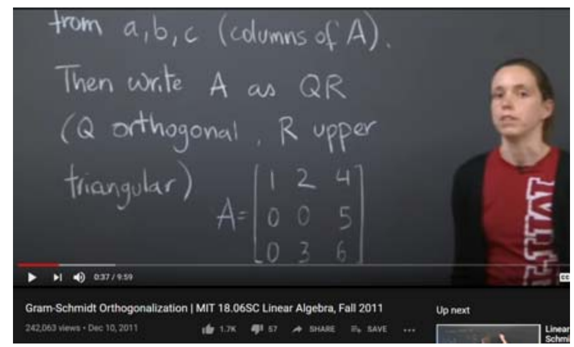
Some videos on QR

1) The bright sight of mathematics “QR on a Square matrix (step by step)” https://www.youtube.com/watch?v=FAnNBw7d0vg



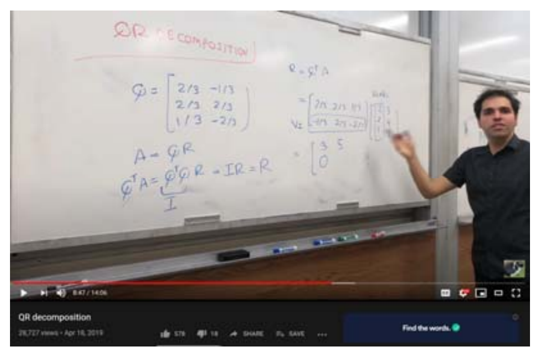
2) MIT TA doing a 3x3 matrix QR

https://www.youtube.com/watch?v=TRktLuAktBQ



3) Dr Peyam doing a 3x2 QR decomposition

https://www.youtube.com/watch?v=J41Ypt6Mftc



Q11) Orthogonal vs Unitary

In some literature, Q is sometimes called an unitary matrix instead of orthogonal matrix. This is because if A is complex, then the resultant Q will be complex. And The real analogue of a unitary matrix is an orthogonal matrix.

See:

https://www.quora.com/What-is-the-difference-between-a-unitary-and-orthogonal-matrix